Nowcasting Net Asset Values: The Case of Private Equity*

Gregory W. Brown† Eric Ghysels‡ Oleg Gredil§

July 21, 2022

Abstract

We estimate unsmoothed private equity Net Asset Values (NAVs) at the weekly frequency for individual funds. Using simulations and large samples of buyout and venture funds, we show that our method yields superior estimates of NAVs relative to simple approaches based on extrapolation of reported NAVs. The market beta of an average buyout [venture] fund is around 1.0 [1.4], and the total risk is 33% [40%] per year. The risk-return profile of the funds varies significantly over time and across funds. Risk-taking and reporting quality appear to persist by manager.

Keywords: Private Equity, Venture Capital, Leveraged Buyouts, Institutional Investors, State Space Models, Nowcasting, Risk Management

JEL Classification: C3, C4, G11, G17, G23, G24, G30

*We are grateful to the Private Equity Research Consortium and the Institute for Private Capital for support and data access. We thank Wendy Hu for outstanding research assistance. We thank the Editor and the two Referees, Mike Aguilar, Barry Griffiths, Ian Roberts, Erkko Etula, Farshid Asl, Peter Shepard, Sophie Shive, Tray Spilker, and Thomas Gilbert, as well as seminar participants at Chapel Hill and Oxford Private Equity Research Symposiums, Goldman Sachs SQAA group, 2020 AFA Annual Meeting, 2020 EFA Annual Meeting, and 2020 NFA Annual Meeting for useful comments and suggestions.

†University of North Carolina at Chapel Hill, Kenan-Flagler Business School and Frank Hawkins Kenan Institute of Private Enterprise, gregwbrown@unc.edu, (919) 962-9250

‡Department of Finance, Kenan-Flagler Business School, Department of Economics, University of North Carolina at Chapel Hill, Frank Hawkins Kenan Institute of Private Enterprise, and Centre for Economic Policy Research (CEPR), eghysels@unc.edu, (919) 962-9810

§Tulane University, Freeman School of Business, ogredil@tulane.edu, (504) 314-7567

1 Introduction

Valuing illiquid assets accurately is hard but necessary for many critical investment decisions made by institutional investors.[1] Private equity (PE) investments are a prime example. In most cases, secondary markets are undeveloped for private equity and likely reflect the marginal utility to trade of an unrepresentative investor.[2] Instead various stakeholders rely heavily on infrequently-reported Net Asset Values (NAVs) provided by fund managers. However, investors observe additional information that can be used to generate unbiased and more timely higher-frequency estimates of the true value of a fund. This paper develops a method of how to use other types of data *jointly* with reported NAVs in a unified framework to (i) learn about risk, return, and reporting quality characteristics of individual funds, and (ii) to estimate unbiased asset values at the relevant data arrival frequency (in our case weekly), i.e., to *nowcast* PE fund "true" NAVs. Nowcasting is the prediction of the present, the very near future and the very recent past. We expand on the methods used in the literature as nowcasting NAVs is more complex than macroeconomic variables like GDP growth (the most studied example). In particular, we show how (i) comparable asset returns and (ii) cash flows between fund limited partners (LPs) and general partners (GPs) can be used to generate nowcasts. Moreover, our method can potentially also be expanded to include other types of information including characteristics of individual assets held by funds, comparable private transactions, and occasional secondary trades.

We model "true" asset values of a fund and, correspondingly, "true" returns as the latent factor in a sparse-data State Space Model (SSM) estimated at (i) high frequencies (relative to quarterly fund reporting) and (ii) at an individual PE fund level. The novelty of our approach rests on two key pillars. One is that we use the individual fund-level data on cash flows and reported NAVs *simultaneously* to identify how NAVs are "smoothed" in order to estimate the fund's systematic and idiosyncratic risk exposures. This is different from existing work featuring the complete fund-level data that either (i) relies on reported NAVs and attempts to remove autocorrelation induced by smoothing via a distributed lag market-model (Woodward and Hall, 2004; Ewens et al., 2013;

[1] See Brito (1977); Longstaff (2001); Gârleanu (2009); Ang et al. (2014) among others.
[2] See Ewens et al. (2013); Brown et al. (2019) and Nadauld et al. (2019).

Goetzmann et al., 2018), or (ii) disregards the NAV information and relies solely on funds' realized cash flow data (see, for example, Driessen et al., 2012; Franzoni et al., 2012; Buchner and Stucke, 2014; Ang et al., 2018). The second pillar is that we unsmooth weekly cumulative returns (rather than quarterly returns, see also Ang 2014). This is more consistent with the microfoundations of the smoothing bias whereby appraisers look back for relevant events *within* the past quarter to arrive at their valuation assessment. We show that NAV smoothing intensity persists by fund manager, as does the risk-taking.

We evaluate the performance of our method not only via simulated data but also using real PE fund data from Burgiss. In particular, we examine out-of-sample realized cash flows versus the levels implied by the model for 2,513 separate funds. For 69% to 89% (depending on the metric) of these funds, the model outperforms a naïve, but commonly-used, approach whereby reported NAVs are extrapolated using cash flows and returns of a comparable public asset. With partial imputation of auxiliary parameters using peer funds, we can estimate fund-level SSMs for 3,911 funds (over 95% of the sample) with a 74-to-94% improvement rate relative to the naïve approach (again, depending on the metric). Peer funds are defined as having the same strategy and industry, a vintage within one year and having the same size tercile as the estimand fund.

Our performance assessment metrics are based on the idea that if the fund's true returns were observed and used to discount fund cash flows (and the true net asset values), the fund to-date Public Market Equivalents (PMEs) (Kaplan and Schoar, 2005) computed using these series would be equal to one. Furthermore, we show that, given fund-level SSM estimates, the in-sample performance is a good predictor of the out-of-sample performance, yielding useful criteria for model selection. Using our estimate of fund NAVs reduces the out-of-sample forecast error of subsequent cash flows by 59–65% relative to the as-reported NAVs of a typical fund and generates filtered weekly returns with near zero autocorrelation and realistic standard deviations (e.g., about 29% per year for a typical buyout fund and 34% for a typical venture fund).

In addition to our methodological contributions, we generate important new findings about fund-level risk and return. For example, we find fund-level systematic risk (beta) to be lower than many other recent studies. The average is around 1.0 for buyout funds and 1.40 for venture funds,

as opposed to the pooled estimates of 1.25 and 1.80, respectively, in Ang et al. (2018) even though we use their estimates to set our profile grid for betas. Part of this difference results from the fact that most studies do not actually estimate the market beta of a fund over its life time but rather the beta of the panel of time-aggregated fund cash flows. The distinction is important when cash flows from funds are clustered in time and contribute greater weight to the panel (e.g. venture in the late 1990s). Panel-based estimates generally will not reflect the beta of an average fund in this case. We provide fund cross-section and time-series index analyses to support this claim. We also show that "nudging" the estimation to feature higher systematic risk exposures, as well as ignoring the heterogeneity in other parameters in the cross-section of funds, results in inferior nowcasting performance in-sample and out-of-sample. In other words, higher systematic risk assumptions are less consistent with the fund-level cash flow realizations.

We find notable variation in systematic and idiosyncratic risk exposures across vintage years. For example, we estimate betas in the vicinity of 2.0 for half of venture funds incepted during the late 1990s, but three-quarters of venture funds incepted during 2000–2006 have beta estimates of less than 1.5. We see an increase in systematic risk-taking by venture funds incepted after 2009. We also observe meaningful trends for the level of idiosyncratic risk of venture funds (which we measure as a multiple of volatility of a matched public benchmark). For example, the risk multiple dipped to twice the benchmark level after the dotcom bust before growing to three times the benchmark level around the Global Financial Crisis and then reversing course again for 2012-2014 vintages. For buyout funds, we note a decrease in systematic risk relative to the levels in the early 1990s and a spike in idiosyncratic risk around 2003-2005. Finally, we note that the trends in PE fund abnormal returns are consistent with those documented in prior literature.

Our paper contributes to recent studies that develop "matrix pricing" for portfolios of PE funds using either secondary market transactions coupled with a selection model as in Boyer et al. (2018) or a replication of PE fund returns using holdings matched to comparable public firms as in Stafford (2022). Our methodology effectively nests both approaches while allowing for fund-level heterogeneity in risk exposures and taking a more structural approach to modeling the fund return process during the periods in which the fund-related data are missing. We also complement

contemporaneous work by Gupta and Van Nieuwerburgh (2021), who use machine learning tools to estimate multi-factor time-varying exposures from a panel of PE fund cash flows as opposed to the bottom-up approach that matches on portfolio company characteristics. Their approach can be used to construct the comparable asset in our approach. Our method also supports using round-level valuations once they are adjusted for differences in the contractual rights, as shown in Gornall and Strebulaev (2020, 2021).

As one of many potential applications of our fund-level approach, we demonstrate how asset allocation decisions using nowcasted NAVs would have been improved around the 2008 financial crisis (Section 5.1). We argue this case study to be instructive for the valuation challenges that PE fund investors face during significant swings in market valuations.

2 Intuition on Methodology

In this section, we illustrate the need for nowcasting in PE fund investing and provide an intuitive overview of our methodology.[3] Consider an investor who manages a portfolio that includes a specific PE fund. On an ongoing basis the investor needs to make portfolio decisions that rely, in part, on estimates of the PE fund's value (and potentially other characteristics like risk, etc). Because the fund is not publicly traded, the investor's inferences about fund value are confounded by low-frequency reporting (e.g., quarterly) with long delays (4-20 weeks) as well as appraisal errors. For example, suppose the investor seeks to evaluate her overall portfolio situation on March 31, 2020 amid the broad equity indicies being down 23% year-to-date. The latest NAV report from the fund arrived in early February and valued the investor's stake at $100 million as of December 31st, 2019. Furthermore, the fund made a distribution on February 7, 2020 of $15 million. This case underscores a typical problem that PE fund investors face – the latest available NAV reports from PE managers are biased and too outdated for real-time portfolio reporting or analysis. Therefore, even if unwittingly, PE fund investors will engage in some sort of nowcasting of the PE fund NAVs.

We use a simulated fund to highlight the contributions of our paper. Panel A of Figure 1 shows a few reported NAVs (red dots), cash distributions (red bars) and unobservable "true" NAVs (blue

[3] Matlab code for the illustrative example discussed in this section is posted on GitHub (available at https://github.com/orgredil/Nowcasting-PE-NAVs/tree/main).

line) for a hypothetical fund around its ninth year of existence. We note some important features of the fund. First, distributions occur periodically and not necessarily at quarter-ends when NAVs are reported. Accordingly, the true fund NAVs drop by the amount of the distribution on the distribution date. Second, the reported NAVs, in general do not correspond to the true NAVs on the reporting dates and instead will be related to some average of stale portfolio asset appraisals, as well as some mean-zero valuation errors unrelated to past or future returns (henceforth, "reporting noise").

2.1 Naïve nowcasting benchmark

One common approach by practitioners, which we call naïve nowcasting, is to "grow" the latest NAV report using a public market rate of return and subtract (add) the value of distributions (contributions) as they occur. So for our example, the market was up by 4% through February 7. Thus, \$15 million distribution would be subtracted from \$104 million and the new estimated NAV of \$89 million would be multiplied by (1-0.23)/(1+0.04) to obtain a March 31 nowcasted NAV of \$65.9 million.

Panel B of Figure 1 shows what the naïve nowcasts would look like for this fund. The higher-frequency values always tie back to the reported NAVs (which are stale) and evolve according to the return pattern of the comparable asset. In practice, naïve nowcasting is even more challenging given reported NAV values are only observed with substantial delay. Therefore, in practice, the naïve nowcasts continue to depend on the NAV two-quarters prior for quite a while after the end of the quarter. While intuitive and simple to calculate, the naïve nowcasting method makes the following strong assumptions:

A1. the NAV report reflects the true asset value (or at least an unbiased estimate),

A2. the returns of the benchmark fully describe those of the fund, and therefore the systematic risk of the fund equals that of the selected public benchmark.

Finally, in the Figure 1 example, the reported NAVs happened to be not particularly far from the true values. More specifically the RMSE for the naïve nowcast in the illustrative example is 0.331 (on average during all weeks 455 through 507).

2.2 State space model nowcasting

The core intuition behind our SSM-based approach proposed in this paper is related to that of the naïve nowcast, namely the asset value nowcast represents a "blend" of fund NAV reports, cash flow realizations, and comparable public asset returns. However, the blend that we propose provides more realistic and unbiased estimates and is an optimal combination of the available information. There are several aspects of our method that are novel and important:

S1. The fund cash flow and NAV reports are used jointly as distinct data points,

S2. We unsmooth asset values (as opposed to periodic returns) at a higher frequency and allow for a time-varying bias,

S3. There is no pooling of fund-specific series but instead a partial imputation of hard-to-identify parameters from peer funds with better data.

Unlike the literature cited in the Introduction, S1 says that we take the fund's distributions and NAV reports as distinct observations, recognizing they each separately contain relevant information. The fundamental intuition behind our approach is as follows: While NAVs are informative, they are constructed with error using stale data and therefore are predictably biased estimates of the true fund value. On the other hand, distributions, because they represent cash flows from actual transactions, provide accurate information on the true value but only represent part of total value. Consequently, jointly observing distributions and NAVs, allows for the identification of fund cumulative return paths (and hence the asset values) via a mapping of cash flows to reported NAVs. So, for example, suppose an investor knew portfolio-level marks right before a distribution was made. In this case, it is possible to compute that distribution's fraction of the unsmoothed portfolio value. Comparing this to the reported NAVs identifies the smoothing function, the residual bias variance, and hence produces the unsmoothed path of returns. Consequently, S1 is fundamentally different from A1 which assumes unbaised NAVs.

Turning to S2, we note that a number of papers have proposed methods to unsmooth returns, see e.g. Geltner (1991, 1993), Getmansky et al. (2004) and Couts et al. (2020). These methods rely on fixed-coefficient moving average representations of the observed return process and therefore do not take into account that the smoothing of PE returns changes through time. Our method

produces, among other things, unsmoothed periodic returns as well, but we model the smoothing of reported NAVs rather than the smoothing of periodic returns. This is an important distinction that allows us to interpret the reported NAVs as a weighted-average of valuation snapshots taken at different points in the past. Most of the weight will come from weeks *within* the past quarter (rather than from values at the end of previous quarters) corresponding to a realistic assumption about how the actual appraisal of fund assets is performed. We examine weekly valuations which we find is a good balance between a desire for higher-frquency valuations and computational complexity.

To further clarify what is novel about S2 and how we unsmooth NAVs, we highlight a few equations of our model (the full model specification introduced in the next section):

$$\bar{r}_{0:t} = \Big(1 - \lambda(\cdot)_t\Big) r_{0:t} + \lambda(\cdot)_t \bar{r}_{0:t-1}, \tag{1}$$

$$\lambda(\cdot)_t := \lambda \cdot (1 - w_t), \tag{2}$$

The first equation determines how $\bar{r}_{0:t}$, the smoothed log returns from inception until t (that the reported NAVs map to), relate to the unsmoothed latent log returns $r_{0:t}$ over the same time span. Thinking for the moment in terms of a constant $\lambda \in (.01, 0.99)$ the first equation represents an exponentially-weighted moving-average scheme. The closer λ is to 1, the more stale are the reported NAVs. Conversely, values closer to zero erase the weight of past returns and the two series, $\bar{r}_{0:t}$ and $r_{0:t}$, become virtually indistinguishable. Note that λ in equation (1) is actually not a constant as $\lambda(\cdot)_t$ pertains to λ at time t, where we use lower case letters followed by $(\cdot)$ to refer to a scalar-valued function involving data. As follows from the second equation, in this case the data are $w_t \in [0, 1]$, which indicate the fraction that the cash flow in week t comprises in the naïve nowcasts of fund asset values. Therefore, the weeks in which a fund makes new investments or distributes capital back to fund investors, receive higher weight (i.e., are "remembered more") in the subsequent NAV reports. Two specific procedures advocated by the American Institute of Certified Public Accountants, namely "Backtesting" and "Calibration" (as previously referenced in footnote 6) provide a microfoundation for this modeling choice, which we find to notably improve the nowcasting performance. The fund example appearing in Figure 1 has $\lambda = 0.9$, yet the weight on the past returns for week 489 is only 0.524 when the distributions comprise 72% of the naïve

NAV nowcast corresponding to $w_t = 0.4178$.[4]

Finally, S3 is also distinct from the literature which pools cross-sectional fund data. We estimate parameters and produce nowcasts on a fund-by-fund basis for two reasons. One is conceptual—we want to test for significant heterogeneity among funds without assuming that such heterogeneity actually does not affect inference about the average fund characteristic. For example, we show that the beta of the average fund is lower than the beta of the index return series because funds with higher betas contribute more to the index variation. The second reason for not to pool fund-level observations is more technical but also important for the estimation quality (and nowcasts' interpretability). Since PE funds incur interim capital calls and distributions, cumulative returns are generally not equal to the asset values scaled by initial investment. This is why we recursively estimate a mapping function M_t between cumulative returns and the asset value of the fund. The mapping function is the same for distributions and NAVs for a given fund, and satisfies:

$$M_t = \exp\{r_{0:t}\} \,/\, V_t, \tag{3}$$

$$M_t/M_{t-1} = (V_t + D_t - C_t) \,/\, V_t, \tag{4}$$

where V_t is the latent true value of the fund assets, while C_t and D_t are, respectively, capital calls and distributions by the fund.[5] Just after the fund makes a new capital call [distribution], M_t decreases [increases] and stays constant until the next cash flow occurrence, because the change in value of fund assets is driven solely by the fund returns between cash flows. Pooling data across multiple funds complicates the estimation of M_t and propagates uncertainty that stems from the funds' heterogeneity in the distributions' pattern and reporting quality. Instead of pooling data across funds, we pool fund-level parameter estimates to partially impute some SSM parameters that are otherwise difficult to identify well at the fund level. Our SSM-based estimates will have jumps both at NAV reporting and at distribution dates just like the naïve nowcasts, however the magnitudes and sometimes even the direction of these jumps are different from those of naïve nowcasts. For example, Panel C of Figure 1 shows that our SMM-based NAVs jump downwards

[4] One critical issue with time-varying data-driven parameters such as $\lambda(\cdot)_t$ is the potential of mis-specification such that using a constant parameter might yield better results for certain applications. A well-known example is the CAPM with time-varying betas versus constant betas, see Ghysels (1998) for further discussion.

[5] Consistent with NAV accounting, we assume that value includes capital calls in period t, but excludes distributions made in that period, i.e., $V_t = V_{t-1}R_t - D_t + C_t$, where R_t is the gross return from $t-1$.

following the NAV report at week 481 but jump upwards at week 494. In both, the direction is seemingly independent from how the reported NAVs compared to the nowcasts just before the date. This happens because the SSM takes into account that the reported NAVs are stale and extracts only a signal about the true NAV values. Finally, we note that the RMSE of our SSM-based nowcasts in this example is 0.165.

2.3 Comparing the naïve and state space model nowcasts

We also examine the errors for both methods, namely the difference between the true NAVs and, respectively, naïve and SSM nowcasts scaled by the true NAVs. As evident from Panel D, at both aforementioned jump occurrences in weeks 481 and 494, the SSM nowcasts move closer to the true values, while exhibiting 53% smaller RMSE right before the NAV reports across the weeks plotted. Similarly with distributions, SSM-based nowcasts jump more (e.g., week 474) or less (e.g., week 497) than the amount distributed, depending on how the distributions compare with the previous nowcast, given the expected fraction of assets that the distributions represent.

To better understand why our method is superior to the naïve method we further explore the implications of Assumptions A1-A2. For example, Assumption A1 is particularly strong in light of the well-documented smoothness of appraisal-based values. In PE settings the appraisals are normally derived from comparable transactions over previous months, historical accounting data, etc., that are by their nature stale.[6] The lags built into the process result in quarterly NAV estimates which are both predictably stale and systematically biased.

Recall that the RMSE for the naïve nowcast in the example is 0.331 and that of our SSM-based nowcasts is 0.165, which is a 50% reduction. From the Assumptions A1-A2 we can think of a decomposition along the following lines: (a) the part of the RMSE attributable to the use of as-reported NAVs and (b) the part of RMSE attributable to the use of comparable asset returns. Let us start with (a) and examine what happens if we use SSM-based NAVs, which means we correct the biases in the NAVs, but maintain the comparable asset returns. This yields a RMSE of 0.292, as opposed 0.331. Regarding (b), if we use SSM-based returns instead of comparable asset ones,

[6] In Section A.7 of the Online Appendix we provide excerpts from the 688-page guide for valuation of venture capital and private equity funds used by appraisal professionals, the American Institute of Certified Public Accountants.

the RMSE is 0.233, which is closer to 0.165 of the fully SSM-based nowcasts. About 90% of this decrease in RMSE is driven by incorporating signals about the idiosyncratic return of the fund relative to the comparable asset (i.e., relaxing Assumption A2), inferred from NAV reports and distributions. Finally, the market β of the fund in Figure 1 is 1.19 while it is 1.0 for the comparable asset, which also contributes to the reduction in RMSE (again relaxing Assumption A2).

2.4 Dealing with real data

Since in real data the true asset values are not observed, we need a method for comparing the quality of our SSM-based nowcasts to the naïve ones. We propose nowcasting performance metrics that (i) take advantage of the fact that we do not need the full history of fund cash flows to estimate the SSM parameters, and (ii) operationalize the intuition that, if discounted at the true returns, fund cash flows must have zero net present value. Put differently, the closer given return series are to the true fund returns, the closer the PME computed with these series should be to 1.0 (as opposed to the market index return series applied in a standard PME calculation). The same logic applies for to-date PMEs (i.e., whereby the last reported NAV value is treated as a terminal distribution in the calculation) where biases in reported NAVs contribute positively to the variation in PME across periods.

The first estimation quality metric we compute is simply the mean squared deviation from 1.0 of the to-date PME computed quarterly for the remainder of a fund's life. For naïve nowcasts, we use the matched benchmark and the reported NAVs. For SSM nowcasts, we use the model-based return and NAV estimates. Even though we consider to-date PME values only after the parameter estimation window, each PME value utilizes the history of fund cash flows that occurred since fund inception. Hence, we refer to this as a partially out-of-sample metric (henceforth, *Hybrid*) and argue that it is a sensible metric for unresolved funds. For a fully out-of-sample metric (henceforth, *OOS*), we compute PMEs for the last period of each fund's life, as if the fund started right after the estimation period and the first subsequent NAV (nowcasted for SSM and as reported for naïve) was the only capital call made by the fund. Our simulations suggest that both metrics adequately characterize the nowcasting performance and that there are substantial efficiency improvements from using SSM. For the example in Figure 1, the OOS errors for SSM and naïve nowcasts are,

respectively, 0.226 and 0.626.

In addition to the Hybrid and OOS RMSE statistics, we examine the nowcasted return series' properties, e.g., correlations, risk loadings in a linear model, etc. We show that, unlike the naïve nowcasts, SSM-based returns do not obscure risk exposures and that observing autocorrelations (even at a high frequency) close to zero in a nowcasted series does not ensure correct inference about the funds' risk.

3 The State Space Model of a Private Equity fund

This section covers the technical details of the SSM representation of a PE Fund, with some supporting material appearing in Online Appendix Sections A.1 through A.4. We start with a summary of the variables in our model listed in the table below. There are three panels, separating the latent variables from the observed high frequency series and finally the infrequently observed ones. Recall from the previous section that we use capital letters to denote the levels of variables (e.g., D_t for distribution amount) and lowercase ones for the natural logarithm thereof.

Summary of Model Variables

Latent at weekly frequency			
V_t :	Asset value of the fund	$r_{0:t}$:	Log returns from inception until t
R_t :	Gross return of the fund	$\bar{r}_{0:t}$:	Smoothed log returns from inception until t
Observed at weekly frequency			
R_{mt} :	Gross return on the market	h_t :	The common factor of conditional variance in idiosyncratic returns of R_t and R_{ct}
R_{ct} :	Gross return on Comparable Asset		
V_t^0 :	Naïve nowcasts of fund NAVs	w_t :	Fraction of $C_t + D_t$ in $V_t^0 + D_t$
Observed at low (e.g. quarterly) or irregular frequency			
NAV_t :	NAVs reported by the fund's manager	D_t :	Distributions from the fund
		C_t :	Capital calls by the fund

The variables in the last panel are observed infrequently. They feature the same time index t, i.e. weekly, with NaN entries for the missing observations and numerical values during the weeks observations take place. We note that R_{ct}, i.e., the average gross return on publicly traded assets comparable to the fund in week t, is assumed to be scalar, meaning that the relevant asset return is a single index. Accordingly, we use a GARCH(1,1)-filtered conditional variance for the idiosyncratic

returns of R_{ct} (from a linear projection on R_{mt}) as a proxy for h_t.[7]

3.1 Model Structure and Parameters

We assume that the following relations describe the fund return and cash flow processes:

$$R_t = \exp\{\alpha + \beta r_{mt} + \eta_t\}, \quad (5) \qquad R^*_{ct} = \exp\{r_t \cdot \beta_c + \psi + \eta_{ct}\}, \quad (6)$$

$$V_t = V_{t-1}R_t - D_t + C_t, \quad V_0 = C_0 - D_0 > 0 \quad (7) \qquad R_{0:t} = \prod_{\tau=1}^{t} R_\tau \equiv V_t \cdot M_t, \quad (8)$$

$$\bar{r}_{0:t} = \Big(1 - \lambda(\cdot)_t\Big) r_{0:t} + \lambda(\cdot)_t \bar{r}_{0:t-1}, \quad (9) \qquad \text{NAV}_t = \exp\{\bar{r}_{0:t} - m_t + \epsilon_{nt}\}, \quad (10)$$

$$D_t = \delta(\cdot)_t \Big(V_t + D_t\Big) \exp\{\epsilon_{dt}\} \quad \text{iff } D_t > 0, \quad (11) \qquad R^*_{ct} = \exp\{r_{ct} - (b - \beta_c \cdot \beta) r_{mt}\}, \quad (12)$$

with $\eta_{ct} \sim \mathcal{N}(0, F_c^2 \cdot h_t)$, $\epsilon_{nt} \sim \mathcal{N}(0, \sigma_n^2)$, $\epsilon_{dt} \sim \mathcal{N}(0, \sigma_d^2)$, $\eta_t \sim \mathcal{N}(0, F^2 \cdot h_t)$, and $\lambda(\cdot)_t := \lambda \cdot (1 - w_t)$.

Lower case letters followed by $(\cdot)$ refer to a scalar-valued function, e.g. involving data, such as $\lambda(\cdot)_t$ pertaining to λ at time t as specified in equation (9). Note that b is the OLS slope of the projection of R_{ct} on R_m at weekly frequency. As with h_t, we consider b as inputs to the model (i.e. estimated separately). The following table summarizes parameters in our model which collectively are referred to as θ:

Fund's risk-return			
β :	systematic risk	F :	h_t-normalized idiosyncratic risk
α :	abnormal return		
Fund's reporting quality		*Fund's distribution process*	
λ :	smoothing intensity	δ :	distribution's intensity trend
σ_n :	reporting noise	σ_d :	distribution's intensity noise
Comparable Asset			
β_c :	slope to the fund's idiosync. return	F_c :	h_t-normalized idiosyncratic risk level to the fund's returns
ψ :	log return intercept to the fund		

We are agnostic about the economic interpretation of the parameter α. For example, it could be compensation for the liquidity risk incurred by the fund investors or excess returns after appropriate

[7] In the Online Appendix, we discuss extensions whereby R_{ct} is a vector with partially missing entries to account for relevant transactions (e.g., M&A activity or venture rounds). The GARCH model is viewed here as a filter in the sense of Nelson (1992) with an error that can be controlled by potentially relying on high frequency data.

assessment of all relevant risks.[8] Also, since equation (5) features log returns, to have an arithmetic CAPM α interpretation, the estimate needs to be adjusted for the returns' variance (see, e.g., Korteweg and Sorensen, 2010; Driessen et al., 2012). Accordingly, we add $0.5(\tilde{\sigma}^2 - \tilde{\beta}(\tilde{\beta}-1)\tilde{\sigma_m}^2)$, where $\tilde{\sigma}$, $\tilde{\beta}$ and $\tilde{\sigma_m}^2$ are OLS estimated from SSM-filtered weekly fund returns projected onto the market. Note that from equation (5), $\tilde{\beta}$ is by construction asymptotically equal to β.

As discussed in Section 2, the NAV smoothing is modeled as pertaining to both, systematic and idiosyncratic components of fund returns, as per equations (5) and (9), and, because t indexes weeks, the model accounts for returns *between* the quarterly report dates. Also, because equation (9) features cumulative returns of the fund, $r_{0:t}$, we actually model a smoothing of NAVs rather than smoothing of periodic returns. Accordingly, the predictable component of the appraisal bias is represented by the function $\lambda(\cdot)_t$ appearing in equation (9) while NAVs are modeled as a weighted-average of valuation snapshots taken at different points in the past. As pointed out previously, most of the weight comes from weeks *within* the past quarter (rather than from values at the end of previous quarters) corresponding to a realistic assumption about how the actual appraisal of fund assets is performed. In addition to the predictable component of the appraisal bias, we allow for random non-persistent valuation errors in NAV reports. These appear as ϵ_{nt} in equation (10) and are assumed to be standard normal with the standard deviation σ_n. For the distribution rule $\delta(\cdot)_t$ in equation (11), we assume:

$$\delta(\cdot)_t := \min(0.99,\, \delta \cdot t_y), \tag{13}$$

where $\delta \in (0, 1)$ is a scalar parameter, and t_y is the time (in years) since fund inception. Constraining the domain of $\delta(\cdot)_t$ is intuitive – until it is known that the fund is resolved, it has to be that D_t is non-negative and $V_t > \delta(\cdot)_t(V_t + D_t)\exp\{\epsilon_{dt}\}$.

The economic interpretation for δ and σ_d is, respectively, the trend and the noise of the distribution density. Funds with higher δ will tend to return capital faster. Overall there is relatively little insight that can be learned from $(\delta,\, \sigma_d)$ though. These are auxiliary parameters relating distributions – which reveal the true unsmoothed value of the fund portfolio – and the reported NAVs, therefore allowing for inference about λ and σ_n.[9] Finally, applying log transformations to

[8] But it does not reflect market timing resulting from the schedule of fund cash flows as studied in Gredil (2022).

[9] If the valuation reports for fund portfolio companies and exit proceeds were observed, then $\delta(\cdot)_t$ would not need

equations (5) through (12), except for (9) which is already in log form, yields a state-space model as detailed in Online Appendix Section A.2.

3.2 Other Estimands of Interest

Given the data observed for week $t = 1, ..., T$ and parameters θ, we can apply the Kalman filter to obtain estimates of fund returns for each week. We then apply the mapping function M_t in equation (8) to obtain the estimate of the fund asset values. We use the notation $\hat{R}_t$, $\hat{R}_{\tau:t}$, and $\hat{V}_t$. Although these estimates are functions of a particular θ and the ending period T, we drop these two arguments to simplify the notation. For each fund and θ of interest, we compute:

(a) Variances and autocorrelations of the filtered weekly returns between $t = 1$ and T using the standard estimators at weekly and quarterly frequency;

(b) $\text{PME}_{0:t}$ are the Kaplan and Schoar PMEs on a to-date basis that utilize filtered returns and asset values along with the complete history of fund cash flows realized up to period t.

(c) $\text{PME}_{\tau:t}$ are PMEs on the to-date basis in which capital calls up to period $\tau < t$ are replaced with $\hat{V}_\tau$ (using θ with data up to τ) and $\hat{R}_{0:t}$ are replaced with $\hat{R}_{\tau:t}$.

While implausibly high (or low) variance and autocorrelations of filtered return in (a) provide straightforward diagnostics of the model misspecification and the sensitivity to different parameterization, the measures in (b) and (c) are the building blocks to appraise nowcasting performance. As discussed in Section 2, these metrics are based on the simple idea that the NPV of cash flows will be zero if true fund returns are used for discounting. Specifically, we define the following three metrics, so that each can be interpreted as a fund-specific pricing error:

1. **In-Sample RMSE** – mean squared difference of ($\text{PME}_{0:t} - 1$) over the period $t = \tau_0, ..., \tau$ where τ_0 and τ are within the span of data used to estimate θ;
2. **Out-of-Sample (OOS) RMSE** – mean squared deviation of ($\text{PME}_{\tau:T}$ -1) such that no fund-specific data beyond week $\tau - 1$ is used to estimate θ;
3. **Hybrid RMSE** – mean squared deviation of ($\text{PME}_{0:t}$-1) over periods $t = \tau, ..., T$ such that

to be estimated. Instead it could be computed as sold assets as a fraction of the previous quarter NAV. Furthermore, under the assumption that the excess return (relative to the overall fund portfolio) is zero during the exit quarter, ϵ_{dt} is also zero, which makes σ_d a redundant parameter. In that case, $d_t + m_t$ maps without uncertainty to the true to-date return $r_{0:t}$.

no fund-specific data beyond week $\tau - 1$ is used to estimate θ. It is a hybrid between in-sample and out-of-sample data because, even though it utilizes the out-of-sample NAVs only, all since-inception cash flows are included.

3.3 Parameter Estimation

We estimate the parameter vector θ for each fund in our sample using a combination of two methods: (i) maximum likelihood (ML) and (ii) partial imputation. In both, we utilize a penalty function in the spirit of Ridge estimators, and an iterative procedure in the spirit of the Expectation Maximization (EM) algorithm (Dempster et al., 1977).

Our method exploits simultaneously cash flow and reported NAV information. It allows us to obtain fund-specific estimates of the parameter vector for a majority of funds independent of information from other funds. Nevertheless, the span of distributions and innovations in NAV reports for many funds are simply insufficient to identify all parameters independently. In such cases we resort to imputation. Another reason for adopting this strategy is that some parameters might not vary much across fund groups, so there are efficiency gains from imputing parameters across peer funds or using estimates reported in the existing literature. We also find that a partial imputation approach for parameters dominates the pooling of the data across funds.

In the following subsections we describe key steps and features of the two approaches we consider: (1) fund-specific estimates and (2) partially imputed methods. In both, we obtain standard error estimates for every parameter in θ using the numerical Hessian method as in Miranda and Fackler (2004). We conclude the section with a synopsis of Monte Carlo simulation results that illustrate parameter estimation efficiency and nowcasting performance across different degrees of model misspecification and data quality. In the Online Appendix Section A.4 we provide further technical details regarding the estimation procedure.

The important feature of PE fund data is that distributions almost never occur on the day of NAV reports. This fact allows us to identify σ_d from σ_n – the non-persistent noise parameter on NAV reports. Meanwhile, the sparsity and irregularity in distributions and NAVs mitigates the effect of the autocorrelation in the observed residuals that arises due to the possible misspecification of both $\delta(\cdot)_t$ and $\lambda(\cdot)_t$. We scrutinize this claim in our simulation and empirical analysis.

3.3.1 Fund-specific estimates

It is well established that identifying just α and β in PE is difficult even when data are pooled across many funds (e.g., see Ljungqvist and Richardson, 2003; Driessen et al., 2012; Ang et al., 2018). To estimate the 10-dimensional parameter vector θ by fund, we therefore proceed in two steps. The goal of the first step is to find the fund-specific α and β that fit the SSM well while satisfying an additional NPV-based restriction. We pursue this goal via a profile likelihood method augmented with a penalty function (see, e.g. Ghysels and Qian, 2019, for details). In the second step, we iteratively estimate the remaining 8 parameters in θ and the value-to-return mapping function $\hat{m}_t$.

To implement the first step, we build a 15-by-15 grid of plausible values for α and β based on range of estimates obtained from the prior literature. Specifically, in case of a buyout fund, β ranges between 0.668 and 1.831 with a mean of 1.25 and corresponding 15 probability quantiles – 0.01, 0.05, .125, ..., 0.875, 0.95, 0.99; guided by estimates in Ang et al. (2018).[10] We center the α-range at the annualized estimate of fund-level excess returns from the Kaplan and Schoar (2005) PME method (see Online Appendix Section A.4 for details). We consider seven equally spaced values up to +/-3.5% per year from each side of the α-range center. For each of the 225 (α, β)-pairs on the grid we (i) obtain the likelihood score corresponding to the MLE estimate of $(\delta, \lambda, F, \sigma_n, \sigma_d)$ in which (α, β) are kept fixed at the respective grid values and r_{ct} is dropped from the observations vector; (ii) evaluate the pricing error that the (α, β)-pair implies against the fund cash flows.

In the second step, we estimate the remaining 8 parameters (collected in the subvector ϑ), via maximum likelihood while keeping α and β at their optimized profile levels. We use R_{ct}*-interpolated NAVs* as the estimate for V_t (see Online Appendix Section A.3 for details) to construct the initial asset-to-value mapping. We then iteratively update the mapping function and re-estimate ϑ using the new mapping until the change from the previous iteration ϑ is below the threshold (see Online Appendix Section A.4 for details).

[10] Following Ang et al. (2018), $\beta \sim \mathcal{N}(1.25, 0.25)$, $\mathcal{N}(1.80, 0.30)$, and $\mathcal{N}(0.77, 0.23)$ for buyout, venture, and real estate funds respectively. For funds classified as generalists with regards to the investment style (i.e., pursuing both mature and early stage targets), we assume that the "literature-imputed" β (and the center of profiling grid) is 1.50.

3.3.2 Partially imputed estimates

For partially imputed estimates we again proceed in several steps. First, we estimate α, β, and λ. However, now we profile α and λ on a 15-by-7 grid, whereby the range for α is determined as before while λ is centered around that of the peer funds median. Recall that λ is essentially the exponential moving-average weight, constrained to be below 1 and above 0. We therefore, define the range of λ in terms of its probit function's z-score distance from the center.

As for β, we treat it as a free parameter in the first step but constrained to vary within just one standard deviation (using again Ang et al. (2018) estimates by fund type) from that of the peer funds (henceforth, β-anchor). We continue applying the penalty function to the profiled likelihood grid. However, now the penalty measures the distance from zero for the autocorrelation of filtered returns (see Section 3.2) obtained from a nowcast with only R_{mt}. In doing so, we mitigate forcing too high or too low smoothing intensity upon the fund from its peers. We find this approach to be more informative about the (β, λ) pair location than the PME-based penalty when the fund exhibits only several quarters with meaningful distributions.

In the second step, we iterate the return-to-value mapping function and the remaining free parameters until convergence as we do with the fund-specific estimates, however we always keep F and σ_n fixed to the median of the peer funds. Therefore, this EM-like procedure involves only 5 parameters (rather than 8 under the fund-specific estimation approach).

In both steps, the peer fund estimates are defined as medians of the respective parameter estimated independently. The next section examines several ways to define peer funds. We include the fund's own fund-specific estimates to compute the peer group median. For β, we additionally consider setting the center of the distribution to be equal to the estimates in the literature by fund type (Ang et al., 2018).

3.4 Simulation Experiment Summary

We conduct a simulation study to examine the performance of our method with regards to both parameter estimation and nowcasting performance. One of the focus areas in this simulation experiment is the consequences of the model misspecification relative to the true data generating

process. For example, we introduce discontinuous jumps in the funds' idiosyncratic return processes (as in Korteweg and Nagel 2020) contrary to the assumption that those are conditionally Gaussian, and examine cases in which the assumed distribution function (i.e., equation (13)) is different from the true one.

For brevity, we defer the details to Online Appendix Section A.6 where we examine a single case step-by-step process and report extensive summaries for a realistic panel of simulated funds. Below we summarize the main takeaways.

1. Key parameters are consistently estimated despite substantial idiosyncratic risk with non-Gaussian jumps and under reasonable misspecification of the distribution function $\delta(\cdot)$; cross-sectional variation in α and β is detectable with realistic sample sizes.
2. Under the assumption that the true parameters are close to those of the peer funds' estimates, partial imputation may reduce the estimation error on β and λ by a factor 1.2 to 1.6.
3. The median OOS nowcast error of 0.126 represents a 42% reduction relative to the naïve nowcast, yielding an improvement for 68% of the funds; partial imputation of parameters also yields a modest improvement in the nowcasting performance.

Overall, the simulations suggest that parameter estimation and nowcasting performance are promising using our methods and that the results are not particularly sensitive to distributional and functional form misspecifications.

4 Empirical Results

In this section, we report results for parameter estimates and nowcasting performance for the sample of private equity funds provided by Burgiss (see Brown et al., 2015, for description). The data include fund-level history of cash flows between each fund and its investors, fund NAVs reports for most quarters, as well as time-invariant data such as fund strategy, vintage year, and industry (based on 8 Global Industry Classification sectors). We use the industry designation to select a comparable asset from the Fama-French 12 value-weighted industry benchmarks. We use the CRSP value-weighted index for market returns.

4.1 PE Fund Sample

We start by describing the general properties of the fund sample. The sample includes all funds that satisfy the following criteria:

- vintage year is between 1983 and 2014,
- operated for at least 24 quarters, and reported NAVs for at least 20 quarters,
- made at least two distributions,
- have at least two peer funds for which we obtain fund-specific estimates (Section 3.3.1).

Table 1 reports basic summary statistics separately for 2,444 buyout and 1,679 venture funds in Panels A and B, respectively.[11] This is a subset of the Burgiss Manager Universe as of March 2021, augmented with additional data on fund industry specialization. We focus on relatively mature funds so we can evaluate nowcasting performance. The peers are defined as funds of the same type (i.e. buyout or venture) with vintage years within a year of the fund of interest, and the same money multiple tercile within the strategy. For feasibility of fund-specific estimates, we additionally require at least five quarters during which the distributions exceed 5% of fund size and require that the latest reported NAV does not exceed one-third of its cumulative distributions.

The filters drop 225 buyout and 89 venture funds from our sample. We note that the excluded funds tend to have lower returns and smaller size. For example, the excluded venture funds had median size/PME of \$50mln/0.76 as opposed to \$173mln/0.88 for the included funds. These differences are also notable for the buyout sample at \$180/0.97 for excluded and \$384/1.01 for included. Nevertheless, once adjusted for vintage averages, the differences in performance are statistically insignificant (t-stat of -1.27 for the combined sample). The excluded funds comprise less than 4% and 3% of capital committed of, respectively, buyout and venture funds.

The number of quarters with meaningful distributions is an important fund characteristic for estimation of the model.[12] The last rows of each panel in Table 1 shows that a quarter of buyout funds have 5 or fewer quarters with distributions that exceed 5% of fund size. For venture funds the corresponding number is only 2. This reduces the feasibility of fund-specific estimates for the

[11] We classify all non-venture equity-focused (i.e. not debt or mezzanine) funds as buyouts except growth equity / expansion capital.

[12] For additional details see the case study in Online Appendix Section A.6.1.

venture sample especially (Online Appendix Table A.2 reports summary statistics for PE Real Estate funds).

For each fund in the sample, we match the comparable asset series, R_{ct}. For 490 funds that Burgiss classifies as generalists or provides no industry information, we define the comparable asset for venture [growth] funds as the value-weighted small growth [value] portfolio downloaded from Ken French's website. For 1,892 funds Burgiss provides us with an industry breakdown of the actual investments made by the fund (top 3 industries), so we value-weight the respective industries. Otherwise we match the closest industry. We utilize industry benchmarks because it is a relatively unexplored dimension in prior research on PE performance (despite the high attention from practitioners) but, most importantly, because our method requires a proxy of idiosyncratic volatility in the public benchmark as well.[13]

4.2 Fund-specific estimates

An important feature of our model is that estimation and nowcasting only uses the data from the individual funds and their matched public benchmarks. However, data limitations can hinder the estimation in some cases. Overall, we are able to obtain fund-specific parameter estimates for 2,513 PE funds or 61% of the sample described in Table 1. Table 2 reports summary statistics for fund-specific estimates of SSM parameters and the assessment of their associated nowcasts.

4.2.1 Parameters

Perhaps the most interesting parameters we estimate are the funds' αs and βs. Panel A of Table 2 shows that the average [median] α of PE funds with fund-specific estimates is 4.7 [3.0]% per year with an interquartile range of 15%. This is not necessarily representative of a typical fund's risk-adjusted return since these funds are likely to have a greater fraction of successful deals that resulted in large distribution amounts happening at different points of their lives. The loading on the public market return is 1.30 on average, ranging from 0.84 at the 10th percentile to 1.68 at the 90th.

[13] Herskovic et al. (2016) show that idiosyncratic volatility changes are significantly less correlated across industries than those of similarly coarse size cohorts, and the common idiosyncratic volatility factor explains more than half of the cross-sectional variation in the ten BM portfolios.

Also of interest is the scale coefficient estimate on our proxy of time-varying idiosyncratic risk. Results in the third row of Panel A report show that the standard deviation of idiosyncratic returns for a typical PE fund is 3.5 times the GARCH(1,1)-filtered idiosyncratic volatility of the matched industry index. However, we note the large dispersion for this parameter. While it is equal or smaller than 1.4 for 10% of the funds, it is greater than 4 for almost a quarter of funds. In fact, for about 5% of funds it hits the upper bound of 10 that we imposed in estimations. This suggests that the idiosyncratic volatility parameter is hard to identify at the individual fund level. However, it seems plausible to assume F is similar across funds within the same strategy, vintage, and size cohort and we utilize this assumption later.

The next two rows of Panel A report summaries of parameters that help us understand the errors in and smoothing of reported NAVs. PE fund NAVs indeed tend to exhibit a very high degree of appraisal smoothing, as follows from the percentile ranks of λ. The median of 0.961 implies that public market valuations from five or more weeks ago comprise 82% ($= 0.961^5$) of a typical fund NAV report. This exponential weight is consistent with the AR(2) coefficients of about (0.5, 0.3) on quarterly return series derived from reported NAVs. Our results suggest substantially less smoothing for the bottom decile of funds, e.g., λ is 0.52 or less, which correspond to a non-detectable persistence at the quarterly frequency.

We observe that the average [median] NAV noise – the standard deviation in NAVs that the SSM cannot attribute to past or future returns – is 6.4% [5.7%]. For some funds it appears to be implausibly high or low, e.g., 14% at the 90th percentile and close to the lower bound of 1% for 10% of funds. Therefore, this parameter also appears hard to identify reliably. Just as with the idiosyncratic risk levels, we subsequently use the fact that these are plausibly similar across peer funds. Naturally they relate to fund investments' type, portfolio diversification and operating period. Panel A also reports estimates for δ and σ_d that govern the fund distribution process. While auxiliary in nature, they need to be estimated for each fund to reflect the heterogeneity in the distribution magnitudes (given the assumed functional form of $\delta(\cdot)$). Here we just note that for the vast majority of funds both parameters are well within the upper (0.9 and 4) or lower (both 0.0001) bounds imposed in the estimation.

Finally, we examine the parameters that govern the mapping of the comparable asset to the latent fund returns at weekly frequency. The last three rows of Panel A report the intercept, slope, and variance scale parameters. To gain intuition regarding these parameters, we plug in (the log of) equation (5) for r_t into equation (6) and denote the intercept [residual] from a projection of r_{ct} on r_{mt} with a_c^m [ϵ_{ct}^m]. It follows then that $\psi = a_c^m - \beta_c \alpha$ is not particularly informative, while $\beta_c = \text{cov}(\epsilon_{ct}^m, \eta_t)/\text{var}(\eta_t)$ is the covariance of the comparable asset and fund returns scaled by the variance of the fund returns, which somewhat limits the parameter's magnitude comparability across funds. Note that for truly comparable assets this covariance should be positive. Therefore, we impose a lower bound of 0.01 on this parameter. The boundary is hit for 27% of funds indicating that the matched benchmark does not help explain those funds' NAV and distributions data. A disproportionally large share of these are funds for which we do not have the industry weights from Burgiss, especially if classified as generalists (therefore, a size & style benchmark is utilized). However, for a typical fund in the sample β_c estimate is 0.18 and the average t-statistic for the parameter is 2.2 (untabulated).

Given our proxy for h_t and the assumption built in the SSM estimation and filtering that $\mathbb{E}[\eta_t \eta_{ct}]$ = 0, it follows that:

$$F_c \approx \sqrt{1 - (F\beta_\epsilon)^2}, \tag{14}$$

where $\beta_\epsilon = \text{var}(\beta_c \eta_t)/\mathbb{E}h_t$.[14] The table shows that for just under 90% of the fund-specific estimates F_c are indeed below 1. The vast majority of funds with $F_c > 1$ also have β_c-estimates at the lower boundary. Therefore, F_c represents another statistic on the matched comparable asset's relevance (as well as the degree of SSM misspecification with given specific fund data) that is more comparable across funds. Besides, F_c yields a useful insight for comparison of different SSM estimation methods. From equation (14) it follows that, for a given fund data and r_{ct}-series, F_c will be lower if upward-biased systematic risk estimates inflate the idiosyncratic risk reflected in F without a compensatory decline in its covariance with the comparable asset. We will use this to compare estimation results across methods in the next section.

[14] From equations (5)–(6) and $r_{ct} = a_c^m + \beta_c^m r_{mt} + \epsilon_{ct}^m$, $\text{var}(\epsilon_{ct}^m) = \text{var}(\beta_c \eta_t) + \text{var}(\eta_{ct}) = \beta_c^2 F^2 \mathbb{E}h_t + F_c^2 \mathbb{E}h_t$, while $\epsilon_{ct}^m \sim \mathcal{N}(0, h_t)$ by assumption.

4.2.2 Properties of filtered returns

We now want to compare the time-series proporties of different estimates. Specifically, we compare the autocorrelations and variances of fund returns obtained from a naïve nowcast that is based on as-reported NAVs with those from the fund-specific SSM estimates. Panel B of Table 2 reports statistics for series at two frequencies – weekly and quarterly. For the naïve nowcast, the quarterly series correspond to NAV-changes adjusted for the value of within-quarter cash flows, discounted at the industry benchmark returns. In contrast, the weekly naïve nowcast assumes the asset values track the industry benchmark on top of the interpolated between-quarter change in PME (see Online Appendix Section A.3). We see that those series exhibit notably positive autocorrelations, which looks higher at quarterly frequency at 0.227 for a median fund. The results for variances suggests that the annualized standard deviation of naïve returns nowcasts for the middle half of PE funds is between 20% and 38%.

The SSM-based return nowcasts are obtained via the Kalman filter at weekly frequency, as explained in Section 3.2, and then aggregated to the quarterly frequency by summation of the log returns within each quarter. Panel B shows that the autocorrelation of the SSM-based returns is only 0.09 for a median fund. At the weekly frequency, we observe effectively zero return persistence for at least three quarters of the sample. Meanwhile, the variance of filtered returns is on average 1.2 times higher than for the naïve nowcasts, suggesting the annualized standard deviations are between 25% and 45% for the middle half of PE funds.

While these results suggest that the SSM-filtered returns are more representative of the true return process than the ones inferred from reported NAVs, we note that for a significant minority of funds the SSM nowcasts exhibit fairly large autocorrelations. For example, one-fifth of the fund sample exhibits autocorrelations of filtered returns in excess of +0.44 or below -0.33 at the quarterly frequency and among those there is a 10-fold prevalence of funds with either $\beta_c < 0.02$ or $F_c \geq 0.999$, whereby updates to r_t are driven solely by the distribution and NAV reports, which gives rise to notable persistence when the backward induction method for filtering is utilized. The remaining cases appear to be driven by the model (e.g., $\lambda(\cdot)_t$ and $\delta(\cdot)_t$) or parameter misspecifications, which usually can be addressed on a case-by-case basis in practice. We also note that usage of the

backward induction with the Kalman filter likely underestimates the variance of the filtered return series somewhat, and, hence, overestimates the magnitude of the correlations.[15]

4.2.3 Nowcasting performance

The out-of-sample RMSEs are similar to the in-sample RMSEs and are positively correlated, as evident from Figure 2. Hence, in-sample pricing errors are a useful predictor for nowcasting performance as suggested by the simulations discussed in Section 3.4. Panel C of Table 2 reports fund-level nowcasting performance metrics introduced in Section 3.2. The SSM-based RMSEs are notably smaller than those from the naïve nowcast. For a typical fund, the Hybrid RMSE drops from 0.281 to 0.045 while the OSS error drops from 0.233 to 0.12 and are sizable across all percentiles. Figure 3 shows that the improvements are largest for funds with relatively large naïve nowcast errors and that the SSM-based nowcasts are materially inferior to the reported NAVs only when the latter happen to be particularly precise – e.g., to the left of log MSE of -5 on the x-axis, which corresponds to the RMSE of less than 0.082. Across the buyout and venture sample, SSM-based nowcasts with fund-specific parameter estimates result in smaller Hybrid [OOS] RMSE than the naïve nowcast for 90% [70%] of individual funds. In Section 5 we show that gains are usually higher when a portfolio of funds is concerned because SSM-based nowcast errors tend to be less correlated than those from the naïve approach.

4.3 Partially imputed estimates by fund type

Next, we examine partially imputed parameter estimates as described in Section 3.3.2 and compare the nowcast performance with those generated by fund-specific estimates. Most importantly, the partial imputation enables us to obtain parameter estimates for approximately 95% of the sample funds and, therefore, provide a more representative picture of the institutional PE fund universe.

We consider three ways to define peer funds. First, in what we will refer to as *peer-imputed* we set F, σ_n as well as the center locations of the β-anchor and λ-grid to the median values of the respective fund-specific parameter values for funds of the same strategy (buyout of venture) incepted in the same or adjacent vintage year. Additionally, if there are three or more peer funds

[15] The fact that the autocorrelation magnitudes are smaller at weekly frequency is also consistent with this explanation since the variance bias accumulates within a quarter.

in the same size tercile, we limit the peer group to those peers only. We then attempt to further narrow the peer group to at least three funds in the same industry of specialization. Second, we take the opposite approach and set the four imputed values to the buyout or venture fund average in what we refer to as *average-imputed*. Finally, we fix the β-values to the strategy-specific estimate from Ang et al. (2018) in what we refer to *literature-imputed*, while identifying all other parameters independently as described in Section 3.3.1.

We then compare the SSM nowcast performance metrics introduced in Section 3.2 under these four different sets of parameters. This exercise sheds light on three important questions: (i) is there a meaningful time-series variation in PE fund risk-return profiles, (ii) how big is the noise reduction resulting from restricting the fund-specific parameters, (iii) are our systemic risk exposure estimates more consistent with the fund-level cash flow observations than in Ang et al. (2018), who encompass the previous literature estimates as priors for their Bayesian MCMC.

4.3.1 Buyout funds

Our findings show that peer-imputed estimates dominate other partially-imputed estimates and benefit from a noise suppression relative to the fund-specific estimates for a majority of funds. There appears to be genuine heterogeneity in parameters across many buyout funds. Forcing higher levels of fund systematic risk does not improve the nowcasts.

The evidence for selected parameter estimates and nowcast performance metrics for the three imputation methods alongside the fund-specific estimates are reported in Table 3. The latter are available for 1,598 funds, while the partial imputation increases the coverage to 2,299 funds (94% of the sample reported in Panel A of Table 1). We first focus on how key parameter estimates, namely α and β, vary across estimation methods. We observe that the average and median α estimates of 4.5-to-5.1% are lower for peer- and average-imputed estimates in comparison to the 6.3% mean in the sample of fund-specific estimates (Panel A). However, these are still notably higher than the near-zero values for the literature-imputed estimates, which also stand out with notably higher βs of 1.25 on average, especially as compared to the range of 0.87-to-1.09 for the middle two quartiles using the peer-imputed method.

The various estimation methods provide consistent results across funds for other model param-

eters. Specifically, in Panel A we examine the smoothing parameter λ, as well as the variance scale on comparable asset parameter, F_c,. The mean and median F_c are very close across all four methods. Still the the literature-imputed estimates yield lowest levels, which is consistent with an upward bias in the systematic risk estimates. As for the NAV smoothing rate, we observe that the means are in a narrow range of 0.835–0.886 across all four methods. The inter-quartile is highest for the average-imputed method possibly reflecting a tighter constraint on the variation of the β parameter relative to that in the peer-imputed method. However, the inter-quartile range is smallest for the literature-imputed method whereby β is also fixed. The literature-imputed estimates also feature the highest median λ of 0.964 that is nonetheless very close to that of the fund-specific method (0.954) despite a 0.15-difference in the median β estimate. These results indicate that there is no "mechanical relation" between the estimated levels of systematic risk and NAVs smoothing intensity.

We also examine results for weekly autocorrelations and quarterly variances of filtered returns and find generally consistent and stable values across the various estimation methods (Panel B). The naïve approach results in similar levels of autocorrelation (6.4%) as observed for the fund-specific estimates reported in Table 2 (which includes venture funds). The average [median] annualized standard deviation of returns is estimated to be 28 [24]% with the naïve approach. The fund-specific returns have very low persistence and return standard deviations of about 34 [29]% per year for an average [median] fund. While the average and the percentile readings of the total fund risk per the peer- and average-imputed methods are very close to those with the fund-specific method, they are characterized by somewhat elevated levels of autocorrelations of filtered returns in the right tail of the cross-section. The 75th percentile of 0.13–0.146 is actually closer to that of naïve nowcasts. This is driven by a larger fraction of funds for which we don't have a good match of the comparable asset – in the subset of funds with observed industry weights the 90th percentile autocorrelation is only 0.06 (untabulated). As for the literature-imputed estimates, the standard deviations of filtered returns are 38% on average, which is notably higher than those of other SSM estimates, while the distribution of autocorrelation coefficients is similar to that of the fund-specific method but is somewhat less symmetric and shifted to the negative domain.

Finally, we examine the goodness-of-fit estimates for the various methods. We find that the partially-imputed estimates have lower nowcast errors and higher improvement rates relative to the naïve nowcast on both the Hybrid and OOS basis, except for the literature-imputed estimates (Panel C of Table 3). Nonetheless, there are notable improvements over the naïve approach from utilizing the SSM method regardless of how systematic risk levels are identified. For example, the median OOS RMSE with literature-imputed β is 0.113, or 28% higher than with peer-imputed estimates, but nonetheless 55% lower than 0.251 for the naïve approach. Therefore, the SSM-method is relatively forgiving to reasonable variations in the model parameters.

4.3.2 Venture funds

We repeat the analysis described above for venture capital funds and report the results in Table 4. We are able to obtain fund-specific estimates for 859 funds, while the partial imputation increases the coverage to 1,621 funds (96.5% of the venture sample).

In contrast to buyout funds, the median venture fund produces negative α regardless of the method. However, at -8.1% per year, the level is much lower with the literature-imputed β of 1.80 than the -2.1–3.0% range for the MLE-estimated βs. The 25th and 75th percentile αs are 5-to-7 percentage points higher than the literature-imputed estimates. These results are consistent with prior findings that the median VC fund underperforms the broad market. The mean αs are negative except for the funds-specific method (and consistently about 4 percentage points higher than the median values). This mean/median discrepancy is consistent with a more positive skew in fund returns compared to the buyout sample.

Similar to the buyout sample, the MLE βs are 0.2-to-0.4 lower than the literature suggests, with the peer-imputed methods yielding the lowest estimates. Interestingly, there appears to be less skewness in systematic risk-taking in the cross-section of venture as compared buyout funds – the median fund-specific [peer-imputed] estimate of 1.62 [1.41] are within 0.02 from their respective means. Also similar to buyout funds, we see that F_cs tend to be lower for the literature-imputed method with a similar pattern for λs – i.e., highest[lowest] dispersion for average- [literature-]-imputed methods even though both feature no variation in β. The average and median λ are 0.05-0.08 higher than in the buyout sample regardless of the method, suggesting an even greater

extent of staleness in reported NAVs.

We observe again patterns similar to buyout funds for weekly return autocorrelations and annualized standard deviations. Results shown in Panel B of Table 4 indicate that, relative to the naïve method, there is a notable and roughly equal reduction in autocorrelations and an increase in the total risk estimates. The differences in the return standard deviations levels across methods are somewhat more salient in the venture sample, e.g., the peer-imputed mean of 40% is 8 percentage points higher than the naïve mean and 4 percentage points lower than the literature-imputed mean. Similarly, we see that peer-imputed parameters result in better nowcasts than fund-specific parameters but keeping F and σ_n fixed and/or nudging higher βs appears counter-productive (Panel C). We also note that, although SSM RMSEs are higher for the venture sample, relative to the naïve method they improve about as much as in the buyout sample.

Finally, as with buyouts, the literature-based estimates of β are consistent with inflated levels of the systematic risk in the filtered series. The F_c-based diagnostic, the leftwards-shifted distribution of the autocorrelation coefficients, and the relative level of the nowcasting errors shown in Panel C confirm this. Overall, the conclusions about the SSM nowcasts outperformance relative to the naïve approach are robust to how the SSM is parametrized.

4.4 Fund-level heterogeneity

4.4.1 The cross-section of fund risk

So far we have shown that our estimates of systematic risk are lower while also more consistent with cash flow realizations of individual funds. This raises the question why other studies have documented higher β estimates. One potential explanation is that funds with higher risk produce larger distributions and take on greater importance in methods used by other studies that combine distributions across funds. Similarly, if funds with more concentrated investments (i.e., lumpier cash flows) tend to have higher βs, then differences in β would be driven by the fact that we generate fund-by-fund estimates (and are looking at the average) and others do pooled estimates for an index-like PE exposure.

We test this conjecture by regressing fund-specific estimates of β on fund characteristics. While controlling for fund industry, geography and subtype fixed effects, we consider (logs of) fund size,

total distributions to size ratio, and the number of distributions made by the fund. Column (1) in Table 5:A shows that, by itself, size is not significantly associated with the fund risk. However, the relation strengthens and turns significantly positive if we control for the market- and industry-return variance observed during the fund's life, as shown in column (3). The estimates in column (3) suggest that doubling the fund's MOIC [size] increases β by 0.08 [0.01], while doubling the number of distributions the fund makes lowers β by 0.034. In the next section, we show that value-weighted indices generated from the funds in our sample have βs that are closer to values documented in other studies.

We also examine the persistence in fund-level risk by manager by regressing the current fund's β estimate on that of the manager's previous fund (while also controlling for the current fund style, geography, and the industry specialization). To reduce the overlap in investment portfolios we require that the previous fund was incepted at least five years before the current fund. The combination of these sample restrictions drops the number of observations by half, yet we see (in column (4)) that the coefficient on the previos fund's β is significantly positive even when we control for its λ, and when we control for all other covaratiates (in column (5)). This is consistent with the amount of risk-taking being a trait of PE managers.

Similar to the analysis for βs, we can examine the fund-level (cross-sectional) determinants of NAV smoothing. To do this we repeat the analysis but with the logs of fund-specific λ-estimates as the dependent variable. The results, reported in Panel B, indicate that fund size [MOIC] positively [negatively] predicts the NAV smoothing intensity as does the smoothing intensity estimate for the previous fund by the same manager. However, the R^2s of these regressions are less than one-tenth of those in Panel A, suggesting that NAV smoothing are more of an idiosyncratic characteristic of the fund and the assets' return history.

This finding is evident from Figure 4 which plots industry and fund subtype fixed effects from column (3) of Table 5 Panel A and that of a similar regression but using the peer-imputed sample.

Finally, we also examine risk levels by industry. Our findings show that the industry effects are consistent with the common belief that non-cyclical consumer goods and industrials exhibit less systematic risk relative to the baseline of the consumer discretionary sector (see Figure 4). In

contrast, telecoms and information technology funds tend to have higher βs. We also examine βs for various fund sub-types (relative to buyouts) by plotting fixed-effects (Panel B). Venture capital funds of all types have significantly higher βs whereas real estate funds have lower βs. Generalists (who do both buyout and venture deals) also have higher βs. Interestingly, late-stage venture and expansion capital (also commonly called 'Growth Equity') funds tend to have more exposure to the market than the early-stage funds.

4.4.2 Trends in PE fund risk

Our results suggests notable cross-sectional variations in PE fund risk-taking and performance. The PE industry is known to be cyclical so it is also interesting to examine time-series variation. Figure 5 displays the time trends in selected fund-specific parameter estimates by vintage year using box-and-whiskers plots.

We start by examining buyout funds (Panel A) and find a notable bump in αs between 2000 and 2006 but no apparent long-run trend toward lower αs as some studies suggest (see, e.g., Phalippou, 2020). In particular, buyout αs appear to bounce back post-2010. We also note that the interquartile range of αs is about 15% and remarkably stable post-1996. Estimates of β also vary substantially through the sample period. Starting in the late 1990s estimates turn lower and are below one for nearly half of the funds incepted between 1998 and 2006. However, relatively few buyout funds have β¡1 from 2007 onwards. Unlike with buyout αs, there is more variation in the bottom and the top quartile of βs than in the median. We see a moderation in fund-level idiosyncratic risk from 2003 onwards (third and fourth panels) especially if measured as a ratio to the idiosyncratic volatility of the comparable asset. Overall, these results are consistent with average buyout PMEs exceeding one across most vintages (see Korteweg and Nagel, 2020, for links between the SDF and β-methods).

We next examine time variation incharacteristics of venture capital funds. There is a large spike in both estimated αs and βs during the dotcom tech-bubble years (Panel B of Figure 5). Top quartile 1996 and 1997 vintage venture funds have estimated αs in excess of 50%. For these and the adjacent vintages, top-quartile βs are in the vicinity of 2.0. As the dotcom bubbble burst we observe a rapid decline in both metrics. The βs drop below 1.5 for the majority of the funds

incepted between 1999 and 2004 before rebounding to the 1.5–1.7 range more recently. Also notable is the collapse in the idiosyncratic risk-taking during 2000-2004 and an increase in the interquartile range post-2010.

5 Selected Applications

To conclude the paper we look at two specific applications. The first pertains to PE allocations around the Great Financial Crisis (GFC) of 2008, and the second examines how we can use our nowcasting model to generate PE return indices.

5.1 PE allocations and nowcasts around the GFC

We beleive the SSM-based nowcasts provide a superior basis for asset allocation decisions, especially after periods of very high or very low market returns. Specifically, our analysis shows that fund investors who assume that the reported NAVs are unbiased will tend to significantly misjudge their current allocation to PE. The GFC provides an excellent opportunity to illustrate this and to examine the performance of the SSM nowcasts around a dramatic market move. For this exercise, we restrict our sample to North American buyout and venture funds that were 5-8 years old as of the first half of 2007 and not fully resolved before 2010. We also drop a few outliers with nowcasted NAVs less than 1/10× or more than 10× reported NAVs. This leaves 175 buyout and 159 venture funds. To measure nowcast performance, we rely on the OOS RMSE metric introduced in Section 3.2 which we recompute for each fund-quarter between Q4 2006 and Q4 2009.[16]

We examine results separately for the buyout and venture subsamples. We start by plotting the following for 2007–2009: average fund NAVs, the average value of the public benchmark corresponding to each fund's industry, and the average nowcasting error using the naïve approach, which takes the reported NAVs as true, as introduced in Section 3. Both NAVs and the public benchmark values are normalized to 1.0 in 2006Q4. The dark solid line in Figure 6:A shows that the public markets rallied through 2007Q3. During that period, the growth in NAVs (gray bars) trailed that of the markets, and was in fact negative in the buyout sampleas those funds were making significant distributions to LPs during this time. In early in 2008, the NAVs of both buyout

[16]For SSM-based nowcasts, we do not re-estimate the parameters or filtered returns each quarter but use the same estimates as in Section 4.

and venture funds fell at a lower rate than the public benchmarks, which suggests that either NAV marks were lagging public market declines and/or outperforming the public markets. The dotted lines show the average naïve nowcast errors across the respective types of funds. We see that naïve nowcast errors are trending downwards as GFC passes (especially for buyouts) and funds approach resolution (reported NAVs become more indicative of the present value of future distributions).

However, the apparent out-performance of PE fund NAVs is due to the changes in the direction of the appraisal bias. This is evident from the bar plots in Panel B of the ratio of reported NAVs to the corresponding SSM nowcasts. In fact, the as-reported PE valuations (and therefore the overall portfolio weights) fell short of the SSM estimates by about 10% as of mid-2007 and then swung to more than +20% at the end of 2008. The dotted lines in Panel B provide measures of the accuracy of the SSM nowcasts and show that they consistently outperform the naïve estimates. The dotted line plots the fraction of funds with SSM nowcasts better than the naïve nowcasts (right axis) suggesting that SSM produces better nowcasts than the naïve method for 70–90% of fund-week observations. The dashed line shows the ratio of SSM-based nowcasts' RMSEs to naïve nowcast standard errors (also right axis) for the portfolio of funds, suggesting a 60–80% lower error if SSM nowcasts were used instead of the naïve nowcasts. This is more than a 40% improvement in average RMSE on a fund-by-fund basis reported in Table 2 and indicates that the SSM-based nowcast errors tend to be less correlated than those from the naïve approach.

Finally, we provide one additional assessment of the value of our fund-level nowcasts for portfolio choice by examining the standard deviation of reported NAV bias during the crisis. Figure 6 Panel C investigates this by plotting the standard deviations of the appraisal bias and nowcast errors across funds in a given quarter instead of the cross-sectional means appearing in Panel B. The plots show that the standard deviation of reported NAV bias spikes during the crisis in 2008 while the SSM nowcast error across funds remains mostly flat (the exception is buyouts 2007Q4). These results suggest that it is important to account for the fund-specific parameters and idiosyncratic return paths preceding the market crash. In other words, one should not assume that appraisal bias is the same across all currently operating funds.

We note that constructing comparable assets out of the Gupta and Van Nieuwerburgh (2021)

risk loadings, which are fund age- and type-specific, might offer some distinct advantages for applications similar to the GFC as they allow for greater granularity in nowcasting at a fund level. In general, such granular benchmarks are likely to beat our more coarse approach of matching a self-declared industry which is kept fixed during a fund's life. It is also possible to use our framework to see how much one gains from constructing comparable assets from the fund's holdings-level data.

5.2 PE index analysis

Our methodology generates time series of individual fund returns and asset value estimates. Therefore, a natural next step is to construct indices out of those individual series. Not only would the resulting series embed the cross-sectional and time-series variation of risk discussed in Section 4.4, but they can also provide an important diagnostic tool for examining the validity of the PE fund representation via SSM that we propose. Specifically, we are interested if, as shown in Couts et al. (2020), the autocorrelations "bleed-up" back at the index level as a result of possible model misspecification or estimation error at a fund-level.

We construct our index from the first week of 1987 through the 13th week of 2021. We use the fund-specific estimates to be consistent with the analysis in Table 5 and fund-weeks between the 13th and 572th weeks since the respective fund's inception. The number of constituent funds by type and the aggregate reported NAVs are plotted in Panel A of Figure 7. The decline in both metrics following 2013–2014 reflects the fact that, while older funds resolve, no new funds are added after 2014. We construct five series for each fund type for this analysis. The first index uses $NAV weights and the comparable asset returns as a public market benchmark (henceforth, 'Comp'); the second index instead uses weekly returns implied by naïve NAV nowcasts (henceforth, 'Naïve'). Both use the one-week lagged naïve nowcasts of fund NAVs to aggregate the series from the individual funds. The third index series is based on the return and NAV estimates from the fund-level models (henceforth, 'SSM'). Finally, we construct two excess return indices relative to the Comp index – one for Naïve and one for SSM. The time series of SSM and Naïve are plotted in Panel B of Figure 7. To enhance contrast, we plot 5-year rolling returns. We then examine the persistence and risk-factor exposure of these index series. As evident from the plots, the SSM indices exhibit properties of transactions-based indices rather than the appraisal-based Naïve indices in the sense

that they reflect market peaks and troughs prior to when these events are seen in Naïve indices.

We examine the statistical properties of the buyout fund index by examining autocorrelations and factor exposures. Table 6 (Panels A1 and A2) summarize the analysis beginning with statistics for the NAV-weighted returns on comparable assets as a benchmark. The index of comparable asssets (Comp) exhibits negative but statistically insignificant autocorrelation, has a market beta of 1.086 and an overall (idiosyncratic) standard deviation of 2.66% (0.63%) per week. The results in the second column of each panel show that, unlike the fund-level averages in Table 3, the corresponding weekly index series exhibits a small but statistically significant negative persistence on the first lag and has a market β of 1.03. This result is driven by the fact that the fund-level return smoothing is diversified away while the Comp series is slightly negatively autocorrelated. We examine this supposition in column (3), which includes higher order lags, and column (6), which examines the AR(1) of excess returns of Naïve relative to Comp. While the autocorrelation and the total return index level is close to zero, it is large and significant for the Naïve excess returns (0.78). The results in column (6) of Panel B show that, consistent with the analytical results in Online Appendix Section A.3, the excess returns for Naïve load negatively on the lagged systematic risk suggesting (incorrectly) low systematic risk exposure.

We now turn to the SSM PE index for buyouts which is the series of primary interest. The results for analysis similar to that for the Naïve PE index are presented in columns (4), (5), and (7). Panel A1 suggests that the SSM series does not exhibit any meaningful persistence. Results presented in Panel A2 indicate that the index series has a market risk factor of about 1.21. This is higher than the 1.15 average for the respective fund-level estimates in Table 3 and underscores the fund heterogeneity effects analyzed in Section 4.4 – specifically, that higher performing funds (which receive greater weight in our index) have higher βs. Finally, consistent with common knowledge that PE sponsors have historically transacted disproportionately in small value-oriented companies, we find significantly positive loadings on the Fama-French SMB and HML factors for both Naïve and SSM index series in columns (3) and (5) respectively. We also note that, unlike Naïve excess returns, SSM excess returns do not load on lagged market return, as follows from columns (6) and (7) of column of Panel A2.

We obtain similar conclusions for the index of venture funds, but a few differences are worth noting (Panels B1 and B2 of Table 6). First, the β of the SSM index series is much higher (1.70). This result is driven by higher loadings from both the Comp (column 1) and excess return series (column 7) and is close to venture β estimates from studies that examine estimates from pooled samples (e.g., Ang et al., 2018). Second, the venture SSM index loads negatively on HML which is consistent with venture companies investing primarily in companies with high growth potential. Finally, we note that the overall standard deviation of the venture SSM index (4.1% per week) is higher than that of buyouts (2.9%), driven partially by higher idiosyncratic risk of excess returns as measured by the RMSEs (reported in columns (7) of panels A2 and B2): 0.8% for venture versus 0.4% for buyouts. These results suggest that our approach of modeling funds as a state space model at high frequency does not suffer from the misspecification symptoms outlined in Couts et al. (2020), who analyze standard ARMA unsmoothing techniques with a focus on commercial real estate funds and hedge funds. This underscores the limitations of the standard approaches that (unlike us) assume time-invariant coefficients and operate at the reported NAV frequency only.

Our findings also raise the question of which βs are more relevant: fund-level averages or index-level? We argue that if the question involves risk-budgeting in a typical portfolio containing only a handful of funds (see, e.g., Gredil et al., 2020), then a conditional expectation via a regression like in Table 5 is more informative.[17] However, if the question is what difference the pro-cyclical capital deployment pattern makes for the risk profile of PE as an asset class, then the index results are more informative. Importantly, these results show that a lack of autocorrelation is not a sufficient condition for the return series to be indicative of the systematic risk of the investment.

6 Conclusion

This paper develops a new method that provides reliable nowcasts of PE fund asset values at high frequency. We shed light on the rich temporal and cross-sectional variation in manager risk and

[17] Notably, previous findings of higher level of systematic risk tend to rely on round-to-round valuation change of venture-backed companies (Korteweg and Sorensen, 2010) and secondary fund trades in buyouts (Boyer et al., 2018). The contracts between portfolio firms and funds as well as between a fund and its investors also attenuate the risk, since entrepreneurs and fund GPs get an option-like payout which reduces the risk on the fund variation in fund cash flows but subsidizes them on the downside due to the state-contingent allocation of cash flow rights (see Kaplan and Strömberg, 2003; Gornall and Strebulaev, 2020, 2021).

reporting-quality characteristics even when only fund-level cash flows and NAVs are utilized for nowcasting. We show how this framework can be extended to incorporate other types of data, such as portfolio-level deals information, secondary transactions with fund stakes, etc. These new insights are of critical importance in light of continued growth of PE and other illiquid assets in the institutional portfolios and the increasingly high regulatory interest to the risks arising from these investments. Our results suggest that the risk-return profile of PE based on the samples from the 1990s and early 2000s may not be representative of the currently operating funds.

Some final comments regarding our method are noteworthy. While our approach is reasonably robust to autocorrelation in the unobserved error terms, the fund idiosyncratic returns may not be conditionally Gaussian. If for some funds the tails are "too heavy" such that the variance is very high (infinite), then obviously the Kalman filter is ill-defined and MLE is infeasible. This might explain why our iterative procedure does not converge or results in implausible asset value estimates for about 5% of funds. Minor departures from normality, however, typically preserve the attractive properties of the Kalman filter (see, e.g., Schick and Mitter, 1994) although inference is no longer efficient but still asymptotically consistent (see, e.g., the discussion regarding QMLE in Gourieroux et al. 1984). In essence, the Kalman filter deteriorates in the presence of large outliers and results in invalid inference. The remedies for these situations (see, e.g., Durbin and Koopman, 2012) are usually either computationally involved or require strong auxiliary assumptions, such as the precise nature of the heavy-tailed distribution. In this paper we rely on the standard Kalman filtering and simply discard the minority of cases whereby the procedure clearly fails. We leave the question of handling outliers and reducing misspecifications in modeling PE funds as SSM for future research.

Overall, we make stronger assumptions than the methods that use only cash flows. It may well be that resulting specifications errors regarding reported NAVs lead to beta underestimation, or the wrong decay patterns in the weights for past valuations. Nonetheless, our SSM parameter estimates produce, on average, better in- and out-of-sample nowcasts.

References

Ang, A. 2014. *Asset management: A systematic approach to factor investing.* Oxford University Press.

Ang, A., B. Chen, W. N. Goetzmann, and L. Phalippou. 2018. Estimating private equity returns from limited partner cash flows. *Journal of Finance* 73:1751–1783.

Ang, A., D. Papanikolaou, and M. M. Westerfield. 2014. Portfolio choice with illiquid assets. *Management Science* 60:2737–2761.

Boyer, B., T. D. Nadauld, K. P. Vorkink, and M. S. Weisbach. 2018. Private Equity Indices Based on Secondary Market Transactions. National Bureau of Economic Research Working Paper.

Brito, N. O. 1977. Marketability restrictions and the valuation of capital assets under uncertainty. *Journal of Finance* 32:1109–1123.

Brown, G. W., O. R. Gredil, and S. N. Kaplan. 2019. Do private equity funds manipulate reported returns? *Journal of Financial Economics* 132:267–297.

Brown, G. W., R. S. Harris, T. Jenkinson, S. N. Kaplan, and D. Robinson. 2015. What Do Different Commercial Data Sets Tell Us About Private Equity Performance? University of North Carolina Working Paper.

Buchner, A., and R. Stucke. 2014. The systematic risk of private equity. Passau University Working Paper.

Couts, S., A. Gonçalves, and A. Rossi. 2020. Unsmoothing Returns of Illiquid Funds. USC Lusk Center of Real Estate Working Paper Series.

Dempster, A. P., N. M. Laird, and D. B. Rubin. 1977. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B* 39:1–22.

Driessen, J., T.-C. Lin, and L. Phalippou. 2012. A new method to estimate risk and return of nontraded assets from cash flows: the case of private equity funds. *Journal of Financial and Quantitative Analysis* 47:511–535.

Durbin, J., and S. J. Koopman. 2012. *Time series analysis by state space methods.* Oxford University Press.

Ewens, M., C. M. Jones, and M. Rhodes-Kropf. 2013. The price of diversifiable risk in venture capital and private equity. *Review of Financial Studies* 26:1854–1889.

Franzoni, F., E. Nowak, and L. Phalippou. 2012. Private equity performance and liquidity risk. *Journal of Finance* 67:2341–2373.

Gârleanu, N. 2009. Portfolio choice and pricing in illiquid markets. *Journal of Economic Theory* 144:532–564.

Geltner, D. M. 1991. Smoothing in appraisal-based returns. *Journal of Real Estate Finance and Economics* 4:327–345.

Geltner, D. M. 1993. Estimating market values from appraised values without assuming an efficient market. *Journal of Real Estate Research* 8:325–345.

Getmansky, M., A. W. Lo, and I. Makarov. 2004. An econometric model of serial correlation and illiquidity in hedge fund returns. *Journal of Financial Economics* 74:529–609.

Ghysels, E. 1998. On stable factor structures in the pricing of risk: do time-varying betas help or hurt? *Journal of Finance* 53:549–573.

Ghysels, E., and H. Qian. 2019. Estimating MIDAS regressions via OLS with polynomial parameter profiling. *Econometrics and Statistics* 9:1–16.

Goetzmann, W., E. Gourier, and L. Phalippou. 2018. How Alternative Are Alternative Investments? The Case of Private Equity Funds. *Yale School of Management Working Paper* .

Gornall, W., and I. A. Strebulaev. 2020. Squaring venture capital valuations with reality. *Journal of Financial Economics* 135:120–143.

Gornall, W., and I. A. Strebulaev. 2021. A Valuation Model of Venture Capital-Backed Companies with Multiple Financing Rounds. University of British Columbia Working Paper.

Gourieroux, C., A. Monfort, and A. Trognon. 1984. Pseudo maximum likelihood methods: Theory. *Econometrica* pp. 681–700.

Gredil, O. 2022. Do Private Equity Managers Have Superior Information on Public Markets? *Journal of Financial and Quantitative Analysis* 57:321—-358.

Gredil, O., Y. Liu, and B. A. Sensoy. 2020. Diversifying Private Equity. *Tulane University Working Paper* .

Gupta, A., and S. Van Nieuwerburgh. 2021. Valuing Private Equity Investments Strip by Strip. *Journal of Finance* 76:3255–3307.

Herskovic, B., B. Kelly, H. Lustig, and S. Van Nieuwerburgh. 2016. The common factor in idiosyncratic volatility: Quantitative asset pricing implications. *Journal of Financial Economics* 119:249–283.

Kaplan, S. N., and A. Schoar. 2005. Private Equity Performance: Returns, Persistence, and Capital Flows. *Journal of Finance* 60:1791–1823.

Kaplan, S. N., and P. Strömberg. 2003. Financial contracting theory meets the real world: An empirical analysis of venture capital contracts. *Review of Economic Studies* 70:281–315.

Korteweg, A., and S. Nagel. 2020. Risk-Adjusted Returns of Private Equity Funds: A New Approach. University of South California Working Paper.

Korteweg, A., and M. Sorensen. 2010. Risk and return characteristics of venture capital-backed entrepreneurial companies. *Review of Financial Studies* 23:3738–3772.

Ljungqvist, A., and M. Richardson. 2003. The cash flow, return and risk characteristics of private equity. National Bureau of Economic Research Working Paper.

Longstaff, F. A. 2001. Optimal portfolio choice and the valuation of illiquid securities. *Review of Financial Studies* 14:407–431.

Miranda, M. J., and P. L. Fackler. 2004. *Applied Computational Economics and Finance.* MIT Press.

Nadauld, T. D., B. A. Sensoy, K. Vorkink, and M. S. Weisbach. 2019. The liquidity cost of private equity investments: Evidence from secondary market transactions. *Journal of Financial Economics* 132:158–181.

Nelson, D. B. 1992. Filtering and forecasting with misspecified ARCH models I: Getting the right variance with the wrong model. *Journal of Econometrics* 52:61–90.

Phalippou, L. 2020. An Inconvenient Fact: Private Equity Returns and the Billionaire Factory. *Journal of Investing* 30:11–39.

Schick, I. C., and S. K. Mitter. 1994. Robust recursive estimation in the presence of heavy-tailed observation noise. *Annals of Statistics* pp. 1045–1080.

Stafford, E. 2022. Replicating private equity with value investing, homemade leverage, and hold-to-maturity accounting. *The Review of Financial Studies* 35:299–342.

Woodward, S. E., and R. E. Hall. 2004. Benchmarking the returns to venture. National Bureau of Economic Research Working Paper.

Table 1. Fund sample summary statistics

This table reports summary statistics for the sample of buyout (Panel A), and venture funds (Panel B) incepted between 1983 and 2008. For the fund to be included in our sample, it has to meet four criteria: (i) have been operating for at least 6 years since inception, (ii) have at least 20 NAV reports, (ii) made at least 2 distributions, (ii) have at least two peer funds which meet criteria for the estimation in which SSM parameters are obtained independently from other funds (see Section 4.1 for details). # of 'Capital Calls', 'Distributions', 'NAV reports' count the respective data point for each fund during at most 12 years of fund operations ('Fund life'). Data beyond 12th year of fund life are omitted. '% Unresolved' is the ratio of fund latest NAV report to the sum thereof with the cumulative distributions. The last two rows of each panel count the number of quarters with non-zero distributions or only when those exceed 5% of fund size.

Panel A. Buyout funds

	Number of funds: 2,444									
	mean	sd	skew	p5	p10	p25	p50	p75	p90	p95
Vintage year	2004.7	7.1	-0.9	1990	1995	2000	2006	2010	2013	2014
Fund size (USD mil.)	931.3	1750.3	5.0	50	78.8	168.1	383.9	873.4	2169.8	3743.6
Fund life (years)	11.6	2.5	-0.6	6.81	7.40	9.52	12.4	14.1	14.1	14.1
IRR(%)	10.3	18.1	1.3	-10.8	-4.69	3.22	10.2	18.7	28.7	36.7
Money Multiple	1.7	1.1	5.3	0.55	0.76	1.18	1.57	2.05	2.66	3.23
PME (v. CRSP VW)	1.1	0.6	4.6	0.34	0.48	0.74	1.01	1.35	1.77	2.09
# Capital Calls	31.0	19.4	1.2	6	10	17	27	41	56	67
# Distributions	28.0	21.5	2.6	5	8	14	23	36	51	66
# NAV reports	21.5	2.2	-1.0	17	17	20	22	23	24	24
% Resolved	81.4	23.2	-1.5	29.1	44.3	71.6	92.7	98.1	99.4	99.6
# Quarters w/ Dist g0	8.8	5.1	0.4	1	2	5	8	12	16	18
w/ Dist >5% of Fund	5.3	3.6	0.9	0	1	3	5	7	10	12

Panel B. Venture funds

	Number of funds: 1,679									
	mean	sd	skew	p5	p10	p25	p50	p75	p90	p95
Vintage year	2003.1	8.0	-0.8	1986	1992	1999	2005	2009	2013	2014
Fund size (USD mil.)	268.9	326.3	4.0	26	39.4	78.9	172.7	340	575.8	816.5
Fund life (years)	12.1	2.5	-1.0	6.79	7.63	10.2	13.5	14.1	14.1	14.1
IRR(%)	10.3	38.7	6.1	-15.2	-9.16	-0.36	8.23	20.2	38.0	54.5
Money Multiple	2.4	3.5	6.9	0.32	0.50	0.97	1.56	2.57	4.64	6.74
PME (v. CRSP VW)	1.3	1.7	6.1	0.20	0.31	0.56	0.88	1.38	2.31	3.54
# Capital Calls	24.0	15.8	1.9	5	8	14	20	31	43	53
# Distributions	18.3	15.3	3.7	3	4	8	15	24	38	46
# NAV reports	21.8	2.2	-1.2	17	18	21	23	23	24	24
% Resolved	70.7	29.2	-0.9	10.5	21.3	51.1	82.2	95.4	98.8	99.5
# Quarters w/ Dist g0	4.8	3.9	1.1	0	1	2	4	7	10	12
w/ Dist >5% of Fund	3.3	2.9	1.2	0	0	1	3	5	7	9

Table 2. Summary of *fund-specific* estimates

This table reports summary statistics parameter estimates and nowcasting performances metrics obtained with fund-level SSMs for the sample of buyout and venture funds described in Table 1. Panel A reports SSM parameters that are estimated independently for each fund as described in Section 3.3.1 using at least (most) four (seven) years. The return autocorrelations and variances in Panel B are computed from the full history of fund returns estimates. The nowcasting performance metrics in Panel C are as discussed in Section 3.2 utilize at least [most] one [five] remaining years of fund life.

	Number of fund-estimates: 2,513							
	mean	sd	skew	p10	p25	p50	p75	p90
Panel A. Parameter estimates								
Main parameters:								
α: Abnormal return (p.a.)	0.047	0.15	2.03	-0.116	-0.048	0.030	0.114	0.210
β: Systematic risk	1.30	0.33	0.18	0.839	1.040	1.307	1.548	1.684
F: Idiosync Volatilty ($\times$)	3.50	2.26	1.51	1.392	2.092	2.903	3.994	6.858
λ: NAV smoothing bias	0.86	0.25	-2.53	0.520	0.899	0.961	0.983	0.990
σ_n: NAV report noise	0.066	0.041	0.59	0.018	0.037	0.057	0.089	0.140
δ: Dist intensity trend	0.014	0.019	13.3	0.004	0.006	0.011	0.017	0.027
σ_d: Dist intensity noise	1.51	0.66	1.09	0.797	1.031	1.405	1.891	2.360
Parameter mapping to Comparable asset:								
ψ: intercept to Fund return (p.a.)	-0.001	0.049	0.13	-0.059	-0.030	-0.003	0.028	0.062
β_c: loading on Fund return	0.16	0.19	2.95	0.010	0.010	0.177	0.217	0.294
F_c: IdVol vs Fund return ($\times$)	0.88	0.16	-2.35	0.707	0.799	0.918	0.998	1.027
Panel B. Filtered return properties								
Autocorrelations:								
Reported NAVs (quarterly)	0.25	0.22	0.47	-0.011	0.090	0.227	0.381	0.552
SSM estimates (quarterly)	0.077	0.30	-0.18	-0.332	-0.112	0.092	0.286	0.445
SSM estimates (weekly)	0.014	0.17	2.09	-0.119	-0.091	-0.033	0.056	0.212
Naive nowcast (weekly)	0.12	0.17	1.68	-0.024	0.011	0.070	0.183	0.353
Variances:								
Reported NAVs (quarterly)	0.035	0.046	2.86	0.006	0.010	0.019	0.038	0.080
SSM estimates (quarterly)	0.038	0.042	2.62	0.011	0.016	0.024	0.039	0.081
Panel C. Nowcasted performance assessment								
In-Sample RMSE	0.062	0.066	2.51	0.009	0.019	0.041	0.080	0.139
Hybrid RMSE SSM	0.070	0.077	2.53	0.009	0.020	0.045	0.092	0.159
Hybrid RMSE Naive	0.44	0.56	4.01	0.061	0.136	0.281	0.514	0.973
OOS RMSE Naive	0.34	0.42	4.73	0.042	0.110	0.233	0.432	0.694
OOS RMSE SSM	0.20	0.28	4.67	0.018	0.052	0.120	0.251	0.459

Table 3. Buyout fund estimates comparison

This table reports summary statistics on fund-level SSM parameter estimates (Panel A), fund quarterly return estimates (Panel B) and nowcasting performances metrics (Panel C) for buyout funds incepted between 1983 and 2008. Section 3.2 defines the nowcasting performance metrics. Row titles indicate the method that parameter estimates were obtained with. Sections 3.3.1–3.3.2 describe the key differences between fund-specific and partially imputed parameter estimations. Section 4.3 describes the three different imputation methods we applied.

	Number of fund-level estimates:				fund-specific—1,654, partially-imputed—2,300			
	mean	p25	p50	p75	mean	p25	p50	p75
Panel A. Selected parameters								
	α (p.a.)				β			
Fund-specific	0.063	−0.016	0.050	0.127	1.146	0.962	1.101	1.307
Peer-imputed	0.046	−0.025	0.045	0.120	0.983	0.868	0.929	1.092
Average-imputed	0.051	−0.023	0.048	0.125	0.969	0.912	0.912	0.912
Literature-imputed	0.011	−0.073	0.007	0.089	1.346	1.250	1.250	1.250
	F_c				λ			
Fund-specific	0.895	0.832	0.943	1.008	0.835	0.871	0.954	0.981
Peer-imputed	0.908	0.850	0.964	1.011	0.886	0.831	0.910	0.968
Average-imputed	0.908	0.819	0.964	1.012	0.839	0.673	0.926	0.955
Literature-imputed	0.871	0.801	0.922	1.002	0.860	0.901	0.964	0.985
Panel B. Fund return properties								
	Weekly return autocorrelation				Annualized Standard Deviations			
Naive nowcast	0.119	0.004	0.064	0.173	0.278	0.206	0.239	0.280
Fund-specific	0.031	−0.080	−0.015	0.072	0.343	0.230	0.288	0.378
Peer-imputed	0.076	−0.054	0.029	0.146	0.333	0.231	0.286	0.367
Average-imputed	0.067	−0.056	0.020	0.129	0.340	0.230	0.289	0.388
Literature-imputed	0.016	−0.084	−0.025	0.048	0.377	0.265	0.317	0.401
Panel C. Nowcast performance assessment								
	Hybrid RMSE				OOS RMSE			
Naive nowcast	0.396	0.141	0.288	0.522	0.335	0.118	0.251	0.454
Fund-specific	0.066	0.019	0.042	0.083	0.176	0.045	0.104	0.213
Peer-imputed	0.059	0.016	0.035	0.070	0.164	0.037	0.088	0.184
Average-imputed	0.068	0.031	0.050	0.081	0.160	0.047	0.099	0.178
Literature-imputed	0.075	0.020	0.046	0.094	0.206	0.046	0.113	0.233

	Hybrid Improv. %	OOS Improv. %	Autocorr. Improv. %
Fund-specific	88.8	71.2	58.1
Peer-imputed	92.2	74.6	56.5
Average-imputed	90.1	74.3	57.3
Literature-imputed	88.9	69.6	59.2

Table 4. Venture fund estimates comparison

This table reports summary statistics on fund-level SSM parameter estimates (Panel A), fund quarterly return estimates (Panel B) and nowcasting performances metrics (Panel C) for venture funds incepted between 1983 and 2008. Section 3.2 defines the nowcasting performance metrics. Row titles indicate the method that parameter estimates were obtained with. Sections 3.3.1–3.3.2 describe the key differences between fund-specific and partially imputed parameter estimations. Section 4.3 describes the three different imputation methods we applied.

	Number of fund-level estimates: fund-specific—858, partially-imputed—1,622							
	mean	p25	p50	p75	mean	p25	p50	p75
Panel A. Selected parameters								
	α (p.a.)				β			
Fund-specific	0.016	−0.094	−0.021	0.061	1.603	1.455	1.621	1.684
Peer-imputed	−0.015	−0.111	−0.030	0.048	1.427	1.342	1.415	1.480
Average-imputed	−0.008	−0.107	−0.024	0.061	1.391	1.339	1.339	1.339
Literature-imputed	−0.059	−0.167	−0.081	0.011	1.800	1.800	1.800	1.800
	F_c				λ			
Fund-specific	0.851	0.768	0.865	0.971	0.913	0.936	0.969	0.984
Peer-imputed	0.839	0.739	0.867	0.980	0.932	0.899	0.951	0.980
Average-imputed	0.832	0.758	0.846	0.977	0.911	0.901	0.963	0.979
Literature-imputed	0.812	0.736	0.822	0.957	0.924	0.953	0.976	0.987
Panel B. Fund return properties								
	Weekly return autocorrelation				Annualized Standard Deviations			
Naive nowcast	0.116	0.011	0.070	0.176	0.326	0.199	0.268	0.386
Fund-specific	−0.017	−0.105	−0.071	0.022	0.405	0.296	0.343	0.431
Peer-imputed	0.001	−0.102	−0.057	0.043	0.403	0.285	0.335	0.423
Average-imputed	−0.009	−0.104	−0.058	0.024	0.395	0.286	0.338	0.420
Literature-imputed	−0.030	−0.108	−0.076	−0.006	0.443	0.336	0.381	0.456
Panel C. Nowcast performance assessment								
	Hybrid RMSE				OOS RMSE			
Naive nowcast	0.531	0.199	0.377	0.634	0.393	0.159	0.314	0.522
Fund-specific	0.084	0.027	0.056	0.114	0.235	0.064	0.147	0.282
Peer-imputed	0.069	0.020	0.042	0.085	0.206	0.055	0.126	0.236
Average-imputed	0.082	0.034	0.056	0.097	0.201	0.070	0.139	0.249
Literature-imputed	0.093	0.026	0.058	0.122	0.237	0.061	0.141	0.281
	Hybrid Improv. %		OOS Improv. %		Autocorr. Improv. %			
Fund-specific	90.7		63.2		56.0			
Peer-imputed	93.8		73.5		56.4			
Average-imputed	93.2		71.1		57.0			
Literature-imputed	91.9		68.6		55.6			

Table 5. Fund characteristics and *Systematic risk* exposure

This table regresses the fund-specific β (Panel A) and the natural logs of λ (Panel B) estimates the fund's αestimates as well as selected characteristics of the fund and the period it was operating. The sample includes PE funds described in Table 1 as well as REPE funds per Online Appendix Table A.2. ***/**/* denotes significance at 1/5/10% confidence level, t-statistics robust to error clustering at vintage level are reported in parentheses.

Panel A. Fund-specific β estimates

	(1)	(2)	(3)	(4)	(5)
log(Size USD)	−0.005	0.005	0.008**		0.013**
	(−1.48)	(1.41)	(2.15)		(2.44)
log($\sum$Distrib./Size)		0.085***	0.082***		0.102***
		(10.25)	(9.72)		(9.00)
log(# Distrib.)		−0.036***	−0.034***		−0.049***
		(−4.49)	(−4.28)		(−4.53)
Industry variance			−2.502***		−2.592***
			(−7.07)		(−5.17)
Market variance			1.359***		0.465
			(3.64)		(0.92)
Previous fund's β				0.032*	0.039**
				(1.66)	(2.00)
Previous fund's λ				0.018	0.005
				(0.82)	(0.26)
Fixed effects:			Industry, Region, Fund type		
Observations	2,811	2,811	2,811	1,492	1,492
R^2	0.781	0.790	0.793	0.799	0.814

Panel B. Fund-specific λ estimates

	(1)	(2)	(3)	(4)	(5)
log(SizeUSD)	0.055***	0.041**	0.044**		0.030
	(3.06)	(2.03)	(2.09)		(1.10)
log($\sum$Distrib./Size)		−0.102**	−0.105**		−0.071
		(−2.17)	(−2.21)		(−1.18)
log(# Distrib.)		0.055	0.053		0.007
		(1.22)	(1.17)		(0.13)
Industry variance			−0.621		1.825
			(−0.31)		(0.79)
Market variance			1.530		0.086
			(0.86)		(0.04)
Previous fund's β				−0.042	−0.041
				(−0.54)	(−0.53)
Previous fund's λ				0.230**	0.234**
				(2.05)	(2.08)
Fixed effects:			Industry, Region, Fund type		
Observations	2,811	2,811	2,811	1,492	1,492
R^2	0.040	0.042	0.042	0.044	0.048

Table 6. Private equity index

This table examines the time series properties and risk factor exposures of buyout (Panels A) and venture (Panels B) $ NAV-estimate weighted indices at weekly frequency. 'Comp' uses the returns on matched industry/style benchmarks, 'Naive' uses the returns implied by naïve NAV nowcasts, 'SSM' uses our fund-specific estimates for both NAVs and returns. 'Excess Returns' are in those of respective index in excess of 'Comp'. See Appendix A.3 for details. The sample of returns is 1987–2019, the sample of funds is described in Table 1. Robust to autocorrelation t-statistics are reported in parentheses, ***/**/* denotes significance at 1/5/10% confidence level.

Panel A1. Autocorrelations of the Buyout funds index

	Total Returns					Excess Returns	
	Comp	Naive		SSM		Naive	SSM
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
lag1	−0.029	−0.080**	−0.082**	−0.029	−0.030	0.780***	−0.020
	(−0.71)	(−2.18)	(−2.21)	(−0.69)	(−0.75)	(10.63)	(−0.44)
lag2	0.036		−0.015		0.032		
	(1.19)		(−0.49)		(1.10)		
lag3	0.033		−0.014		0.025		
	(1.23)		(−0.46)		(0.95)		
lag4	−0.030		−0.051*		−0.029		
	(−1.13)		(−1.74)		(−1.12)		
lag5	−0.001		−0.024		−0.005		
	(−0.04)		(−0.72)		(−0.15)		
lag6	0.022		0.001		0.031		
	(0.58)		(0.04)		(0.91)		
lag7	−0.015		−0.033		−0.014		
	(−0.57)		(−1.19)		(−0.48)		
Constant	0.234***	0.343***	0.391***	0.343***	0.322***	0.017	0.092***
	(3.21)	(5.70)	(4.66)	(4.48)	(3.80)	(1.27)	(6.49)
T	1,773	1,779	1,773	1,779	1,773	1,779	1,779

Panel A2. Risk factor exposures of the Buyout funds index

	Total returns					Excess returns	
	comp	Naive		SSM		Naive	SSM
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Market	1.086***	1.032***	1.022***	1.223***	1.212***	−0.061***	0.134***
	(64.11)	(57.69)	(91.04)	(108.92)	(99.15)	(−7.10)	(8.07)
SMB			0.226***		0.184***		
			(11.39)		(13.61)		
HML			0.230***		0.114***		
			(11.61)		(5.28)		
Market lag						−0.066***	0.005
						(−6.79)	(0.72)
Constant	0.060***	0.084**	0.076*	0.068***	0.065***	0.096**	0.066***
	(3.43)	(2.14)	(1.89)	(4.90)	(6.37)	(2.35)	(5.84)
T	1,780	1,780	1,780	1,780	1,780	1,779	1,779
RMSE	0.635	0.914	0.813	0.437	0.338	0.745	0.396
St.Dev.	2.661	2.620		2.944		0.774	0.509

Table 6. Private equity index – (*Continued*)

Panel B1. Autocorrelations of the Venture funds index

	Total Returns					Excess Returns	
	Comp	Naïve		SSM		Naïve	SSM
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
lag1	−0.035	−0.067*	−0.066*	−0.056	−0.057	0.906***	−0.067*
	(−1.03)	(−1.93)	(−1.86)	(−1.51)	(−1.54)	(60.82)	(−1.85)
lag2	0.009		−0.010		0.025		
	(0.34)		(−0.32)		(0.91)		
lag3	0.046		0.035		0.035		
	(1.51)		(1.04)		(1.20)		
lag4	−0.031		−0.033		−0.024		
	(−1.19)		(−1.15)		(−0.92)		
lag5	0.001		0.007		−0.010		
	(0.04)		(0.22)		(−0.30)		
lag6	0.026		0.035		0.027		
	(0.77)		(0.96)		(0.81)		
lag7	−0.032		−0.019		−0.016		
	(−0.98)		(−0.56)		(−0.55)		
Constant	0.243***	0.387***	0.378***	0.420***	0.397***	0.012	0.160***
	(3.15)	(5.55)	(5.51)	(4.06)	(3.52)	(1.64)	(4.21)
T	1,773	1,779	1,773	1,779	1,773	1,779	1,779

Panel B2. Risk factor exposures of the Venture funds index

	Total Returns					Excess returns	
	Comp	Naïve		SSM		Naïve	SSM
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Market	1.148***	1.125***	1.089***	1.699***	1.674***	−0.049***	0.527***
	(49.45)	(41.97)	(59.27)	(84.77)	(132.60)	(−6.20)	(27.57)
SMB			0.171***		0.094***		
			(6.02)		(6.09)		
HML			−0.392***		−0.315***		
			(−9.96)		(−17.59)		
Market lag						−0.057***	−0.011
						(−6.27)	(−1.16)
Constant	0.052**	0.114**	0.130***	0.053**	0.066***	0.133***	0.062***
	(2.09)	(2.21)	(2.64)	(2.37)	(3.47)	(2.95)	(2.75)
T	1,780	1,780	1,780	1,780	1,780	1,779	1,779
RMSE	1.041	1.313	1.142	0.758	0.573	0.722	0.812
St.Dev.	2.924	2.981		4.114		0.743	1.498

Figure 1. PE fund's asset values at weekly frequency

Panels A though C plot a hypothetical PE fund's distributions, "true" NAVs (blue line), reported NAVs (red dots) in weeks 455, 468, 481, 494, and 507, cash distributions (red bars) in weeks 474, 489, 497, and 505, naïve nowcasts (dotted lines in panel (B)) and SSM-based nowcasts (dotted lines in Panel C). Panel D plots -1 plus the ratio of the nowcasts to true NAVs.

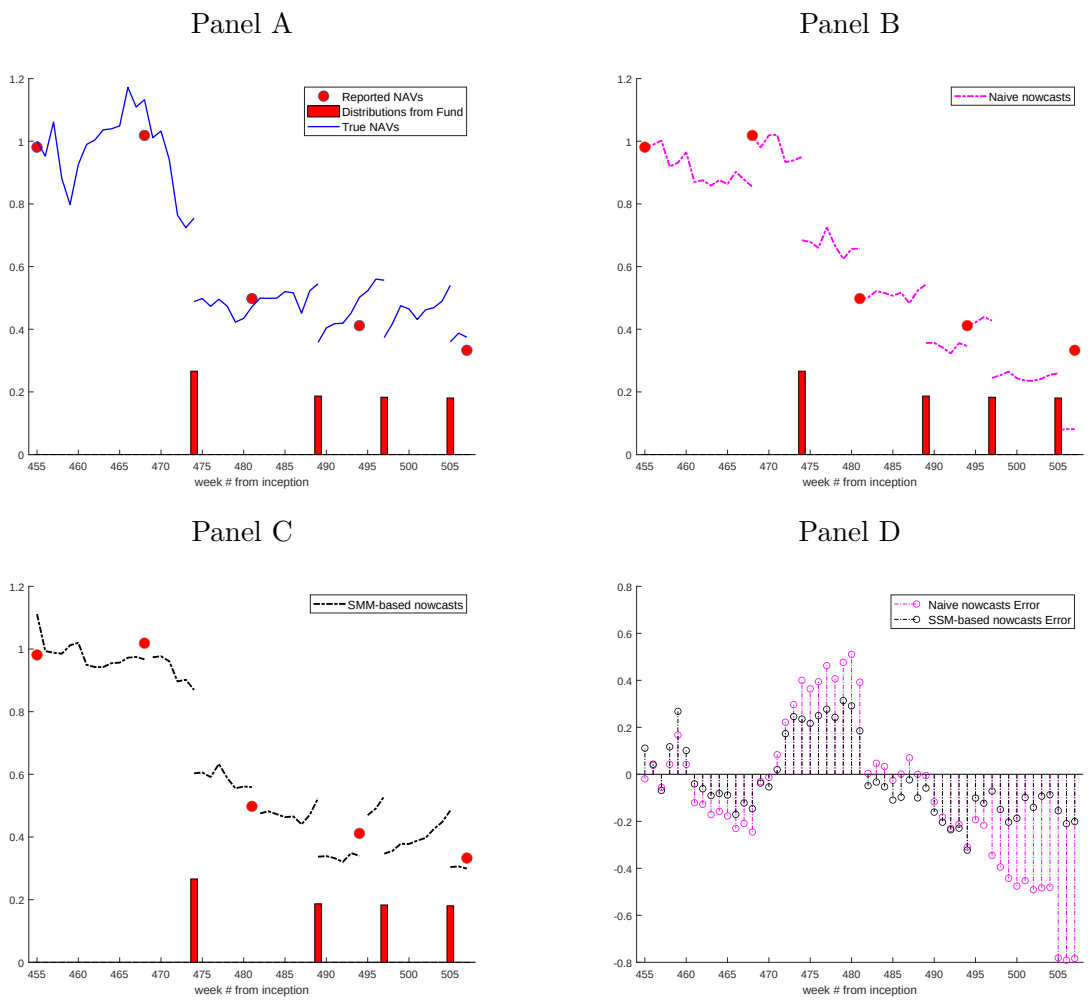
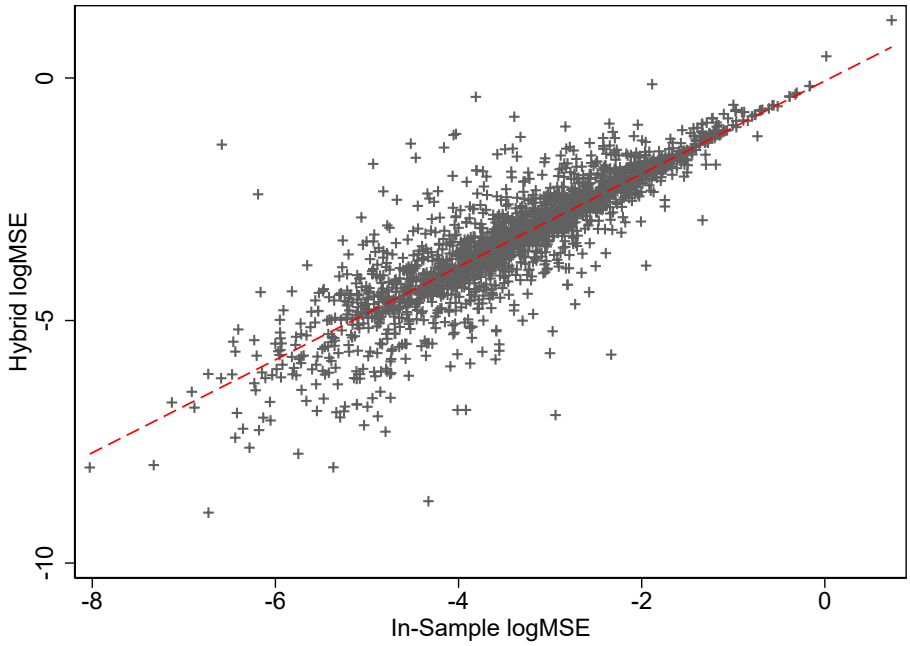


Figure 2. Predictability of nowcasting performance

This figure compares the logs of in-sample nowcasting error (x-axis) with those out-of-sample (y-axis) as defined in Section 3.2 using fund-specific estimates summarized in Table 2. A positive slope indicates that that SSM nowcasting precision is more predictable. Table 1 describes the sample.

Panel A: Hybrid metric



Panel B: Out-of-Sample metric

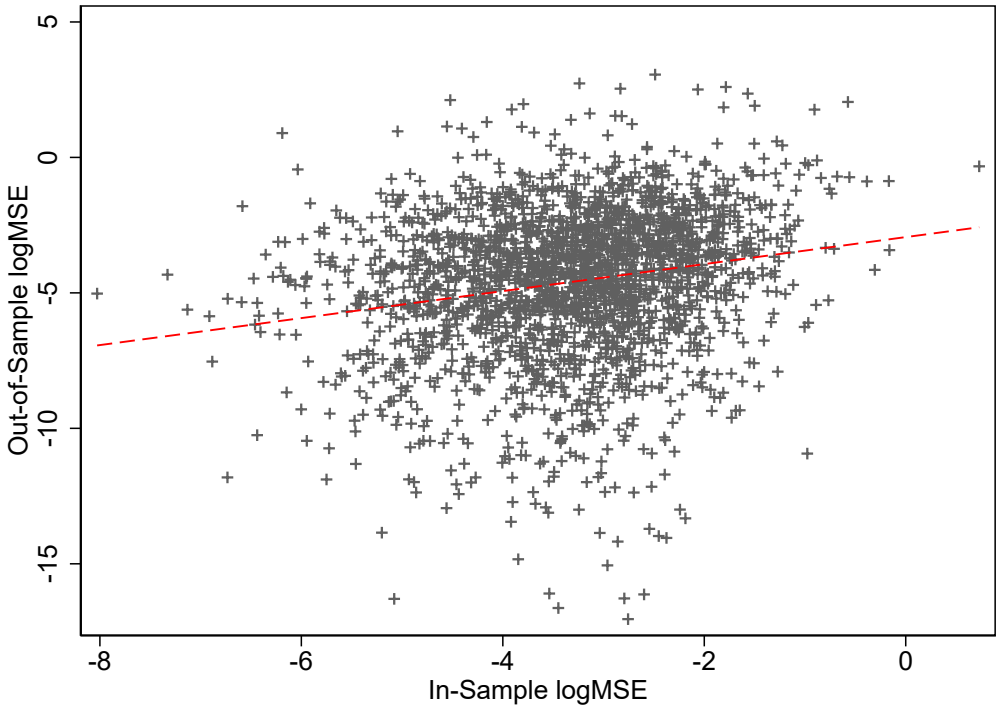
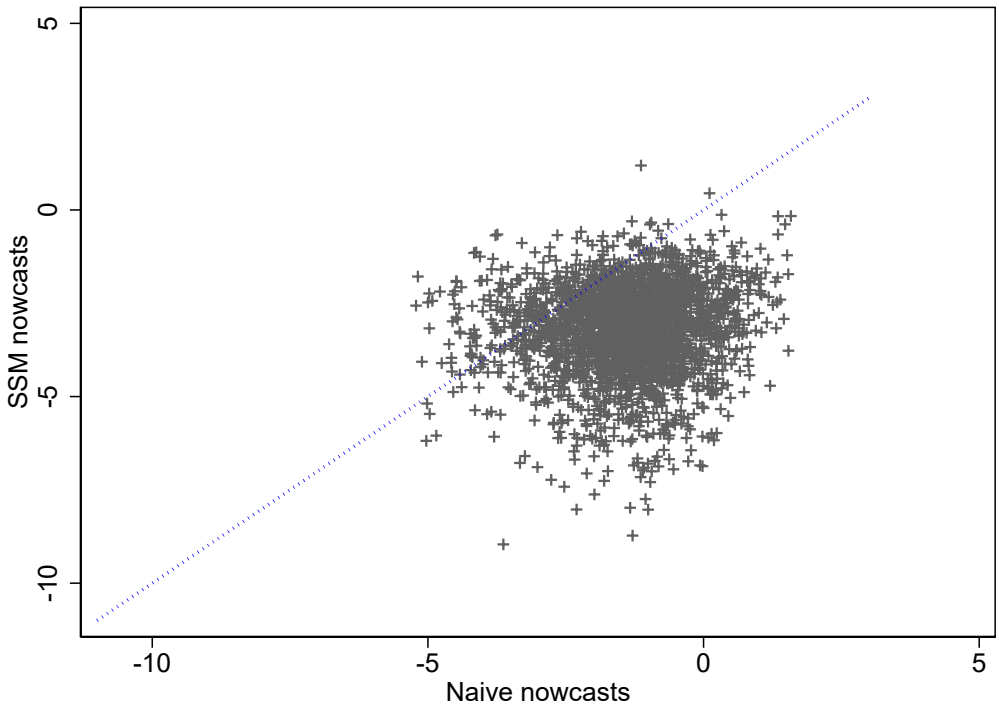


Figure 3. Nowcasting performance assessment

This figure compares nowcasting errors from SSM (y-axis) using fund-specific estimates summarized in Table 2 with those from a naïve method based on as-reported NAVs and matched benchmark returns. The metrics are defined in Section 3.2 and are based on the idea that the NPV of fund cash flows has to be zero if true returns are used to discount them.

Panel A: Hybrid metric



Panel B: Out-of-Sample metric

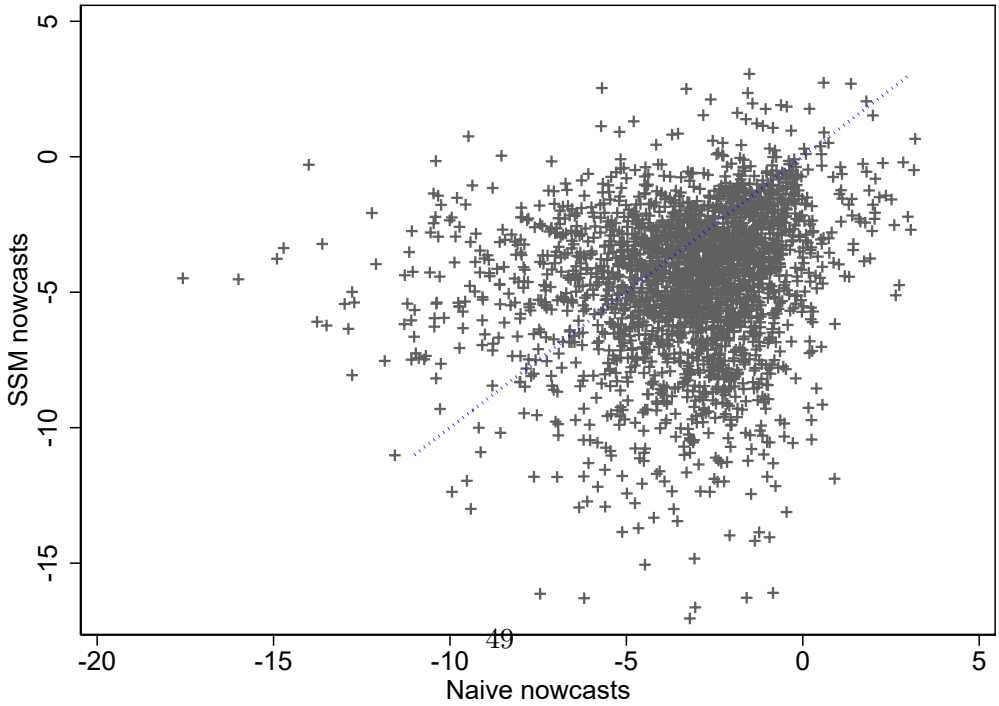
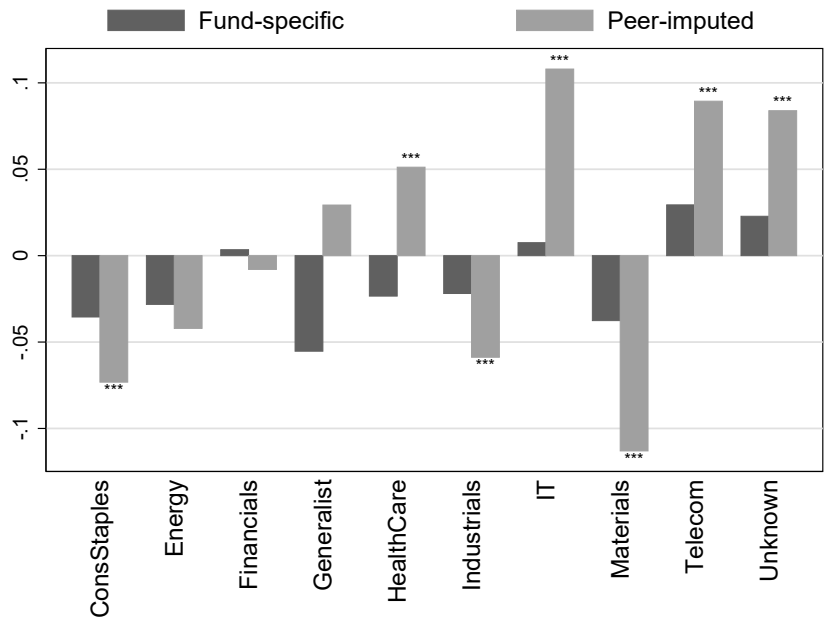


Figure 4. Fund characteristics and *Systematic risk* exposure

This figure plots fixed effect estimates of regression of fund-level β estimates on fund characteristics corresponding to specification 1 of Table 5. The sample includes PE funds described in Table 1 as well as REPE funds per Online Appendix Table A.2. ***/**/* denotes significance at 1/5/10% confidence level with inference robust to clustering at vintage level. The baseline effects are, respectively, 'Consumer Discretionary' and 'Buyout'.

Panel A: Industry fixed effects



Panel B: Fund type fixed effects

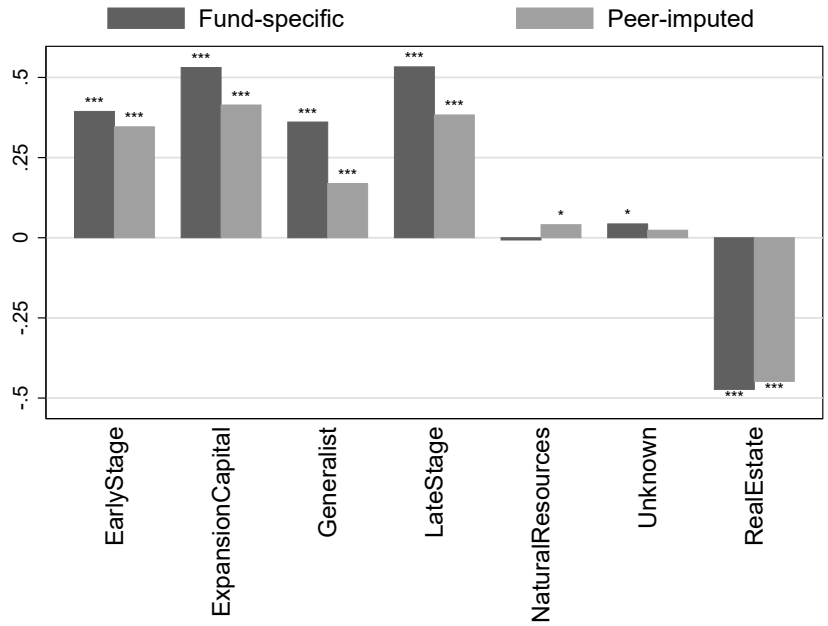


Figure 5. Trends in fund risk and returns

This figure plots the time series and cross-sectional variation of selected fund-specific parameter estimates as described in Sections 3.3.1 and 4.2. The "boxes" reflect the inter-quartile range and the median of the respective parameter over buyout (Panel A) or venture (Panel B) funds incepted in the year indicated on the x-axis. Outlier values are not depicted.

Panel A: Buyout funds

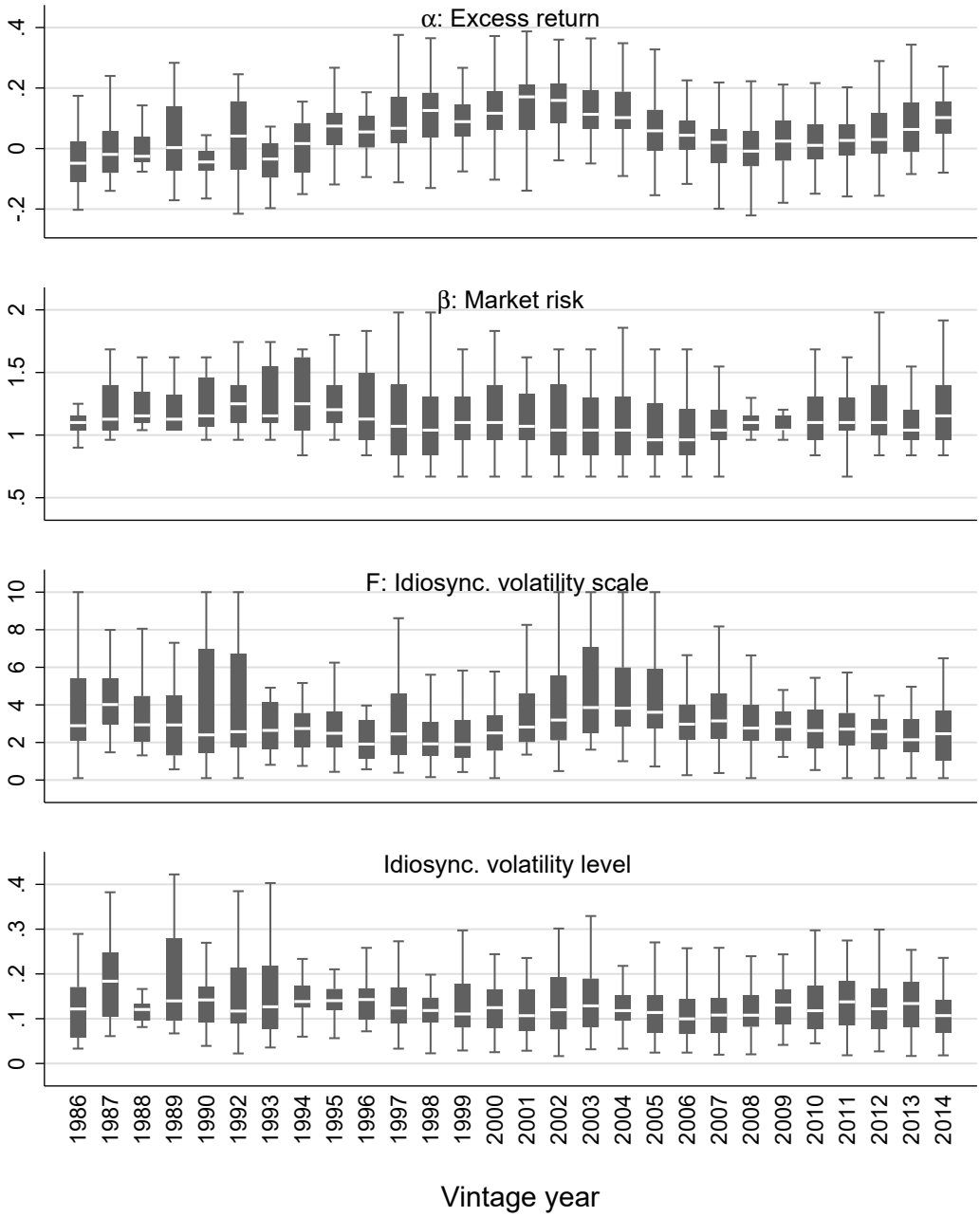


Figure 5. Trends in fund risk and returns – (*Continued*)

Panel B: Venture funds

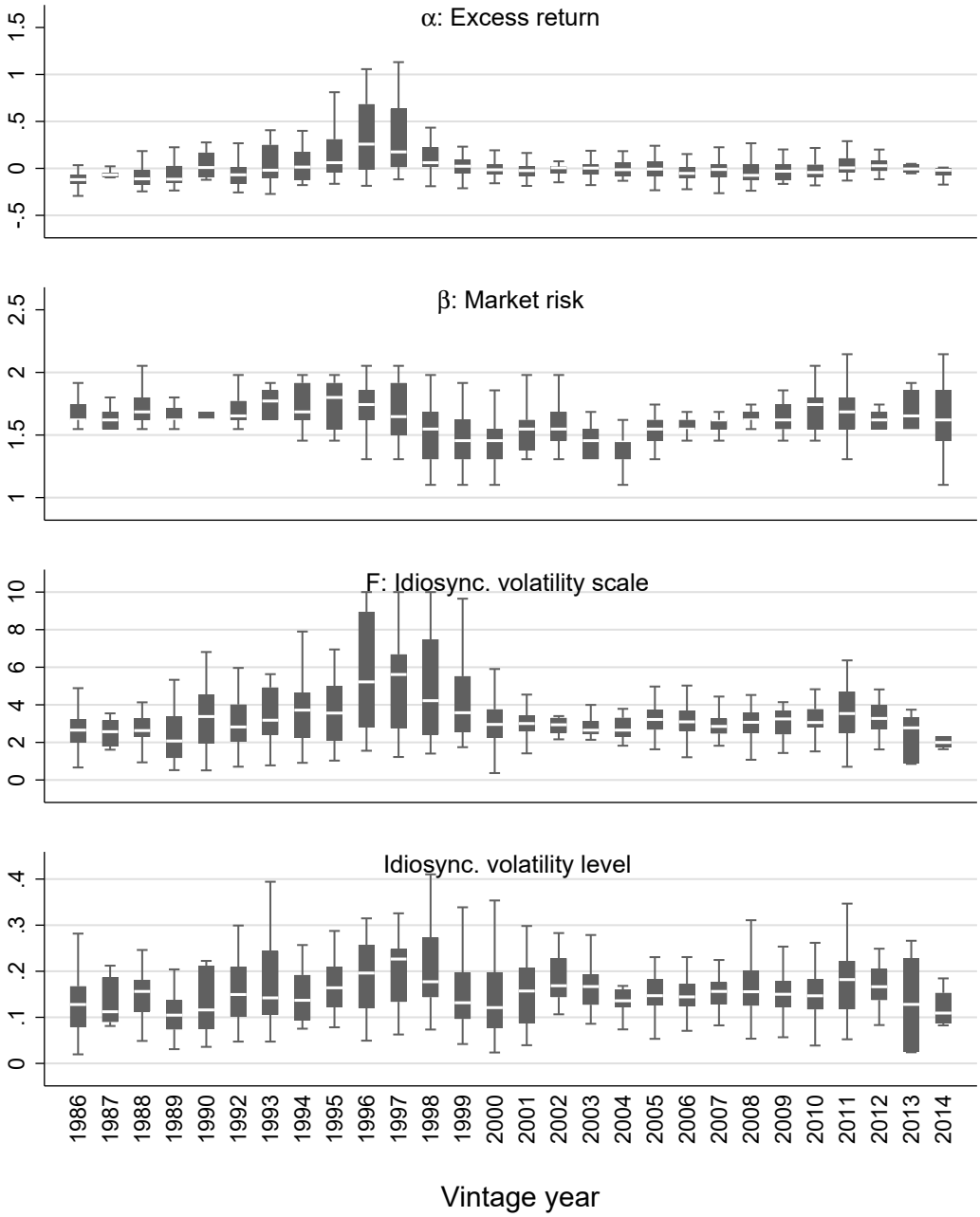
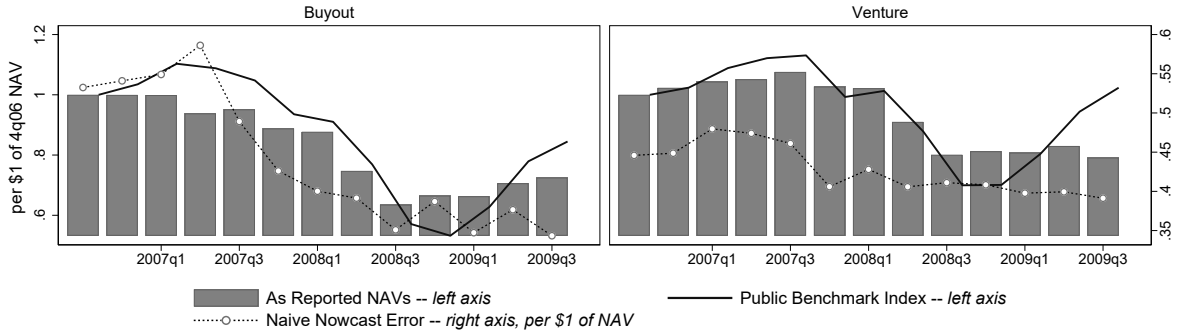


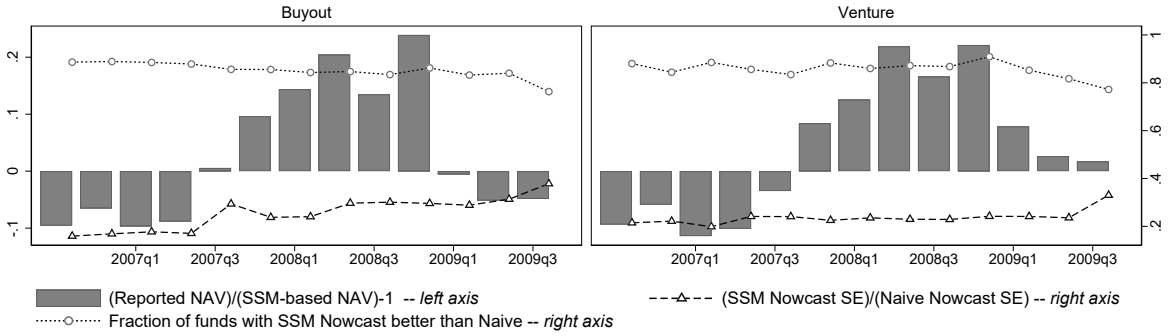
Figure 6. NAV bias and Nowcasting around the GFC

This figure illustrates the application of SSM for asset allocation decisions concerning PE around dramatic shifts in public market valuations such as the financial crisis of 2008. The subsample for this exercise includes 175 buyout funds (left-hand charts) and 159 venture funds (right-hand charts). All funds are U.S.-focused and satisfy the following criteria: (i) were between 5 and 8 years old as of the beginning of 2007, (ii) were not fully resolved before the beginning of 2010, and (iii) had the ratio of the reported NAVs to SSM-estimated values between 0.1 and 10.0 during 2007. Panel A plots the average as-reported NAVs and industry benchmark cumulative returns, both normalized by the level Q4 2006, against the mean nowcasting error on the out-of-sample basis using the naïve approach (Section 3.2). Panel B shows the time series estimates of the average appraisal bias inherit in the reported NAVs and the extent of SSM-based nowcast improvement relatively to the naïve approach. Panel C shows the cross-sectional variation in the NAV bias estimates and the SSM-based nowcast error during the period between Q4 2006 and Q4 2009. Section 5.1 provides details.

Panel A: Reported NAVs and naïve nowcasts



Panel B: SSM-based inference – means across funds



Panel C: SSM-based inference – standard deviations across funds

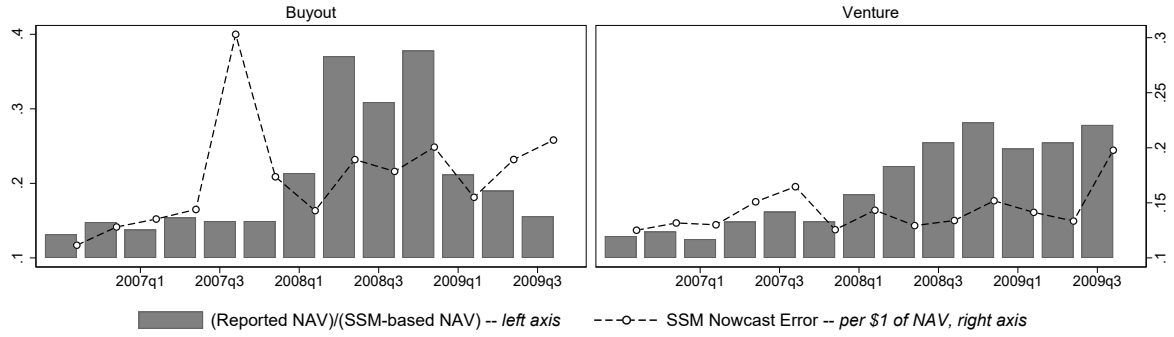


Figure 7. Private equity index

This figure plots five-year rolling returns of $NAV-weighted index for buyout and venture funds. The index is constructed from weekly estimates of the individual fund-level returns starting from the first week of 1987 through the 13th week of 2021. Panel A reports the fund count and as-reported aggregate NAVs. Panel B reports 5-year rolling returns for SSM-based and naïve estimates.

Panel A: Composition

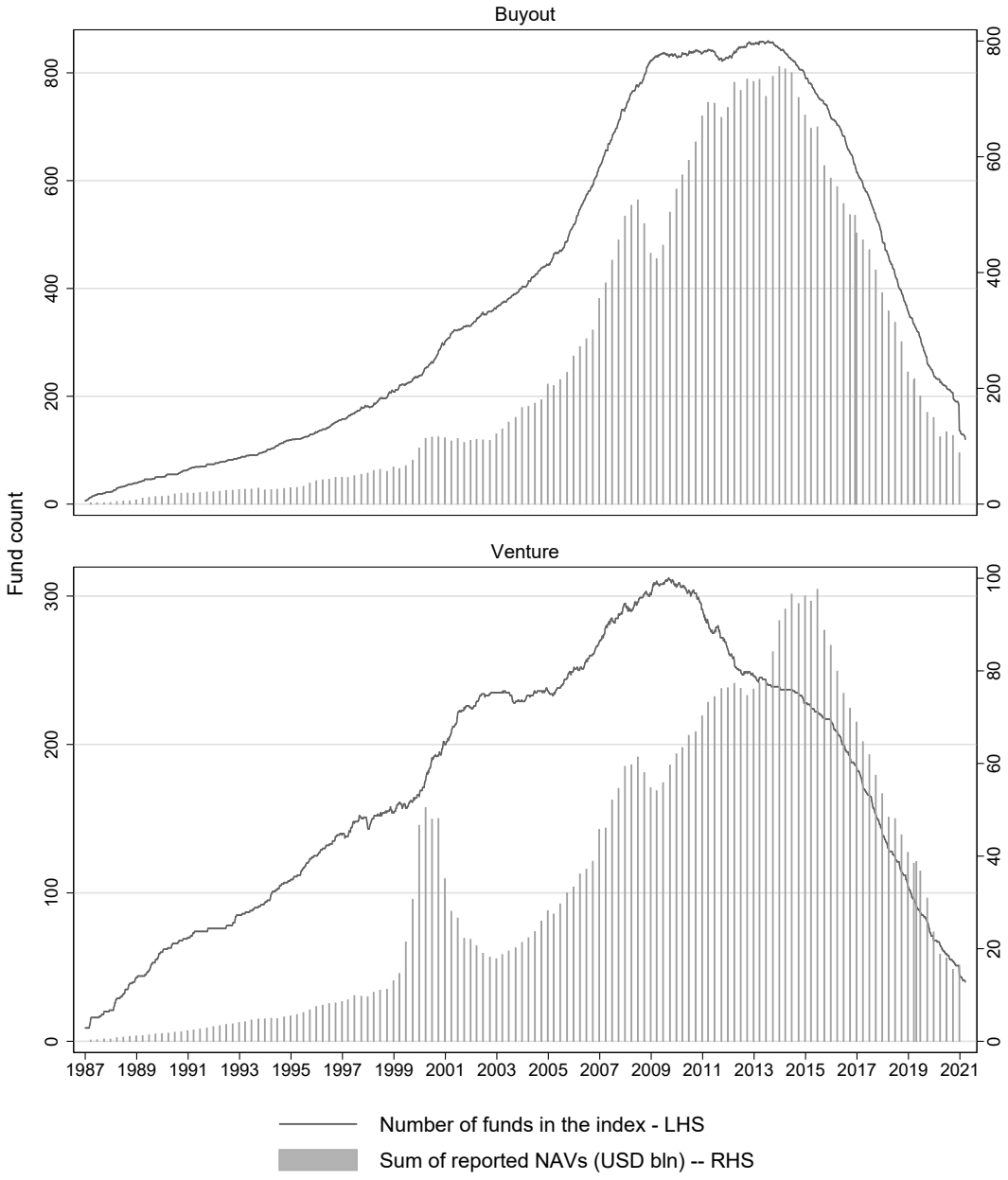


Figure 7. Private equity index – (*Continued*)

Panel B: Rolling 5-year returns

Online Appendix to

Nowcasting Net Asset Values: The Case of Private Equity

Gregory W. Brown* Eric Ghysels† Oleg Gredil‡

This version – July 21, 2022

Contents

A.1 Review of State Space Models i

A.2 SSM Representation of PE Fund - Details ii

A.3 Naïve Nowcasts: construction and properties v

A.4 Estimation details vii

A.5 Why unsmoothing at high frequency is essential viii

A.6 Monte Carlo experiment x

A.7 Excerpts from AICPA Practitioner's Guide xix

List of Figures

A.1 Unsmoothing frequency matters . . . ix
A.2 Case study: Data for a hypothetical fund . . . xii
A.3 Case study: Profiled Likelihood . . . xiii
A.4 Case study: Nowcasting results . . . xiv
A.5 Monte Carlo experiment: cross-section of α,β and nowcasting performance . . . xviii
A.6 PE index risk loadings — time series . . . xxiii

List of Tables

A.1 Monte Carlo experiment: parameter estimates and nowcasting performance . . . xvii
A.2 Real Estate Private Equity (REPE) funds . . . xxiv

*University of North Carolina at Chapel Hill, Kenan-Flagler Business School and Frank Hawkins Kenan Institute of Private Enterprise, gregwbrown@unc.edu, (919) 962-9250

†Department of Finance, Kenan-Flagler Business School, Department of Economics, University of North Carolina at Chapel Hill, Frank Hawkins Kenan Institute of Private Enterprise, and Centre for Economic Policy Research (CEPR), eghysels@unc.edu, (919) 962-9810

‡Tulane University, Freeman School of Business, ogredil@tulane.edu, (504) 314-7567

A.1 Review of State Space Models

Following a slightly modified notation from Durbin and Koopman (2012), the general linear Gaussian SSM can be written as

$$\begin{aligned} y_t - \tilde{\Gamma}_t x_t &= \tilde{Z}_t s_t + \epsilon_t, , \qquad \epsilon_t \sim \mathcal{N}(0, \tilde{H}_t), \\ s_{t+1} &= \tilde{G}_t s_t + \tilde{V}_t \eta_t, , \qquad \eta_t \sim \mathcal{N}(0, \tilde{Q}_t), \qquad t = 1, ..., T, \end{aligned} \tag{A.1}$$

where y_t is a p-by-1 vector of data observable at period t with some (or all) entries possibly being missing, x_t a n-by-1 vector of observable regressors that explain the variation in y_t, s_t a m-by-1 vector of unobserved state that explains the residual variation in the data (i.e., $y_t - \tilde{\Gamma}_t x_t$) up to mutually and temporarily uncorrelated vectors of normally distributed shocks ϵ_t and η_t. The first equation in (A.1) is referred to as *observation* or *measurement equation* while the second equation, which describes the evolution of the latent state, is referred to as *state* or *transition equation*. The initial state value s_0 is either known or drawn from a normal distribution as well while being independent from ϵ_t and η_t.

The matrices ($\tilde{\Gamma}_t, \tilde{Z}_t, \tilde{G}_t, \tilde{V}_t$) and ($\tilde{H}_t, \tilde{Q}_t$) can all be time-varying but deterministic functions of observed data and/or some parameters θ. Moreover, $\tilde{G}_{t-1}$ and $\tilde{Z}_t$ can depend on $y_1, ..., y_{t-1}$. It is intuitive to think about $\tilde{Z}_t$ and $\tilde{H}_t$ as measuring sensitivity of the observed data to, respectively, the state level and to the noise normalized to unit variance.

The important advantage of the SSM approach to a problem is that, given θ, the system (A.1) enables extracting the expected value estimates, conditional on observables, of the latent state series s_t. The most famous application of this method for Gaussian SSMs is the *Kalman filter* (Kalman, 1960) in which the projection theorem for multivariate normally distributed random variables leads to the following standard results derived recursively from $t = 1$ to $t = T$:

$$\begin{aligned} s_{t+1}^{filter} &= \tilde{G}_t s_t^{filter} + \tilde{K}_t \Big(y_t^{resid} - \tilde{Z}_t s_t^{filter} \Big) = \mathrm{E}\Big[s_t \Big| Y_t \Big] \\ \tilde{F}_t &= \tilde{Z}_t \tilde{P}_t \tilde{Z}_t' + \tilde{H}_t = \mathrm{var}\Big(y_t^{resid} - \tilde{Z}_t s_t^{filter} \Big| Y_{t-1} \Big) \\ \tilde{P}_{t|t} &= \tilde{P}_t - \tilde{P}_t \tilde{Z}_t' \tilde{F}_t^{-1} \tilde{Z}_t \tilde{P}_t = \mathrm{var}\Big(s_t^{filter} \Big| Y_t \Big) \\ \tilde{P}_{t+1} &= \tilde{G}_t \tilde{P}_{t|t} \tilde{G}_t' + \tilde{V}_t \tilde{Q}_t \tilde{V}_t' \end{aligned} \tag{A.2}$$

where $\tilde{K}_t = \tilde{G}_t \tilde{P}_t \tilde{Z}_t' \tilde{F}_t^{-1}$ is the Kalman gain, and $y_t^{resid} := y_t - \tilde{\Gamma}_t x_t$ while Y_t denotes the set of the past observations, $(y_1^{resid\,\prime}, \ldots, y_t^{resid\,\prime})'$. The equations in (A.2) are also called the *Forward recursion* as they estimate of state s_t increments forward each period as new information becomes available. However, depending on the problem under consideration, it may be appropriate to condition on the observations from future periods τ, $t < \tau < T$, as well to form the estimate of the state value as of t. These are known respectively as *Kalman smoother* and *backward recursion*. They can be defined as an adjustment to the estimates from *Forward recursion*:

$$s_t^{smooth} = s_t^{filter} + \tilde{P}_t b_{t-1} = \mathrm{E}\Big[s_t \Big| Y_T \Big], \tag{A.3}$$

where $b_{t-1} = \tilde{Z}_t' \tilde{F}_t^{-1} \Big(y_\tau^{resid} - \tilde{Z}_\tau s_\tau^{filter} \Big) + \Big(\tilde{G}_t - \tilde{K}_t \tilde{Z}_t \Big)' b_t$ and: $\tilde{P}_t b_{t-1} = \sum_{\tau=t}^{T} \mathrm{cov}\Big(s_t^{filter}, y_\tau^{resid} - \tilde{Z}_\tau s_\tau^{filter} \Big) \tilde{F}_\tau^{-1} \Big(y_\tau^{resid} - \tilde{Z}_\tau s_\tau^{filter} \Big)$ since η_t and ϵ_t are mutually and temporarily independent.

A final word about estimation, which uses the maximum likelihood principle. Suppose we collect all the parameters governing the matrices $\tilde{\Gamma}_t$, $\tilde{Z}_t$, $\tilde{G}_t$, $\tilde{V}_t$, $\tilde{H}_t$, $\tilde{Q}_t$, $\tilde{G}_{t-1}$, $\tilde{Z}_t$, $\tilde{Z}_t$, and $\tilde{H}_t$, is denoted $\theta \in \Theta$. We will denote the Gaussian likelihood given the data and parameter vector by $\mathcal{L}((y_t, x_t), t = 1, \ldots, T; \theta)$ (see again Durbin and Koopman (2012) for details of the likelihood fn specification).

A.2 SSM Representation of PE Fund - Details

We proceed with casting the equations (5) through (11) in the paper as a SSM per equation (A.1):

$$y_t = [d_{t-1}\ nav_{t-2}\ r_{ct}]', \qquad \tilde{\Gamma}_t = I_{3\times 3}, \qquad x_t = [\hat{m}_{t-1}\ \hat{m}_{t-2}\ (\beta_c \cdot \beta - b) r_{mt}]', \tag{A.4}$$

$$\tilde{Z}_t = \begin{bmatrix} 0 & 1 & 0 & log\left(\frac{\delta(\cdot)_t}{1-\delta(\cdot)_t}\right) \\ 0 & 0 & 1 & 0 \\ \beta_c & 0 & 0 & \psi \end{bmatrix}, \qquad s_t = \begin{bmatrix} r_t \\ r_{0:t-1} \\ \bar{r}_{0:t-2} \\ 1 \end{bmatrix}, \qquad \tilde{H}_t = \begin{bmatrix} \sigma_d & 0 & 0 \\ 0 & \sigma_n & 0 \\ 0 & 0 & F_c\sqrt{\hat{h}_t} \end{bmatrix}, \tag{A.5}$$

$$\tilde{G}_t = \begin{bmatrix} 0 & 0 & 0 & \alpha + \beta r_{mt} \\ 1 & 1 & 0 & 0 \\ 0 & \left(1 - \lambda(\cdot)_{t-1}\right) & \lambda(\cdot)_{t-1} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \tilde{V}_t = \begin{bmatrix} F \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \tilde{Q}_t = \sqrt{\hat{h}_t}, \tag{A.6}$$

where d_t is set to missing values whenever $D_t = 0$ (i.e., all but 20-30 weeks), nav_t is set to missing values whenever the NAV report from GPs is not available (i.e., normally 12 weeks out of 13).

Let us first discuss the state vector s_t in equation (A.5). The first entry of the state vector is defined as $r_t = \alpha + \beta r_{mt} + \eta_t$, corresponding to the log of equation (5) in the paper. The last element of the state vector s_t is a constant, while the other two simply accumulate the realizations of r_t since the fund inception at $t = 0$ until $t-1$ or $t-2$, respectively denoted $r_{0:t-1}$ and $\bar{r}_{0:t-2}$, and then produce the exponentially moving average thereof so that they could be related to the observed fund distributions and NAV reports, respectively. Note that this introduces nonstationarity into the state space models, since the cumulative sums are trending series. This is a topic worth discussing. As highlighted by De Jong (1988) and Kohn and Ansley (1989), among others, SSMs can be applied to a nonstationary process provided that the initialization of the Kalman filter is handled properly. Yet, we do *not* face this problem, because the start values for $r_{0:t-1}$ and $\bar{r}_{0:t-2}$ are in both cases r_0, a stationary series with well defined unconditional mean and variance. This implies that the initialization of the Kalman filter is standard.

Next we discuss the vector of observables, denoted by y_t and appearing in equation (A.4). The only entry without missing values for any weeks is the comparable asset return, r_{ct}, which ideally should be a portfolio of the publicly traded assets matched to the fund holdings as of the beginning of week t. The weekly observed process r_{ct} relates directly to that key state variable, i.e. the fund latent return series, r_t, through the loading β_c with a heteroskedastic unobserved error, $\psi + \eta_t$. The other two elements of y_t, namely d_{t-1} and nav_{t-2}, reflect the up-to-period t history of fund cash flows and, therefore, cannot be mapped to the fund's since-inception returns directly. Instead, they are converted to returns via the mapping function m_t which is implicitly defined by equations (8) and (10) in the main text. These mapping functions are step-functions that increment whenever the fund makes distributions or capital calls (as follows from equation (7) in the paper), note that it is undefined for the last period since V_T is zero - in this case, m_{T-1} should be used):

$$m_t - m_{t-1} = log\Big((V_t + D_t - C_t)/V_t\Big). \tag{A.7}$$

However, since the true fund asset values are unobserved, the vector of regressors x_t contains the correspondent lags of the best feasible estimates of m_t that are filtered from an auxiliary SSM parametrized with the same θ as the PE fund SSM and described in the section A.2.1.

The other latent process featuring in both $\tilde{H}_t$ and $\tilde{Q}_t$ is the stochastic variance factor h_t, which scales distributions of idiosyncratic return shocks: η_t and η_{ct}. We obtain a proxy for it from a weekly frequency GARCH(1,1)-filter of comparable asset returns R_{ct} orthogonalized with respect to the risk-factor returns R_{mt}. We estimate the GARCH models separately and ignore the estimation uncertainty caused by the plug-in method we use for volatility in the model. One justification for this method is that we could use higher frquency data (daily or even intra-day) to mute the impact of estimation error. Both the sample size and sampling frequency would be larger and therefore small compared to T. For the same reason, we also take the slope of the projection of R_{ct} on R_{mt} as data. It appears as coefficient b in equation (A.4).

A.2.1 Returns-to-Value mapping function

This section describes the auxiliary model that filters the value of a mapping function for the fund cumulative return series obtained from the fund SSM (henceforth FSSM) described in the previous section. This is also a SSM but its latent state is the mapping function itself (rather than the fund returns). Accordingly, we refer to it as the mapping SSM (henceforth FSSM). As discussed in the main text (section 3.3), because the mapping value affects the observation vector in FSSM, we run an iteration loop between FSSM and MSSM until the changes in parameters in FSSM are small enough relative to the previous iteration. This ensures that the return nowcasts are consistent with the asset value nowcasts. This section uses the same notation as the main text and the previous section.

The identification of the mapping function is given by the main text's equations (10)–(11) and equation (A.7). Specifically, given an estimate of unsmoothed ($\hat{r}_{0:t}$) and smoothed ($\hat{\bar{r}}_{0:t}$) cumulative returns from FSSM, we can write the estimates of the log of the mapping function as:

$$m_t^n = \hat{\bar{r}}_{0:t} - nav_t \quad , \tag{A.8}$$

$$m_t^d = \hat{\delta}(\cdot)_t + \hat{r}_{0:t} - d_t \quad , \tag{A.9}$$

while the estimate of a log change in the mapping function is given by

$$\Delta m_t^* = \log\Big(\frac{\hat{V}_t + D_t - C_t}{\hat{V}_t}\Big), \tag{A.10}$$

where $\hat{\delta}(\cdot)_t$ reflects the distribution intensity parameter estimated in the FSSM, and $\hat{V}_t$ is the estimate of the fund asset values at weekly frequency.

Accordingly, the latent true mapping process m_t can be characterized by the following system:

$$m_t^n = m_t + \epsilon_{nt}, \quad \epsilon_{nt} \sim \mathcal{N}(0, \hat{\sigma}_n^2) \quad , \tag{A.11}$$

$$m_t^d = m_t + \epsilon_{dt}, \quad \epsilon_{dt} \sim \mathcal{N}(0, \hat{\sigma}_d^2) \quad , \tag{A.12}$$

$$m_t = m_{t-1} + \Delta m_t^* + \epsilon_{vt}^* \quad , \tag{A.13}$$

where $\hat{\sigma}_n$ are $\hat{\sigma}_d$ are noise variance parameter estimates from FSSM, and ϵ_{vt}^* is the innovation in the mapping function, which is effectively the unobserved error on the true change in the mapping function and the estimate from equation (A.10).

Denote the ratio between the true asset value V_t and its best feasible estimate as $e^{\epsilon_{vt}}$, such that $V_t = \hat{V}_t e^{-\epsilon_{vt}}$. It then follows that:

$$\begin{aligned} \epsilon_{vt}^* &= \log\Big(\frac{\hat{V}_t e^{-\epsilon_{vt}} + D_t - C_t}{\hat{V}_t e^{-\epsilon_{vt}}}\Big) - \log\Big(\frac{\hat{V}_t + D_t - C_t}{\hat{V}_t}\Big) \\ &= \epsilon_{vt} + \log(\hat{V}_t e^{-\epsilon_{vt}} + D_t - C_t) - \log(\hat{V}_t + D_t - C_t) \\ &\approx \epsilon_{vt} + \log\Big(1 + \frac{(D_t - C_t)\epsilon_{vt}}{\hat{V}_t + D_t - C_t}\Big) \quad . \end{aligned} \tag{A.14}$$

Accordingly, we set the standard deviation of ϵ_{vt}^* as $\sigma_v \cdot I_t$, where is I_t is computed as follows:

$$I_t = \begin{cases} \Big(1 + \frac{D_t - C_t}{\hat{V}_t + D_t - C_t}\Big) & \text{if net distributions during } t\text{: } D_t > C_t, \\ \Big(1 + \frac{C_t - D_t}{\hat{V}_t + D_t}\Big) & \text{if net capital calls } t\text{: } C_t > D_t, \\ 0 & \text{otherwise} \quad . \end{cases} \tag{A.15}$$

Note that σ_v is the only parameter to be estimated via MLE in the MSSM. This parameter can be interpreted as a standard deviation of the error for estimated asset values over the true ones for weeks in which net cash flows from the fund are close to 0. The key estimand of interest here is m_t

series. Using the generic form of an SSM in (A.1) we can write equations (A.11) through (A.13) as:

$$y_t = \begin{bmatrix} m_t^n & m_t^d \end{bmatrix}' \qquad \tilde{\Gamma}_t = I_{2\times 2} \qquad x_t = \begin{bmatrix} 0 & 0 \end{bmatrix}' \quad . \tag{A.16}$$

$$\tilde{Z}_t = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix} \qquad s_t = \begin{bmatrix} \Delta m_{t+1} \\ m_t \\ 1 \end{bmatrix} \qquad \tilde{H}_t = \begin{bmatrix} \hat{\sigma}_n & 0 \\ 0 & \hat{\sigma}_d \end{bmatrix} \quad . \tag{A.17}$$

$$\tilde{G}_t = \begin{bmatrix} 0 & 0 & \Delta m_{t+1}^* \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \tilde{V}_t = \begin{bmatrix} I_{t+1} \\ 0 \\ 0 \end{bmatrix} \qquad \tilde{Q}_t = \sigma_v \quad . \tag{A.18}$$

Note that even though m_t increments only when $D_t \neq C_t$ and the sign of Δm_t^* is positive [negative] whenever $D_t > [<] C_t$, it is possible that the change in m_t estimate is positive [negative] even when $D_t < [>] C_t$ still (i.e. if ϵ_{vt}^* offsets Δm_t^*). Albeit rarely, this situation occurs when the filtered the noise on observed NAVs and distribution function are too large and the divergence of the fund returns relative to comparable benchmark. Accordingly in these situations, the estimation procedure falls back to the previous mapping.

A.2.2 NAV smoothing and noise

Equations (1) and (10) in the paper assume that the NAV appraisal process is governed by a parametric function $\lambda(\cdot)_t$ and a mean-zero noise ϵ_{nt} parametrized by time-invariant σ_n. It is important to note that smoothing pertains to both the systematic and idiosyncratic components of fund returns and accounts for the path of returns *between* the quarterly report dates as t indexes weeks. Following equation (2) in the paper, $\lambda \in (0.01, 0.99)$ is the structural parameter to be estimated while $w_t \in [0, 1]$ is an observed value reflecting the cash flow's fraction of the fund assets in period t. Below we provide some examples.

Consider a period when distribution D_t fully resolves the fund so that residual NAV is known to be zero. For this case, $\tilde{w}_t = D_t/(N\hat{A}V_t + D_t) = 1$. Consequently, from equations (1) and (2) in the paper, it follows that the "smoothed" return to-date equals the true one regardless of λ:

$$\begin{aligned} \bar{r}_{0:t} &= \Big(1 - \lambda(1 - \tilde{w}_t)\Big) r_{0:t} + \lambda(1 - \tilde{w}_t)\bar{r}_{0:t-1} \\ &= \Big(1 - \lambda \cdot 0\Big) r_{0:t} + \lambda \cdot 0 \cdot \bar{r}_{0:t-1} \end{aligned}$$

Suppose that $w_t = 0.5$ instead, which would be the case if C_t doubles the fund capital invested or D_t resolves the fund by half (i.e., $D_t = N\hat{A}V_t$). If the fund GPs put only half of the weight on past performance (i.e. $\lambda = 0.5$), then the smoothed return to-date would equal:

$$\begin{aligned} \bar{r}_{0:t} &= \Big(1 - 0.5(1 - 0.5)\Big) r_{0:t} + 0.5(1 - 0.5)\bar{r}_{0:t-1} \\ &= 0.75 \cdot r_{0:t} + 0.25 \cdot \bar{r}_{0:t-1} \end{aligned}$$

as opposed to $0.5 \cdot r_{0:t} + 0.5 \cdot \bar{r}_{0:t-1}$ for a period without any cash flows and $0.65 \cdot r_{0:t} + 0.35 \cdot \bar{r}_{0:t-1}$ had the GPs been smoothing at the $\lambda = 0.9$ rate (given $w_t = 0.5$). Given $\lambda = 0.9$ and conditional on absent cash flows during the quarter (i.e., 13 weeks), the quarter-ago valuations would be contributing 25.4% ($= 0.9^{13}$) to the current fund NAV reports.

Equation (1) in the paper therefore models the fund asset appraisal process to reflect both the observed return paths (for r_{mt} and r_{ct}) and the unobserved idiosyncratic fund returns, η_t, so as to be most consistent with (i) the realized cash flow pattern and (ii) the history of NAV reports. All else equal, funds that made substantial distributions recently would be assumed to have less "stale" NAVs. Meanwhile, the weight of returns occurring before period τ, $0 < \tau < t$ in NAV_t is limited by the fraction of fund capital invested before period τ. While under equation (1) the influence of the past valuations decays at an exponential rate, it is also straightforward to incorporate "longer

memory events" in our framework by temporarily increasing $\lambda(\cdot)_t$ after specific events such as an M&A transaction or IPO of a comparable firm.

The Gaussian NAV residual noise parametrized by σ_n is orthogonal to the observed returns and the distribution process. While the temporal independence assumption appears implausible for actual PE funds, we argue that the sparseness in the NAV observation series (i.e., once every 13 weeks at best) and the irregularity of fund cash flows (and, hence, w_t values) mitigate the consequences of possibly non-zero temporal correlation structure in the residuals for the identification of the fund model parameters. The entry corresponding to the NAV observations in y_t is simply missing for most observations and almost never overlaps with a non-missing value for d_t. To better illustrate this, we interpolate ϵ_{nt} between the values drawn for the adjacent quarterly NAV reports to obtain the within quarter estimate of $N\hat{A}V_t$ in a simulation exercise reported in Section A.6. In addition, we investigate the effect of $\lambda(\cdot)_t$ being misspecified relative to the data generating process.[1] One can also consider making σ_n time-varying to reflect the stage of current fund holdings and differences in GP reporting incentives (e.g., see Brown et al., 2019). The latter can be modeled as a conditional mean of the distribution of ϵ_{nt}.

A.2.3 Cash flow process

According to equations (11) and (13) in the paper, the fund distribution process is governed by the parametric function $\delta(\cdot)_t$ and a mean-zero noise ϵ_{dt} parametrized by time-invariant σ_d. By construction, they are capturing the variation in the fund distributions that cannot be explained by the asset return path, on the one hand, and the periodically observed NAV series, on the other hand. Also note that fund capital calls do not relate to either $\delta(\cdot)_t$ or ϵ_{dt} but only affect the asset-to-value mapping function, m_t.

While equation (13) in the paper certainly misspecifies $\delta(\cdot)_t$ relatively to the true conditional expectation (and, therefore inducing potentially heavy temporal dependencies in the residuals), we argue that irregularity in distribution occurrence mitigates the violations of the SSM identifying assumptions discussed in the beginning of this section. As with the NAV smoothing process, we scrutinize this conjecture in our simulations discussed below.

Another insightful extension to the baseline setup in the paper would be utilization of prices (or quotes) from the secondary market transactions with the fund stakes (e.g., see Nadauld et al., 2019). Like the fund distributions (in fact, they are equivalent from the LPs perspective), these prices would map to the fund unsmoothed returns, but one-to-one:

$$log(\text{SecondaryPrice}_t) + m_t = r_{0:t} + \epsilon_{pt} \quad , \tag{A.19}$$

whereby ϵ_{pt}, the price distortion effects, can be either assumed to be zero, or modeled accordingly to incorporate time-varying demand and supply variation in these still nascent markets (e.g., see Albuquerque et al., 2018). Similarly, one can incorporate venture deals round valuation data if the fraction of reported fund-level NAV that the deal corresponds to is known.[2]

A.3 Naïve Nowcasts: construction and properties

Denote the period t cumulative value of distributions and capital calls by $\mathcal{D}_{0:t}$ and $\mathcal{C}_{0:t}$, so that $\mathcal{D}_{0:t} = \mathcal{D}_{0:t-2}R_{c,t-1}R_{ct} + D_{t-1}R_{ct} + D_t$, where R_{ct} is the comparable asset gross return. Also, let $\mathfrak{P}_{0:t}$ denote the Public Market Equivalent of the fund against the R_{ct} series computed in period t using the asset value estimate, N_t, for that period. Hence,

$$\mathfrak{P}_{0:t} = \frac{N_t + \mathcal{D}_{0:t}}{\mathcal{C}_{0:t}} = \frac{N_t + \mathcal{D}_{0:t-1}R_{ct} + D_t}{\mathcal{C}_{0:t-1}R_{ct} + C_t} \quad . \tag{A.20}$$

[1] If the econometrician also observes dates and amounts with regards to the individual investments in the fund portfolio, $\lambda(\cdot)_t$ can be further refined to reflect the return histories during the remaining investments only. Therefore, the 'first-in-first-out' assumption is no longer needed.

[2] The contractual terms would also need to be adjusted accordingly (see Gornall and Strebulaev, 2020).

This implies that, given an estimate of $\mathfrak{P}_{0:t}$ for any date t:

$$\tilde{V_t}^i = \mathfrak{P}^i_{0:t}\mathcal{C}_{0:t} - \mathcal{D}_{0:t} \quad , \tag{A.21}$$

which amounts the naïve method of nowcasting NAVs. In this equation, $\mathfrak{P}^i_{0:t}$ is a log linearly interpolated $\mathfrak{P}_{0:t}$ between (i) the adjacent end-of-quarter values computed using equation (A.20) using as-reported NAV values, (ii) weeks with capital calls, and (iii) weeks, in which D_t was large enough to imply negative residual asset values (if interpolated between (i) and (ii) points only).

For weeks (ii), the PME value is estimated as $(\mathfrak{P}^i_{0:t}\mathcal{C}_{0:t-1} + C_t)/\mathcal{C}_{0:t}$. Not doing this adjustment would amount to an assumption that the $(\mathfrak{P}^i_{0:t} - 1)$ excess return is instantaneously realized on the capital just called. For weeks (iii), the PME value is estimated as $(\mathcal{D}_{0:t-1} + 1.1D_t)/\mathcal{C}_{0:t}$—i.e. the residual NAV is assumed to 10% of the latest distribution if subsequently reported NAVs exceed 1% of the fund size. Otherwise, we assume the fund terminates on the week of the distribution. While producing plausible estimates of NAVs (and the corresponding returns) for 99% of fund-weeks, the above-described procedure uses the hindsight of what the end-of-quarter PME values will be based on then-reported NAVs. We therefore use only the end-of-quarter values for comparisons with SSM-based nowcasts.

Next, we analyze the properties of the weekly return series implied by this naïve method. Denote the growth in PME, $\mathfrak{P}^i_{0:t}/\mathfrak{P}^i_{0:t-1} - 1$, with Δ_t and the change in NAV estimate from $t-1$ to t with $\Delta.N_t$. It then follows that:

$$\Delta.\tilde{V_t}^i = \underbrace{(1+\Delta_{t+1})\mathfrak{P}_{0:t}(\mathcal{C}_{0:t}R_{c,t+1} + C_{t+1}) - (\mathcal{D}_{0:t}R_{c,t+1} + D_t)}_{N_{t+1}} - \underbrace{\mathfrak{P}_{0:t}(\mathcal{C}_{0:t} + \mathcal{D}_{0:t})}_{N_t} \quad . \tag{A.22}$$

Without loss of generality, consider a week such that $C_{t+1} = D_{t+1}$. Rearranging (A.21) we obtain the following expression for the fund excess return ($\equiv eR^c$) relatively to the comparable asset R_c:

$$\begin{aligned}\frac{\Delta.N_{t+1}}{N_t} + 1 - R_{c,t+1} &= \frac{\mathfrak{P}_{0:t}\mathcal{C}_{0:t}\Delta_{t+1}R_{c,t+1} + (\mathfrak{P}_{0:t}\mathcal{C}_{0:t} - \mathcal{D}_{0:t})(R_{c,t+1} - 1)}{\mathcal{C}_{0:t} + \mathcal{D}_{0:t}} - R_{c,t+1} - 1 \\ &= \frac{\Delta_{t+1}R_{c,t+1}}{N_t/\mathfrak{P}_{0:t}\mathcal{C}_{0:t}} \quad ,\end{aligned} \tag{A.23}$$

as from equation (A.20) it follows that $(1 - \mathcal{D}_{0:t}/\mathfrak{P}_{0:t}\mathcal{C}_{0:t}) = N_t/\mathfrak{P}_{0:t}\mathcal{C}_{0:t}$, which can be though of as a *scaling* factor to the Δ_t to undo the modulation in the growth of PME as the fund gets resolved (see Brown et al., 2019). Therefore, the expected excess return over the next week is equal to:

$$E_t\left[eR^c_{t+1}\right] = E_t\left[\Delta^S_{t+1}\right]E_t\left[R_{c,t+1}\right] + \mathrm{cov}_t(\Delta^S_{t+1}, R_{c,t+1}) \quad , \tag{A.24}$$

where $\Delta^S_t \equiv \Delta_t/(N_{t-1}/\mathfrak{P}_{0:t-1}\mathcal{C}_{0:t-1})$, and the covariance term is negative if the reported NAVs do smooth the systematic risk component (present in $R_{c,t+1}$ as well) at high frequency (see section A.5).

As an result of the interpolation, Δ^S_t is highly persistent and rarely changes sign within a quarter. However, the nowcasted returns of a fund may not exhibit significantly positive autocorrelation if $\mathrm{var}(eR^c_t)$ is small in front of $\mathrm{var}(R^c_t)$, or if R^c_t happen to be negatively enough autocorrelated during the fund's life. The variance of excess returns increases in the magnitude of inter-quarter changes in PME against the R^c_t series and decreases in the frequency of the sign change thereof. The latter occurs when short-term smoothing dominates—the covariance term in equation (A.24) is large enough to offsets Δ^S_t.

Note that Δ^S_t is zero (and, hence, so is eR^c_t) whenever the adjacent PME estimates equal one. This would be the case if R^c_t are replaced with the true fund return series, for example. Note also that, insofar as much of the variation in Δ^S_t are fund-specific while R^c_t are highly diversified (and therefore tightly correlated) indexes, pooling the individual funds' naive return nowcasts will tend to decrease the variance of excess returns relatively to that of the average R^c_t. This fact, however,

does not make the index return series more informative about the systematic risk exposure because the negative correlation of the excess return with the systematic risk factor strengthens at the index level (as the idiosyncratic part of Δ_t^S is diversified away).

Finally, we note that the bias-governing equations (9–10) of our SSM specification predict Δ_t^S to be negatively correlated with the market, so long as R_{ct} has a positive beta with the market and smoothing is strong enough. Assuming for simplicity that smoothing is constant, i.e., $\lambda(\cdot)_{t+1} = \lambda(\cdot)_t = \lambda$ and $C_t = D_t$, the change in the log of naive NAV nowcasts can be written as follows:

$$\Delta.log\,\tilde{V}_t^i = (1-\lambda)r_{t+1} + \lambda \cdot \Delta.\bar{r}_{0:t-1} + \Delta.\epsilon_{n,t+1}\} \quad , \tag{A.25}$$

where $\Delta.\epsilon_{n,t+1}$ is the change in appraisal bias unrelated to the exponential smoothing of the since inception return history. Given the mapping between r_t and r_{ct} from equation (6), it can be shown that eR_t^c should load negatively on the market if $\lambda > 1 - \beta_c$.

A.4 Estimation details

α-range center for profiling. For center of the α-range, α_{p0}, we compute IRRs from the discounted cash flows as do Gredil, Griffiths, and Stucke (2014):

$$\alpha_{p0} = \arg\min_{\alpha} \sum_{t=0}^{T-1} \left((C_t - D_t)e^{\alpha\cdot\tau + \sum_{\tau=t+1}^{T-1} f(\beta_{p0})_\tau}\right) + D_T - C_T + NAV_T \quad , \tag{A.26}$$

where T is the end of the parameter estimation period, β_{p0} is the fund-type specific β-estimate from Ang et al. (2018), while

$$f(\beta)_t = \beta \cdot r_{mt} - \frac{1}{2}\beta(1-\beta)\sigma_m^2 \quad , \tag{A.27}$$

and σ_m^2 is the market return variance estimate at weekly frequency from 0 through T.

We approximate fund duration with a weighted average life when IRR solutions are ambiguous. In work contemporaneous to ours, Korteweg and Nagel (2020) show that using average beta estimates improves inference about fund-level abnormal performance relative to the SDF-based interpretation of PME (Korteweg and Nagel, 2016) even if fund βs are heterogeneous. Therefore, to obtain the center of the α-grid, we use the center of the β-grid.

Penalty function. For each combination of α and β on the profile grid (i.e., grid cell ij), we obtain the maximum likelihood estimates of ϑ_{ij}^{pout} – the remaining 8 parameters in θ, and standardize the log-likelihood scores by subtracting the grid-wide minimum and dividing by the grid-wide standard deviation to obtain $sLL(\alpha_i, \beta_j) \equiv sLL_{ij}$. From each sLL_{ij}, we then subtract a penalty function $sPMEerr_{ij}$ which is the grid-wide standardized $PMEerr_{ij}$ defined as:

$$PMEerr_{ij} = \left| \frac{\sum_{t=0}^{T-1} D_t e^{\alpha_i\cdot\tau + \sum_{\tau=t+1}^{T-1} f(\beta_j)_\tau} + D_T + NAV_T e^{bias(\alpha_i,\beta_j,\vartheta_{ij}^{pout})_T}}{\sum_{t=0}^{T-1} C_t e^{\alpha_i\cdot\tau + \sum_{\tau=t+1}^{T-1} f(\beta_j)_\tau} + C_T} - 1 \right| \tag{A.28}$$

to obtain pLL_{ij}. Note that, unlike in equation (A.26), here reported NAV as of T is adjusted by the exponent of $bias(\alpha_i, \beta_j, \vartheta_{ij}^{pout})_T$, which is defined in equation (A.29) below.

Picking the supremum value of pLL_{ij} on the grid identifies the corresponding (α_p, β_p)-pair which we use as fixed data in the second step in whereby we estimate remaining parameters, $\vartheta \in \theta$, as well as the mapping-to-value functions and fund return nowcasts.

Mapping-to-Parameter iterations. Because our latent state is fund returns and future fund distributions inherently relate to today's returns, we argue that the *backward recursion* method is the most natural choice in our context (i.e. nowcasting). We therefore drop the superscript *smooth* and instead superscript it the per period and cumulative returns estimates with the set of SSM

parameters that they were derived from, e.g. r_t^θ, $r_{0:t}^\theta$, $\bar{r}_{0:t}^\theta$. It follows from equations (8) and (10) in the paper that the expected appraisal bias and the the unbiased net asset values (i.e., V_t) for any week t are:

$$bias(\theta)_t := r_{0:t}^\theta - \bar{r}_{0:t}^\theta \tag{A.29}$$

$$V(\hat{m}_t;\theta)_t = \exp\{r_{0:t}^\theta - \hat{m}_t\} \tag{A.30}$$

However, the mapping $\hat{m}_t$ is also a function of parameters θ (see Section A.2.1). This makes equation (A.30) a fixed point, since $\hat{m}_t$ enter the observation vector x_t via equation (A.4), therefore, affecting the parameter values under which the fund SSM log-likelihood is maximized. Therefore, starting with the $\hat{m}_t$ implied by the naïve nowcasts, and the corresponding ϑ estimates, we set a loop over i:

$$\begin{aligned} m(i)_{t=1,\dots,T} &= m(\vartheta_{i-1}^{argMax}) \\ \vartheta_i^{argMax} &= \arg\max_{\vartheta} LL_p(\vartheta; m(i)_{t=1,\dots,T}) \end{aligned} \tag{A.31}$$

in which $m(i)_t$ and ϑ_i^{argMax} are re-estimated sequentially until the average change in ϑ_i^{argMax} from ϑ_{i-1}^{argMax} is smaller than 5% on average across all ϑ elements, whereby LL_p denotes the SSM likelihood function conditional on $\theta \notin \vartheta$ being fixed at values determined in the first step.

A.5 Why unsmoothing at high frequency is essential

Our empirical results in Section 4 as well as simulations in Section A.6.2 show that the choice of SSM parameters, and β in particular, affect nowcasting performance. The standard approach to unsmoothing reported returns is to utilize an ARMA model (Geltner, 1993; Getmansky et al., 2004) or regress the reported returns on the distributed lag of market returns (Ewens et al., 2013) at the reported NAV frequency (i.e., quarterly for standard PE funds). This section shows that if return smoothing occurs at a higher frequency—i.e., appraisers weigh on transactions' and public asset prices *within* the past quarter—the standard approach will not sufficiently unsmooth reported returns and likely underestimate the systematic risk exposure of PE funds. The Accounting and Valuation Guide (see Section A.7) instructs appraisers to weigh more on relevant valuations observed closer to (but before) the "measurement" date.

To illustrate this issue, we simulate a fund and market (r_{mt}) returns at daily frequency for 10 years (2,520 business days): $r_t = 2.0 \cdot r_{mt} + e_t$, where e_t and r_{mt} are drawn independently from a normal distribution with an annualized standard deviation of 0.20. We then cumulate the fund returns and smooth the resulting series using equation (9) of the main text but at a *daily* frequency and assuming a constant λ of 0.95. Both series are plotted in Panel A of Figure A.1 and the extent of smoothing appears to be inconsequential. We then compute quarterly returns from both smoothed and unsmoothed series and estimate a quarterly AR(1) model on the smoothed ones, which yields an autoregressive coefficient close to zero (0.03). This is not very surprising, since the smoothing scheme implies a near-zero $[0.002 = (1 - 0.95) \cdot 0.95^{63}]$ weight of returns that occurred more than 63 days ago in the smoothed cumulative returns (which are just log-NAVs if there were no interim cash flows during the fund's life). Accordingly, the residuals from the AR(1) are not much closer to the true quarterly returns than the smoothed returns themselves as shown on Panel B of Figure A.1. The variances of the smoothed and the residual series are, respectively, 26% and 24% downward-biased (relative to the unsmoothed) and so are the β estimates obtained via OLS on the quarterly market returns (at 1.46 and 1.43, respectively). This shows that the intra-quarter smoothing cannot be accounted for without modeling returns at a higher frequency.

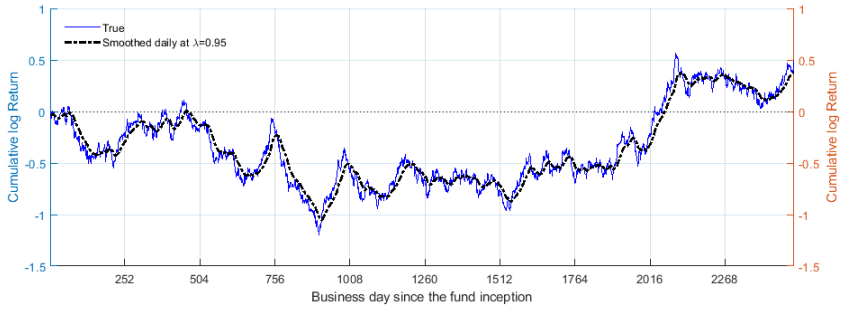
We have attempted estimating the fund SSM at a daily frequency but decided on the weekly frequency instead for two main reasons. First, given the parsimonious exponential-moving-average specification we utilize, the daily frequency is likely not to capture very well "longer-term memories" in NAVs. Panel C of Figure A.1 shows that a λ of 0.85 or less assumes virtually zero weight to valuations before the last month of the quarter. Whereas the fund panel regressions of quarterly

returns suggest that the weights are non-zero for as far back as as 2-3 quarters (Ewens et al., 2013). Second, for a typical 10-year fund life, a model estimated at a daily frequency will not only encounter a large computational burden but will have only about 60 (i.e., 40 NAVs plus 20 distributions) out of 2520 entries in the data vector will feature any fund data. We therefore view the weekly frequency as a reasonable compromise: (i) about a quarter of data observations are from the fund's data (rather than the comparable asset), (ii) there are no "blind spots" with regards to short and medium-term memories, and (iii) for private equity, weekly is still "high frequency". Note that the typical λ-parameter of 0.92–0.96 at the weekly frequency implies a weight of 1–1.5% for valuations older than 2 quarters.

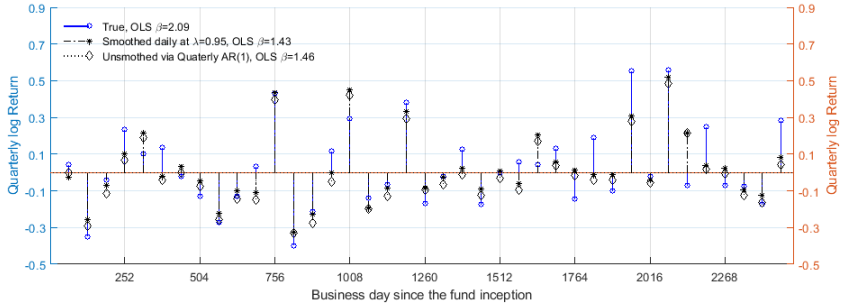
Figure A.1. Unsmoothing frequency matters

This figure reports data for a hypothetical fund returns, which has a market β of 2.0 that are smoothed on a cumulative basis at a daily frequency with an exponential weight of 0.95. Panel A plots cumulative return series sampled at daily frequency. Panel B plots end-of-quarter smoothed series value. Assuming the fund makes no interim cash flows, the cumulative returns map 1-to-1 to NAVs. Panel C plots returns sampled at a quarterly frequency at the end of the respective quarter based on the series from Panel A as well as those obtained by desmoothing the quarterly series via a AR(1) time series model in spirit of Getmansky et al. (2004). The legend of the panel figure also indicates the corresponding OLS estimate of β to the market return series at quarterly frequency. See Section A.5 for discussion.

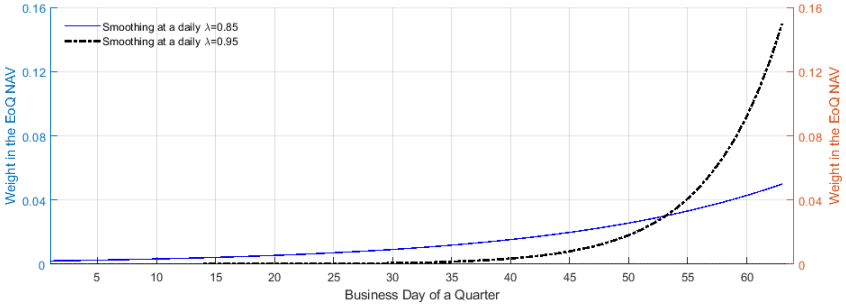
Panel A: Cumulative return sampled daily



Panel B: Quarterly returns sampled quarterly



Panel C: Memory length of smoothed NAVs



A.6 Monte Carlo experiment

In this section, we demonstrate our estimation methodology and its nowcasting performance using simulated data at weekly frequency. We begin with a single-fund case study and then present evidence aggregated over a random sample of such funds.

A.6.1 Single-fund case study

This particular fund operated for 563 weeks. It made 13 capital calls over the first 193 weeks for the total of \$4 (including \$1 at inception, i.e. week 0) and 20 distributions of various amounts for a total of \$3.78 at irregular intervals, starting on 109th week. This case is the same fund discussed in the main text per Figure 1.[3] While broadly representative, it is an interesting case because neither market nor fund exhibited particularly strong trend in returns on average during these 563 weeks, which eliminates the source of "easy gains" for the SSM method relative to the naïve method described in Section A.3 and allows focusing on more subtle properties of the SSM method.

Panel A of Figure A.2 plots these data over fund life-weeks, along with the "True Asset Values" (i.e. V_t in equation (7))—unobservable with real fund data—as well as the quarterly NAV reports that reflect appraisal smoothing as in equation (2) at λ of 0.90 and the reporting noise volatility, σ_n, of 5.0%. We can see that this process generates both upward and downward biases in reported NAVs (relatively to V_t), depending on the recent path in fund returns.

Panel B rescales the variables by the fund asset value so that 'D/(D+V)' correspond to $exp\{\delta(\cdot)_t + \epsilon_{dt}\}$ in equation (11) while 'N/V-1' is the overall reporting distortion, $exp\{\bar{r}_{0:t} - r_{0:t} + \epsilon_{nt}\} - 1$ in equations (8–10) in the paper. The panel also plots similar statistics for weekly nowcasts obtained via the naïve method, as described in Section A.3.

We used the first 455 weeks as the parameter estimation period. This corresponds to the end of the quarter during which the fund made the 80th percent of its total distribution. Figure A.3 depicts the SSM likelihood scores and the pricing error via 3-dimensional mesh plots on Panel A. Both are normalized by subtracting the grid-wide mimimum and dividing by standard deviation. It follows from Panel A that the SSM likelihood scores do peak with respect to β around the value of 1.3-1.4 but hardly exhibit any curvature along the α-axis. In contrast, the flat region in the surface of pricing errors computed per Equation A.28 is off-diagonal. This is typical with an upward-trending market return high cumulative returns can be "equally well explained" by either high slope or large intercept. In this particular case the trend is quite modest, about 3% per year during the estimation period. Generally, it is highly unlikely that the "valley" on SSM likelihood to align perfectly with that for the pricing error. Therefore subtracting these pricing errors from the SSM likelihood scores—i.e., "penalizing the SSM likelihood"—tends to result in a sharp peak across both dimensions, as plotted on Panel B. In our case, the peak corresponds to the (α, β) pair of (0.06%,1.297)—close to the true values of (0.09%,1.19).

As discussed in the main text, sections A.2 and A.4 of this Appendix, having identified the pair of α and β, we iteratively estimate the asset-to-value mapping and the remaining parameters. For this case study fund, this mapping-to-parameter convergence took 2 iterations and resulted in the following $(\alpha, \beta, \vartheta)$-vector, whereby standard errors were obtained using numerical differentiation (see, e.g., Miranda and Fackler, 2004):

	α	β	δ	σ_d	λ	σ_n	F	F_c	ψ	β_c
Point Estimate	.0006	1.295	.0347	0.114	0.819	0.037	1.186	0.970	.0009	0.252
Standard Error	.0002	0.599	.0012	0.037	0.153	0.051	0.385	0.072	.0017	0.182

Figure A.4 examines the quality of the case fund return and NAV estimates by contrasting the SSM-based output to the naïve method. Panel A compares the cumulative return errors and the improvement in the value-to-return mapping following the SSM-based refinement. Panel B compares quarterly RMSE for asset values estimates and the to-date PMEs to compare periodic return estimates. The small ticks on the x-axis for both panels indicate the weeks with positive

[3] The exhibits discussed here can also be reproduced by the Matlab code—https://github.com/orgredil/Nowcasting-PE-NAVs/tree/main—that also reproduces Figure 1.

distributions from the fund. These are the dates when the value-to-return mapping steps up in Panel A. The panel shows that, even though naïve nowcasts deviate notably from the true asset values (per Figure A.2), they still provide a reasonable guess of the value-to-return mapping (dotted black line) since both cumulative returns and NAVs are smoothed to same extent. Nonetheless the above-described iterative refinement procedure moves the mapping (dash-dot black line) closer to the true one, which is plotted as a solid black line.[4] The panel also shows that the cumulative return error tends be closer to zero for the SSM method than for the naïve method, with difference growing over time and when fund returns vary more. This corresponds to the period between weeks 500 and 562 during which the cumulative market and fund returns were -26% and -52% respectively. Consequently, a large gap between the naïve and SSM methods emerged.

Panel B of the figure shows that SSM-based weekly NAV estimates is closer to the true asset values than the naïve estimates on average during a given quarter in all but 7 quarters (i.e. whereby the red diamonds are closer to zero than the red stars). These quarters tend to be early in the fund's life, before any distributions are made, and are characterized by unusually low RMSE for the naïve method. Additionally, the panel plots the to-date log PME estimates that features the true asset values but uses return estimates to grow the preceding cash flows to the NAV date. If the return estimates were error-free, we would expect the back dash-dot line to be a flat at zero. While evidently this is not the case (the fund exhibits considerable idiosyncratic risk relative to the comparable asset), the level and the variance of the dotted series that feed the raw comparable asset series are notably larger. This suggests that the weekly return series from SSM are superior estimates of the true fund returns than the comparable asset returns even though they represent the backbone of the SSM-based estimates.

We emphasize that the metrics examined in Figure A.4 are infeasible with the real-life funds whereby true asset values are not observable. This underscores the need for the alternative nowcasting metrics introduced in the paper. Yet, the intuition behind those metrics is closely related. For example, to compute the feasible Out-Of-Sample error (OOS error), we take as a denominator the NAV estimate for week 462 (7th week after the parameter estimation window end) obtained using the observations through that week only. We then discount to week 462 all cash flows that occur after week 462 with the SSM-based return estimate for the respective week obtained using the observations for all weeks. The sum of the discounted cash flows is the numerator. The ratio can be thought of as PME that uses week 462 NAV estimate as the only capital call. In this Case it is equal to 1.026, as opposed to 1 if we used true asset value for week 462 and discounted cash flows with true returns, and 1.251 if we used naïve nowcast for week 462 and discounted the cash flows with the comparable asset. Thus, subtracting one produces an economically interpretable error that jointly reflects (i) how far periodic return estimates are from the true returns, (ii) how far the NAV estimate is from the true values. In this Case, we reduction in the OOS error happen to have come solely from (i). As evident from Panel B Figure A.4, SSM NAV estimates during weeks 456–468 happen to be inferior to the naïve nowcasts.

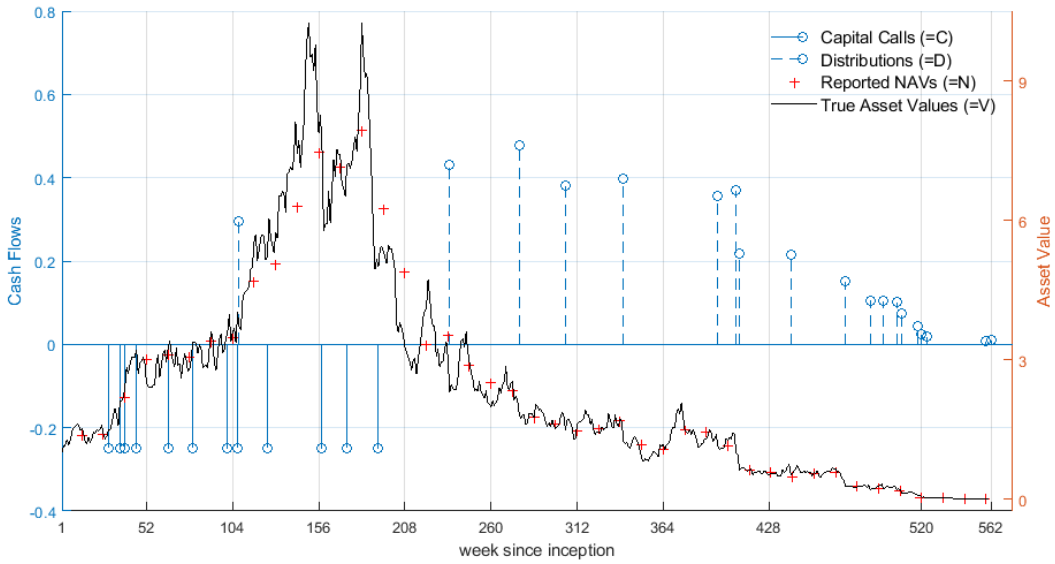
Note also that the magnitude of the OSS error depends on the level of NAVs as of the end of the parameter estimation window and the variance of subsequent returns. This underscores the importance of a simple and consistent rule for selecting the estimation and out-of-sample period when the goal is to evaluate the model performance in a large sample of actual funds. In practice, the choice of the periods can be refined on case-by-case basis to reflect the fund-specific life length and the cash flow pattern.

[4] A complete elimination of the mapping error would require an implausibly frequent distributions and/or comparable asset return perfectly mimicking the actual fund returns. We note that fixing the mapping at the true values does not necessarily improve nowcasting performance, since there remains an inconsistency with the fund return series' estimate.

Figure A.2. Case study: Data for a hypothetical fund

This figure reports data for a hypothetical PE fund to illustrate our method applications. See Section A.6.1 for description.

Panel A: Cash flows and reported NAVs normalized by first capital call



Panel B: Cash flows and reported NAVs normalized by Asset Values

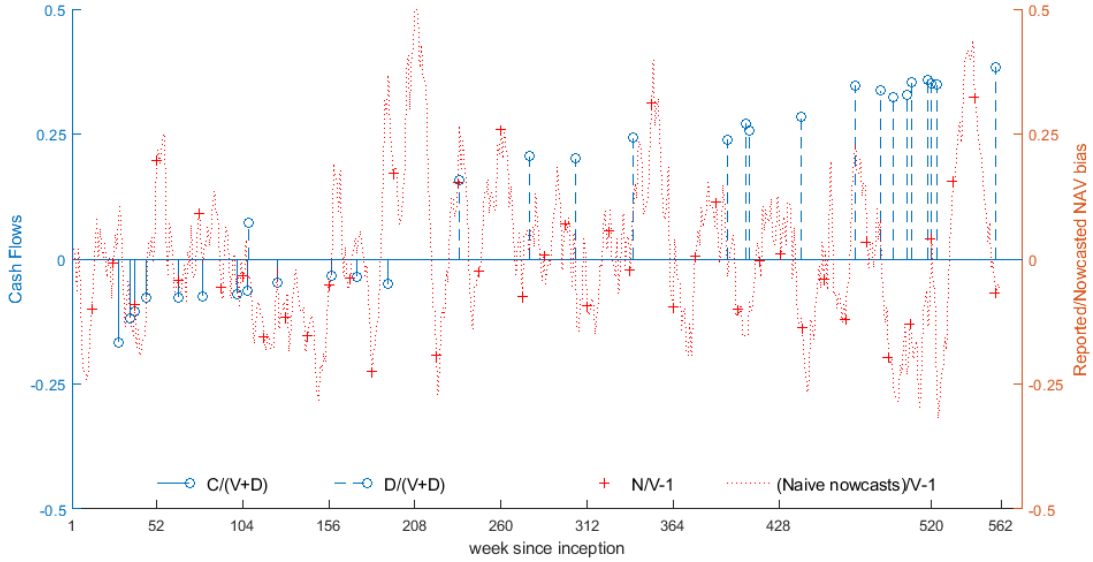
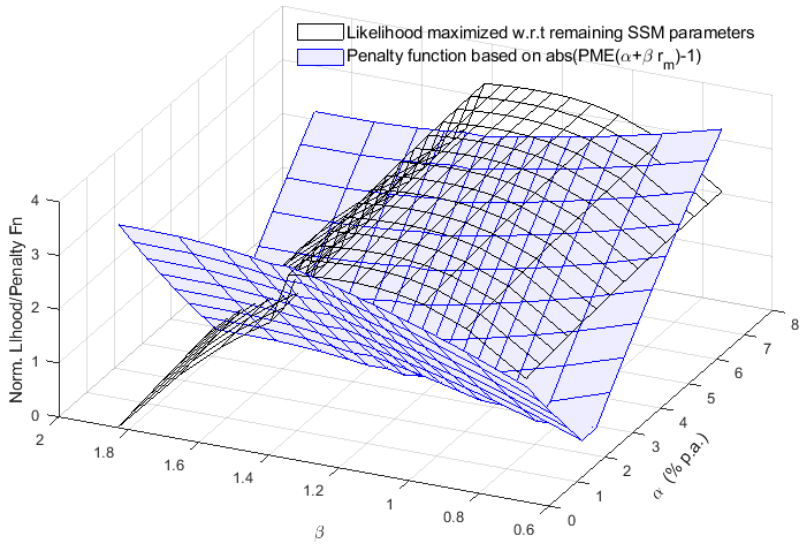


Figure A.3. Case study: Profiled Likelihood

This figure illustrates selected parameters' identification for a hypothetical PE fund to illustrate our method applications. See Section A.6.1 for description.

Panel A: SSM likelihood and the Penalty function



Panel B: Penalized likelihood

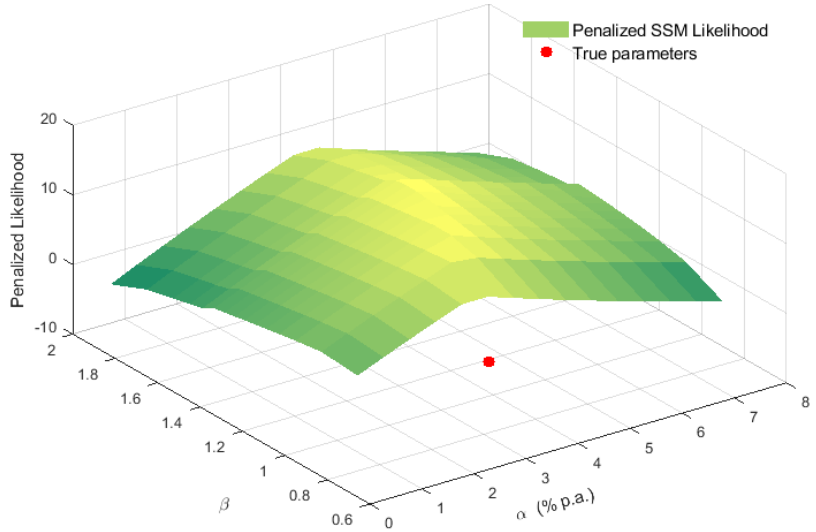
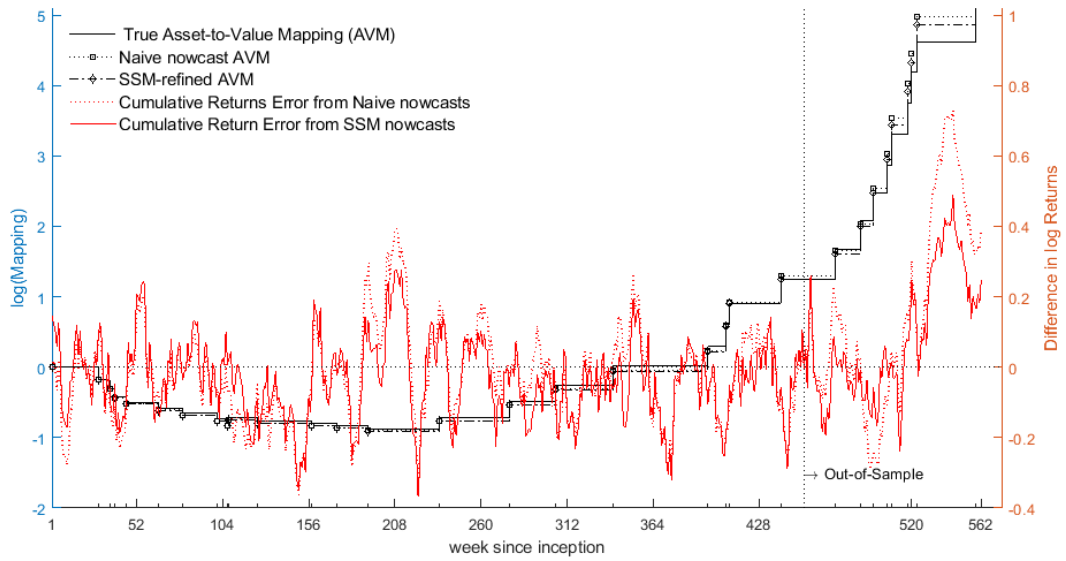


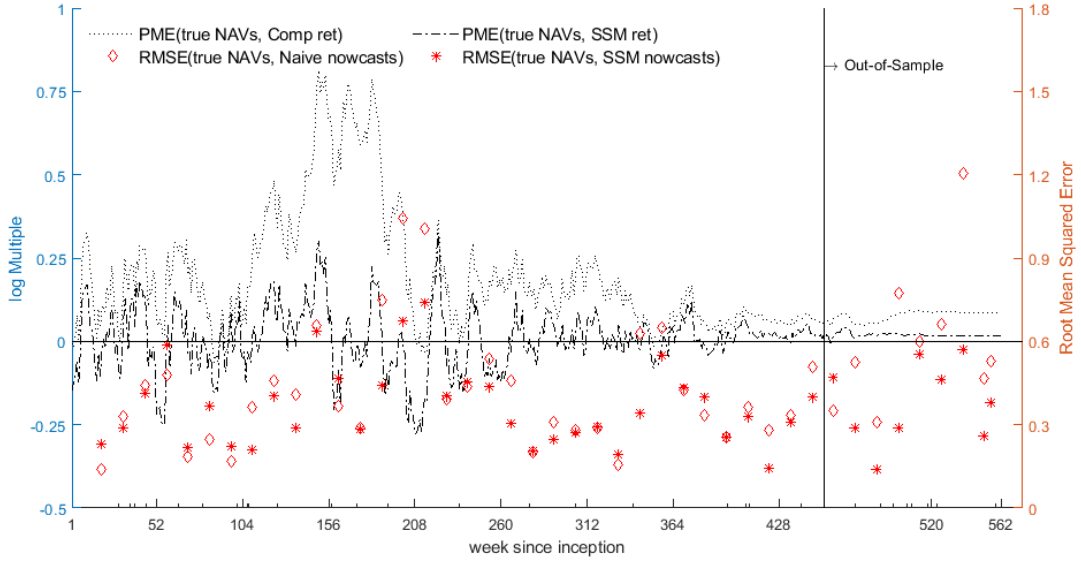
Figure A.4. Case study: Nowcasting results

This figure demonstrates nowcasts obtained from a SSM for a hypothetical PE fund to illustrate our method applications. See Section A.6.1 for description.

Panel A: Return estimates and Mapping to Asset Values



Panel B: Asset Value estimates and PMEs



A.6.2 Simulated panel of funds

Setup. We simulate a panel of 25 fund vintage years with 40 funds per vintage, each operating for at most 676 weeks, $\beta \sim \mathcal{N}(1.250, 0.050^2)$, weekly $\alpha \sim \mathcal{N}(0.0, 0.00220^2)$, and weekly return series with the unconditional volatility of 42% per annum (p.a.) constructed as follows.

First, we simulate market returns R_{mt} from a normal distribution with a mean of 6% p.a. and a stochastic volatility process resulting in the unconditional volatility of 20% p.a.[5] We then construct 5 industries with returns R_{ct}, as the sums of the market return scaled by β_m^c, equal to average draw of β across 200 funds (=8*25) in the respective industry *minus* 0.15, and the industry idiosyncratic shock e_t governed by a persistent volatility process h_t similar to (but independent from) that for R_{mt} and resulting in the unconditional volatility of the idiosyncratic return of 9% p.a. We then construct funds' returns as

$$R_t = 1 + \alpha + \beta(R_{mt} - 1) + 1.5e_t + \eta_t - \bar{\xi} + \xi_t \tag{A.32}$$

, where η_t is independent from e_t but drawn from the normal distribution with the variance proportional to the same h_t (as with e_t) such that $\text{var}(1.5e_t + \eta_t)/\text{var}(e_t) = 12.25$[6]; while ξ_t is a 10% jump in the fund asset value drawn from a Poisson distribution occurring with a 2% probability on a given week, and $\bar{\xi}$ is the average jump realized during the fund's life $\tau = 0, 1, ..., T$.

We construct fund cash flows and asset values as follows. Capital calls and distributions are drawn independently from a Poisson distribution parametrized such that there is on average 13 capital calls lasting through the 225th week and 18 distributions per fund starting with 105th week. The first capital call is equal to 1 with subsequent capital calls (conditional on occurrence) equal to 0.25. The distribution size is drawn randomly as per equation (13) in the paper with δ set to 0.02 while $\sigma_d = 1.00$. The distribution amounts are determined sequentially with fund true asset values following equations (7–11). Fund life ends (i.e., $t = T$) on either 676th week (about 40% of the panel) or the week in which its true asset values drop below 1% of the initial capital call with the last distribution set to equal V_T (and the latter replaced with zero). The quarterly observed NAVs are constructed using equations (9–10) in the paper by setting λ to 0.90 and drawing the observation noise, e_{nt}, from $\mathcal{N}(0, \sigma_n^2)$, σ_n=0.05 independently for each fund-quarter.

We then estimate the SSM, filter returns and NAV series as described in Section A.1 for each one of the simulated funds. The naïve nowcasts are constructed from R_{ct} and the quarterly NAVs, as described in Section A.3 and are used to compute the initial asset-to-return mapping estimate. Importantly, we do not use the series of h_t in the SSM estimation, since these are not directly observable in real data either. Instead we estimate GARCH(1,1) models for each of the five industry return series on the residuals $\hat{e}_t$ from the projection of simulated R_{ct} on R_{mt} (which also yield the OLS estimate of β_m^c). As with real data, to produce fund-specific estimates of h_t, we use the matching industry GARCH coefficients but the fund-specific estimate of unconditional variance of $\hat{e}_t$ over that fund's life.

Results. Table A.1 reports estimation results for the main parameters of interest and the pricing error criteria using the holdout fund-weeks as discussed in Section 3.2. First seven columns of each panel report the means and the medians of parameter estimates across the 1,000 simulated funds as well as the first, second, and third quartile of the root mean squared error of the fund estimate relatively to the true parameter value. The last four columns report Hybrid (H) and Out-of-Sample (OOS) RMSEs, denoted with subscript Nai and SSM. The sample of 1,000 simulated funds is the same across panels (except for panel C). Additionally, Figure A.5 plots selected SSM estimands with the cross section of their true values (or naive nowcasts) at the individual fund level for the baseline simulation per Panel A of the table.

Panel A reports our *Baseline*-case that features fund-specific estimates (Section 3.3.1 of the main text) obtained under the assumption that the distribution function $\delta(\cdot)_t$ is not observed (i.e. the deal-level portfolio marks are not utilized to compute it) but estimated parametrically as per

[5] $log(\sigma^2)_{t+1} = v_0 + 0.9961 \cdot log(\sigma^2)_t + 0.0277 v_{t+1}$, where v_t is the standard normal shock on week t.

[6] This implies the volatility scale parameter $F \approx \sqrt{12.25} = 3.5$.

Equation (13). The panel reveals that estimated parameters are typically close to the true values, exhibiting minimal bias, in particular for α, β, λ, and δ. The parameters governing the noise of the distribution process σ_d and that of the NAV reports σ_n tend to be estimated with an upward bias of 10-to-30% of the true parameter value. Meanwhile the idiosyncratic risk parameter, F, tends to be estimated with a downward bias of 10-15%. Also, as with σ_n, we note that the 75th percentile RMSE for F is in the order of 50% of the true parameter value. Nonetheless, the panel demonstrates the advantage of the SSM method over the naïve method as the SSM-series produce median Hybrid and OOS RMSE that are 11-to-55% smaller than those per the naïve method. We note that the gap narrows for cases where error nowcasting error is large—i.e. the 75th percentile OSS RMSE of 0.258 is only 29% lower than the corresponding 0.361 value as the per the naive method. We also note that even though the median SSM nowcast errors funds are notably closer to zero than as per the naive method.

Figure A.5 plot individual fund estimates for this Baseline case. Panels A and B of the figure show that our fund-level estimates permit studying variation in α and β across funds. The estimates that we obtain via the profiled likelihood method (see Section A.4) are significantly correlated with the respective true values, especially in case of α whereby the slope is almost exactly 1. Whereas panels C and D of the figure show how individual fund's nowcasting errors (measured by logs of MSE for better visualization) compare between naïve and SSM methods via scatter plot. It is evident that (i) the majority of the points are below the 45-degree line, which indicates that the SSM nowcast is better in a given fund-case; (ii) there is no robust relation between the level of naïve and SSM errors, especially on the OOS basis. Consequently, the fund-cases where SSM fails to outperform are mostly those in which naïve nowcasts happened to be atypically good—e.g. lower than the median 0.126 OOS RMSE per SSM (corresponds to point -4.14 on x-axis of Panel D). Nonetheless, the panels shows that the SSM nowcasts are better than naïve for 68[87]% of simulated funds according to OOS[Hybrid] metric unconditionally on the quality of the naïve nowcasts.

In Panel B of Table A.1, we assume the distribution function $\delta(\cdot)_t$ is known and features no uncertainty $\sigma_d = 0$. As discussed in Section A.2.3, this corresponds to a situation in which one can observe the fund individual investments' valuation reports and actual exit terms. Unsurprisingly, this helps with estimation precision, especially, with respect to the variance parameters—F (σ_n) are no longer meaningfully biased while the estimation precisions, as measured by RMSEs, improve for each parameter by up to a factor of 2. Interestingly, the presence of jumps in the fund return process and the measurement errors embedded in the h_t-proxy do not prevent consistency in estimating the volatility scale parameter via MLE which assumes conditionally Gaussian returns. These increases in precision and decreases in biases result in about 15-to-30% lower Hybrid and OOS errors when compared to the Baseline: e.g., median OOS RMSE drops from 0.126 to 0.104. We note that the errors do not shrink to zero since the noisy NAVs and a potential for significant deviation from the comparable asset continue to bind.

In Panel C, we misspecify $\delta(\cdot)_t$ in the data generating process by assuming that $\delta(\cdot)_t$ is proportional to the time since last distribution (rather than the time since inception) as in Korteweg and Nagel (2016). This results in a different distributions schedule and NAV series for each fund. Therefore the nowcasting performance metrics for the naïve method are slightly different (but very similar to) from these in Panels A and B. We however observe no meaningful deterioration in either consistency or precision with regards key parameter estimands (except for δ and σ_d itself) and only a mild increase in the SSM-based nowcasting errors, especially when compared to the counterparts obtained via the naïve method.

In Panel D, we implement the partial imputation approach discussed in Section 3.3.2. To generate the β anchor, we compute the mean between the true β of the fund and the 1.25. We then set F and σ_n to equal 3.0 and 0.068, respectively, which corresponds to the mean parameter estimates per panel A of the table. The results suggest that higher precision on β is potentially attainable using this method. It may however come at a cost of a slight downward bias in α estimates in the order of about a quarter of the sample standard deviation in true α. Still, eliminating the estimation noise in F and σ_n—which Panel A suggest to be difficult to estimate reliably for a large fraction of the sample—results in an improved nowcasting performance. We note that the percentage decrease in the 75th percentile of OOS RMSE from 0.257 to 0.228 is about twice as large as for the 25-to-50th

percentiles, while the gains from utilizing portfolio-level information per Panel B were more salient for the half of funds with lower estimation errors. Intuitively, the imputation method reduces the scope for parameter-driven misspecification (so long as the imputed values are close enough to the true ones) but limits the extent that potentially better fund-specific information can drive larger gains in precision.

Table A.1. Monte Carlo experiment: parameter estimates and nowcasting performance

This table reports SSM estimates on simulated fund data described in Section A.6.2. Each row reports seven key parameters of interest as well Hybrid (H) and Out-of-Sample (OOS) nowcasting metrics as discussed in Section 3.2. Subscripts Nai and SSM denote the naïve and SSM-based nowcasts, respectively. 'Est. mean' and 'Est. median' refer to the simulated fund sample mean and median estimate of the respective parameter or metric. The means for for the last four columns are 5%-trimmed to reduce the effect of the outliers. 'RMSE p25/p50/p75' refer to the percentile of the Root Mean Squared Error relative to the true parameter value (or 1 for Hybrid/OOS columns). Panel A reports the results for the baseline estimation approach, Panel B treats distribution function as known, Panel C misspecifies the estimated distribution function relatively to the simulated data. Panel D reports results for the partially-imputed method. The true parameters' means are (0.0000, 1.250, 0.020, 0.90, 3.50, 1.00, 0.050).

Panel A: Baseline

	α	β	δ	λ	F	σ_d	σ_n	H_{Nai}	H_{SSM}	OOS_{Nai}	OOS_{SSM}
Est. mean	-0.0001	1.258	0.0228	0.877	3.051	1.1189	0.0679	0.036	0.033	-0.039	0.033
Est. median	-0.0001	1.278	0.0205	0.906	2.673	1.0660	0.0685	-0.115	0.010	-0.080	-0.020
RMSE p25	0.0005	0.091	0.0018	0.015	0.691	0.0786	0.0104	0.178	0.019	0.106	0.059
RMSE p50	0.0012	0.184	0.0040	0.034	0.990	0.1900	0.0211	0.357	0.041	0.218	0.126
RMSE p75	0.0021	0.271	0.0072	0.068	1.386	0.3509	0.0363	0.568	0.093	0.361	0.258

Panel B: Known $\delta(\cdot)_t$

	α	β	δ	λ	F	σ_d	σ_n	H_{Nai}	H_{SSM}	OOS_{Nai}	OOS_{SSM}
Est. mean	-0.0001	1.254	-	0.901	3.480	-	0.0545	0.036	0.027	-0.039	0.059
Est. median	-0.0000	1.251	-	0.907	3.138	-	0.0551	-0.115	0.004	-0.080	-0.012
RMSE p25	0.0005	0.074	-	0.009	0.320	-	0.0052	0.178	0.015	0.106	0.042
RMSE p50	0.0011	0.159	-	0.018	0.674	-	0.0112	0.357	0.034	0.218	0.104
RMSE p75	0.0020	0.247	-	0.034	1.049	-	0.0186	0.568	0.077	0.361	0.209

Panel C: Misspecified $\delta(\cdot)_t$

	α	β	δ	λ	F	σ_d	σ_n	H_{Nai}	H_{SSM}	OOS_{Nai}	OOS_{SSM}
Est. mean	-0.0001	1.263	0.0117	0.884	3.078	1.6965	0.0698	0.044	0.046	-0.038	0.058
Est. median	-0.0002	1.278	0.0092	0.905	2.630	1.6209	0.0682	-0.113	0.012	-0.071	-0.007
RMSE p25	0.0005	0.087	0.0081	0.015	0.701	0.3931	0.0107	0.182	0.020	0.100	0.062
RMSE p50	0.0010	0.174	0.0110	0.031	0.999	0.6209	0.0202	0.363	0.048	0.229	0.133
RMSE p75	0.0018	0.263	0.0132	0.058	1.332	0.9290	0.0337	0.593	0.121	0.389	0.260

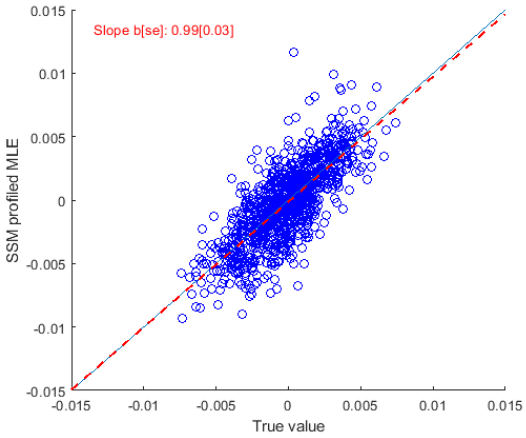
Panel D: Imputed F and σ_n

	α	β	δ	λ	F	σ_d	σ_n	H_{Nai}	H_{SSM}	OOS_{Nai}	OOS_{SSM}
Est. mean	-0.0006	1.237	0.0228	0.884	-	1.1196	-	0.036	0.025	-0.039	0.041
Est. median	-0.0005	1.243	0.0206	0.890	-	1.0710	-	-0.115	0.006	-0.080	-0.019
RMSE p25	0.0006	0.093	0.0018	0.010	-	0.0786	-	0.178	0.017	0.104	0.056
RMSE p50	0.0012	0.116	0.0039	0.030	-	0.1883	-	0.357	0.039	0.218	0.118
RMSE p75	0.0022	0.139	0.0071	0.065	-	0.3485	-	0.568	0.084	0.361	0.228

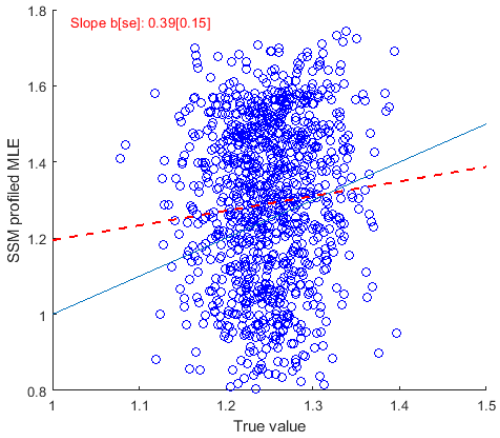
Figure A.5. Monte Carlo experiment: cross-section of α,β and nowcasting performance

This figure reports SSM estimates on simulated fund data described in Section A.6.2. The solid blue and red dashed lines are, respectively, the 45-degree and the least-squared fit lines. Panel A (B) plots the true values of fund αs (βs) on the x-axis against the SSM-based estimates on y-axis. A positive association and low/or variance across y-axis indicate good performance of our method. The 'Slope b[se]:'-text on the respective panel is followed by the coefficient and the standard error from a regression of the estimated values onto the true values. Panels C and D plot the log of mean squared errors obtained using the naïve method on the x-axis against SSM-based estimated fund returns on y-axis. Panel C (D) uses Hybrid (Out-of-Sample) metric discussed in Section 3.2. Observations below the 45-degree line indicate outperformance of our method over a naïve approach that simply uses public benchmark returns to model fund NAVs. The 'Improve Rate:'-text on the respective panel is followed by the fraction of observations below the 45-degree line.

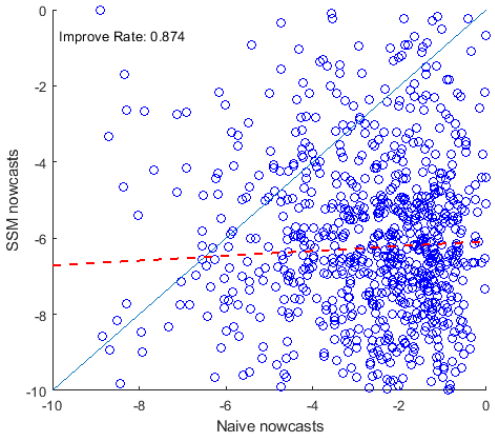
Panel A: α estimates



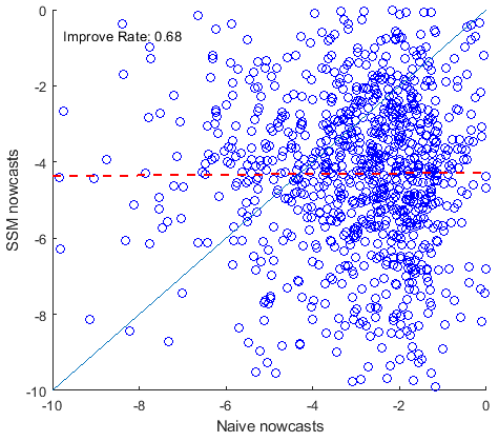
Panel B: β estimates



Panel C: log MSE Hybrid



Panel D: log MSE OOS



A.7 Excerpts from AICPA Practitioner's Guide

The Guide we review has been developed by the American Institute of Certified Public Accountants (AICPA) PE/VC Task Force.[7] This 688-page document provide specific instruction and examples "for preparers of financial statements, independent auditors, and valuation specialists regarding the accounting for and valuation of portfolio company investments of venture capital funds, private equity funds and other investment companies." The document states that its content has been "reviewed and approved by the affirmative vote of at least two-thirds of the members of the Financial Reporting Executive Committee (FinREC), which is the designated senior committee of the AICPA authorized to speak for the AICPA in the areas of financial accounting and reporting."

The guide defines three valuation techniques:

> [p.6] –The *market approach* bases the value measurement on market data (for example, valuing an enterprise based on values for comparable public companies or similar transactions or valuing interests based on transactions in similar instruments). ...
>
> –The *income approach* seeks to convert future economic benefits into a present value for the enterprise or the interests in the enterprise.
>
> –The *asset approach* estimates the value of an enterprise or the interests in the enterprise based on the principle that the equity value is equivalent to the values of its individual assets net of its liabilities.

As we will show below, the first two approaches are consistent with how we model smoothing. The third approach does not. However, the guide advocates against relying relaying on the *asset approach* whenever the first two (i.e., Comps and DCF) are feasible:

> [p.151] Of the three approaches to valuing an operating enterprise and the interests in the enterprise under a going concern premise of value, the asset (or asset-based) approach, under most circumstances, is considered to be the weakest from a conceptual standpoint.
>
> [p.55] A fair value measurement assumes that the transaction to sell the asset or transfer the liability takes place either:
>
> a. In the principal market for the asset or liability
>
> b. In the absence of a principal market, in the most advantageous market for the asset or liability.
>
> ... the principal market is "[t]he market with the greatest volume and level of activity for the asset or liability." Also, as defined in the FASB ASC Master Glossary, the most advantageous market is "[t]he market that maximizes the amount that would be received to sell the asset..."

Between Comps and DCF, the guide appears to slightly favor comparable, especially if a trade-sale is deemed a more likely exit. Empirically. the exits for the vast majority of buyout and venture deals are trade-sales (see, e.g., Braun et al., 2017). Furthermore, the guide advocates against tempering with those historical multiples. This makes it even more likely for reported NAVs to be weighted-average snapshots of valuations corresponding to the past transaction dates, which is consistent with smoothing the cumulatative (rather than per period) returns:

> [p.56] Note that even at interim measurement dates, it would be important to consider the ultimate exit strategy for the investment because market participants in the sponsor-to-sponsor market would consider their expected exit strategy when evaluating the position.

[7] The version we reviewed is dated June 1, 2019. See https://www.aicpa.org/interestareas/frc.html for additional information.

> [p.57] As such, even in situations in which the market for a particular asset is deemed not to be active, relevant prices or inputs from this market would still need to be considered in the determination of fair value. It would not be appropriate to default solely to a valuation model's value based on unobservable inputs (a level 3 measurement), when observable inputs other than quoted prices (level 2 information) is available. However, being able to transact in a particular market (as earlier discussed in paragraphs 2.18–.19) is a key consideration in identifying the appropriate inputs used to estimate fair value.
>
> [p.241] FASB ASC 820-10-35-36 states that "[v]aluation techniques used to measure fair value shall maximize the use of relevant observable inputs and minimize the use of unobservable inputs." Therefore, to the extent that transactions in an active market and secondary market transactions meet the other characteristics required to be relevant indications of fair value, the task force believes it may be appropriate to prioritize these transactions over model-based estimates of fair value.
>
> [p.136] The task force recommends considering the differences between the portfolio company and the selected guideline public companies or transactions directly when selecting the multiples, rather than applying a premium or discount to some arbitrary or formulaic calculation. Calibration should be used to select multiples that are consistent with the transaction price, as long as the transaction price reflects fair value at initial recognition.

However, smoothing is implicit in recommendations about the DCF as well:

> [p.230] In the next measurement period, it typically would not be appropriate to ignore the required rate of return implied by the transaction and assume that the WACC for the company would suddenly fall to be consistent with the median of the guideline public companies. Instead, at subsequent measurement dates, the valuation would consider the portfolio company's progress and changes in observable market data (such as changes in the equity risk premium or betas for the selected guideline public companies) to estimate the fair value under current market conditions.

The guide explicitly (and repeatedly) states that more recent transactions should have larger weight in the appraisal process but advises against using information beyond the measurement date. This provides a microfoundation for modeling NAVs as exponentially moving average of the past cumulative returns.

> [p.135] It may not be appropriate to use guideline company transactions that took place during periods in which economic conditions were not the same as they are at the measurement date, without appropriate adjustments. There are no bright lines but, in general, the older the transaction, the less relevant the information. It is common practice to compare as many guideline company transactions as can be identified during a relevant recent historical time period.
>
> [p.227] The fund would consider the time that has elapsed between the transaction date and measurement date as one factor in assessing the relative reliability of the calibrated valuation model using the observable public stock price and the calibrated valuation model using the fund's direct valuation models.
>
> [p.228] the task force believes one would need to consider the time that has elapsed between the transaction date and measurement date. The point at which a transaction becomes "stale" will vary depending on specific facts and circumstances.
>
> [p.54] Therefore, a fair value measurement considers market conditions as they exist at the measurement date (not at some point in the future), information which is known or knowable at the measurement date, and is intended to represent the current value of the asset or liability, not the potential value of the asset or liability at some future date.

The guide recommends two practices "Calibration" and "Backtesting" to be consistent in accounting for differences between the portfolio company and the comparable asset and in updating of those assumptions. They too produce smoothing. They also provide a microfoundation for how we model $\lambda(\cdot)_t$ as a decreasing function of the fraction the new capital calls and distributions comprise in the reported NAVs (see Section A.2.2).

> [p.207] Most valuation techniques described in this guide require significant unobservable inputs and, therefore, performing a valuation analysis requires judgment. In the absence of relevant transaction data, selecting these inputs typically involves using observable comparable data and making adjustments to capture differences between the portfolio company and the guideline public companies or transactions. Calibration is used when a transaction price that represents fair value is available and provides evidence supporting the unobservable inputs used in each selected valuation technique as of the transaction date. The calibrated inputs may be compared with observable comparable data, providing an initial indication of the adjustments needed to capture the specific characteristics of the investment.
>
> [p.229] Continuing this example, suppose that after 6 months, the company is performing as expected. LTM EBITDA has improved to $55 million, and forward EBITDA has improved to $64 million. Furthermore, the median multiple for the guideline public companies has improved to 9x the LTM EBITDA instead of 8x, and 7.5x forward EBITDA instead of 7x. After considering the company's recent performance and positioning, the fund concludes that market participants would still expect the company to outperform the guideline public companies but not to the same extent. To update the valuation, the fund selects a multiple of 10.5x LTM EBITDA, indicating a value of $577.5 million, and 8.5x forward EBITDA, or $544 million. Averaging these indications of value, the fund concludes on an enterprise value of $560.75 million.
>
> [p.234] The PE/VC Task Force believes that it is a best practice for investment companies whose financial statements are based upon fair value estimates to perform periodic (for example, quarterly) backtesting on investments that have had subsequent realizations, liquidity, or other significant events, comparing the implied value from the transaction to the fair value estimate from the most recent analysis as well as valuations from other prior periods that may be deemed relevant. Backtesting provides an ongoing feedback loop that could enhance the rigor of the fund's valuation processes for periodic fair value estimates and may contribute to the fund's system of internal control over financial reporting.

The guide seems to emphasizes the importance of industry, even relative to the size&style, in how to select comparable assets and transactions:

> [p.125] As a foundation for estimating value, and before diving into the details on any particular valuation approach or method, it is critical to start by understanding what the business is. Is it a service business? Manufacturing? What industry does it operate in? What is its customer base? What is the stage of its development? What is the portfolio company's strategy and expected ultimate exit market? See chapter 1, "Overview of the Private Equity and Venture Capital Industry and Its Investment Strategies," and chapter 3, "Market Participant Assumptions," for further discussion of these issues.
>
> Similar operational characteristics, such as the following:
>
> – Same industry or sector (the North American Industry Classification System or the Standard Industrial Classification code)
>
> – Similar lines of business
>
> – Geographic reach (for example, domestic versus international versus multinational)
>
> – Similar customers and distribution channels
>
> [p.128] Similar financial characteristics, such as the following:

– Similar size (for example, revenues, assets, or market capitalization, if subject is public)

– Similar profitability (for example, earnings before interest, taxes, depreciation, and amortization [EBITDA], operating margin, contribution margin)

– Similar anticipated future growth in revenues and profits

– Similar asset-base (for example, manufacturing versus service business)

[p.131] A nonfinancial metric is often industry-specific and would ordinarily be used when the nonfinancial metric is generally accepted in the industry and would be considered by market participants.

Figure A.6. PE index risk loadings — time series

This figure reports coefficients from the weekly regressions of the SSM-based return indices on Fama-French risk factors. Each regression estimated separately for a given calendar year. We consider four portfolios: (i) buyout funds, (ii) venture funds, (iii) venture funds with the early stage focus, and (iv) venture funds with a late stage focus. The indices is constructed from weekly estimates of the individual fund-level returns starting from the first week of 1987 through the last week of 2019. The sample includes 3,142 funds vintaged between 1983 and 2011.

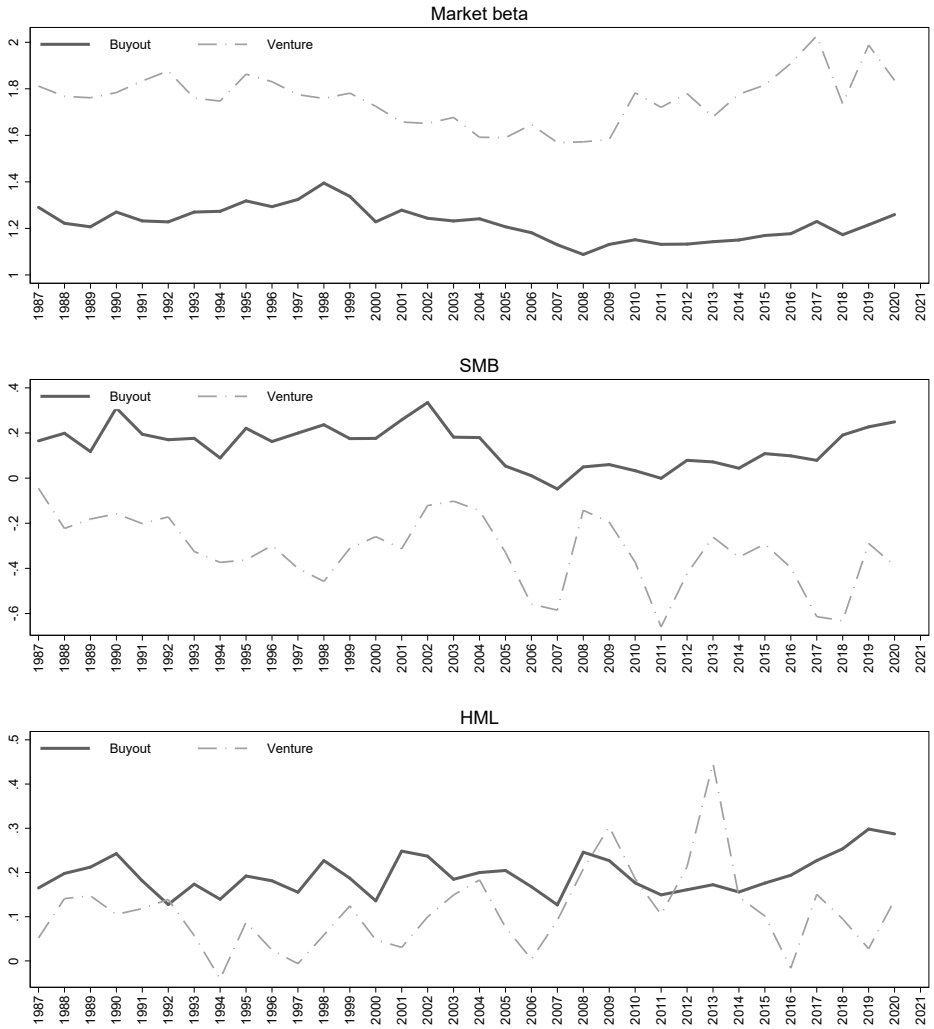


Table A.2. Real Estate Private Equity (REPE) funds

This table reports summary statistics for the sample of 602 Real Estate funds in our sample, selected SSM parameter estimates, filtered returns, and nowcasting performance metrics. See Tables 1 and 3 for definitions and details.

	Number of funds: 726									
	mean	sd	skew	p5	p10	p25	p50	p75	p90	p95
Vintage year	2006.0	5.6	-0.7	1995	1998	2004	2007	2010	2013	2014
Fund size (USD mil.)	635.1	970.0	6.5	67.6	100	204.1	397.4	713.2	1241.6	1829.3
Fund life (years)	10.7	2.5	-0.2	6.54	7.06	8.60	10.8	13.1	14.1	14.1
IRR	0.1	0.1	-1.1	-0.14	-0.084	0.0003	0.076	0.14	0.20	0.25
Money Multiple	1.3	0.6	1.0	0.39	0.60	1	1.34	1.62	2	2.27
PME (v. CRSP VW)	0.9	0.4	0.9	0.23	0.35	0.62	0.88	1.12	1.44	1.68
# Capital Calls	23.5	18.9	1.9	6	8	11	17	29	52	65
# Distributions	32.7	24.8	1.9	6	9	16	27	43	62	78
# NAV reports	21.1	2.4	-0.7	17	17	19	22	23	24	24
% Resolved	86.3	20.3	-2.0	37.1	57.6	82.8	96.3	98.9	99.5	99.6
# Quarters w/ Dist.> 0	10.7	6.3	0.2	1	2	6	10	15	20	21
w/ Dist > 5% of Fund	5.7	4.3	0.9	0	1	2	5	8	12	14

	mean	p25	p50	p75	mean	p25	p50	p75
Key parameter estimates:								
	α				β			
Fund-specific	0.036	0.007	0.064	0.129	0.799	0.655	0.748	0.884
Peer-imputed	−0.021	−0.042	0.032	0.097	0.694	0.587	0.660	0.799
Average-imputed	−0.017	−0.046	0.027	0.092	0.724	0.664	0.664	0.696
Literature-imputed	−0.036	−0.026	0.045	0.096	0.770	0.770	0.770	0.770
	F_c				λ			
Fund-specific	0.816	0.756	0.844	0.950	0.861	0.901	0.963	0.986
Peer-imputed	0.995	1.014	1.041	1.056	0.909	0.856	0.914	0.976
Average-imputed	1.006	1.016	1.042	1.056	0.873	0.806	0.947	0.969
Literature-imputed	0.810	0.755	0.823	0.949	0.871	0.914	0.963	0.982
Filtered return properties:								
	Weekly Autocorrelation				Annualized Standard Deviations			
Fund-specific	−0.027	−0.102	−0.058	0.024	0.209	0.152	0.185	0.234
Peer-imputed	0.071	−0.066	0.026	0.170	0.157	0.119	0.141	0.179
Average-imputed	0.080	−0.042	0.024	0.150	0.166	0.125	0.147	0.191
Literature-imputed	−0.026	−0.092	−0.047	0.004	0.203	0.159	0.187	0.212
Nowcast performance assessment:								
	Hybrid RMSE				OOS RMSE			
Fund-specific	0.076	0.021	0.048	0.095	0.180	0.045	0.097	0.225
Peer-imputed	0.062	0.018	0.039	0.074	0.127	0.028	0.070	0.164
Average-imputed	0.059	0.018	0.037	0.069	0.116	0.026	0.063	0.140
Literature-imputed	0.134	0.020	0.040	0.087	0.260	0.047	0.083	0.182

	Hybrid Improv. %	OOS Improv. %	Autocorr. Improv. %
Fund-specific	83.6	65.8	53.0
Peer-imputed	89.5	76.9	53.4
Average-imputed	89.0	76.8	56.0
Literature-imputed	87.2	66.9	52.9

References for Internet Appendix

Albuquerque, R. A., J. Cassel, L. Phalippou, and E. J. Schroth. 2018. Liquidity Provision in the Secondary Market for Private Equity Fund Stakes. *Oxford University Working Paper* .

Ang, A., B. Chen, W. N. Goetzmann, and L. Phalippou. 2018. Estimating private equity returns from limited partner cash flows. *Journal of Finance* 73:1751–1783.

Braun, R., T. Jenkinson, and I. Stoff. 2017. How persistent is private equity performance? Evidence from deal-level data. *Journal of Financial Economics* 123:273–291.

Brown, G. W., O. R. Gredil, and S. N. Kaplan. 2019. Do private equity funds manipulate reported returns? *Journal of Financial Economics* 132:267–297.

De Jong, P. 1988. The likelihood for a state space model. *Biometrika* 75:165–169.

Durbin, J., and S. J. Koopman. 2012. *Time series analysis by state space methods.* Oxford University Press.

Ewens, M., C. M. Jones, and M. Rhodes-Kropf. 2013. The price of diversifiable risk in venture capital and private equity. *Review of Financial Studies* 26:1854–1889.

Geltner, D. 1993. Temporal aggregation in real estate return indices. *Real Estate Economics* 21:141–166.

Getmansky, M., A. W. Lo, and I. Makarov. 2004. An econometric model of serial correlation and illiquidity in hedge fund returns. *Journal of Financial Economics* 74:529–609.

Gornall, W., and I. A. Strebulaev. 2020. Squaring venture capital valuations with reality. *Journal of Financial Economics* 135:120–143.

Gredil, O., B. E. Griffiths, and R. Stucke. 2014. Benchmarking private equity: The direct alpha method. Tulane University Working Paper.

Kalman, R. E. 1960. A new approach to linear filtering and prediction problems. *Journal of Basic Engineering* 82:35–45.

Kohn, R., and C. F. Ansley. 1989. A fast algorithm for signal extraction, influence and cross-validation in state space models. *Biometrika* 76:65–79.

Korteweg, A., and S. Nagel. 2016. Risk-Adjusting the Returns to Venture Capital. *Journal of Finance* 71:1437–1470.

Korteweg, A., and S. Nagel. 2020. Risk-Adjusted Returns of Private Equity Funds: A New Approach. University of South California Working Paper.

Miranda, M. J., and P. L. Fackler. 2004. *Applied Computational Economics and Finance.* MIT Press.

Nadauld, T. D., B. A. Sensoy, K. Vorkink, and M. S. Weisbach. 2019. The liquidity cost of private equity investments: Evidence from secondary market transactions. *Journal of Financial Economics* 132:158–181.